

Options Pricer and Portfolio Risk Engine

This notebook presents the project in the same order that a reviewer would read it: data, features, forecasting, pricing, validation, Monte Carlo, portfolio construction, scenario risk and Greek P&L attribution.

The reusable implementation lives in `src/options_risk_engine` ; the notebook contains the run-specific orchestration, displays and interpretation points.

1. Environment and reproducible run configuration

Central project settings

The next code cell contains the run assumptions that are intended to be changed between executions. Later notebook cells reference these named values rather than redefining the same assumptions.

This keeps the data, forecasting, pricing, Monte Carlo, scenario, attribution and final-ranking stages aligned with one configuration.

```
In [1]: from pathlib import Path
import logging
import os

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from IPython.display import display

# =====
# CENTRAL PROJECT SETTINGS
# Edit run assumptions here rather than throughout the notebook.
# =====

# -----
# 1. Project location and valuation date
# -----

PROJECT_ROOT = Path(
    "/Users/sam/Progressive-Projects/"
    "Projects/Option Pricer N Risk Engine/src"
)

# Leave as None to use the current date. Use a fixed ISO date
# when reproducing a previously stored market snapshot.
VALUATION_DATE_OVERRIDE = None

DATABASE_PATH = (
    PROJECT_ROOT
    / "data"
```

```

    / "options_risk_engine.db"
)

os.chdir(PROJECT_ROOT)

from options_risk_engine.config import (
    FIGURE_DIR,
    SCHEMA_PATH,
    TABLE_DIR,
    LOG_DIR,
    ProjectConfig,
    ensure_project_directories,
)
from options_risk_engine.logging_config import configure_logging
from options_risk_engine.database import SQLiteStore

# -----
# 2. Underlying universe and historical data
# -----

# Equity underlyings included in the complete workflow.
SYMBOLS = (
    "AAPL",
    "MSFT",
    "META",
    "WMT",
    "GOOGL",
    "AMZN",
    "HCA",
)

# Desired option maturity in calendar days.
TARGET_DTE = 45

# Historical market-data period and sampling frequency.
HISTORY_PERIOD = "5y"
HISTORY_INTERVAL = "1d"

# Common continuously compounded annual risk-free rate.
RISK_FREE_RATE = 0.0375

# Maximum historical observations used by the final volatility
# estimate. Approximately five trading years at daily frequency.
FORECAST_LOOKBACK = 1260

# -----
# 3. Volatility-model evaluation
# -----

# True runs the full forecasting benchmark. False is useful for
# quicker pricing and risk-engine development runs.
RUN_FORECAST_BENCHMARKS = True

# Fraction used by the separate historical-mean holdout check.
FORECAST_TEST_FRACTION = 0.20

```

```

# -----
# 4. Initial valuation and contract-screening assumptions
# -----

# Minimum forecast-value advantage over the ask required for
# the initial Buy label. 0.05 means 5%.
INITIAL_BUY_EDGE = 0.05

# Numerical tolerance used by put-call parity checks.
PARITY_TOLERANCE = 1e-6

# Liquidity and moneyness rules used by the highlighted initial
# Buy table. These do not change the broader recommendation
# labels retained for Sections 18-21.
INITIAL_MAX_SPREAD_PCT = 0.15
INITIAL_MIN_ASK = 0.50
INITIAL_MIN_OPEN_INTEREST = 100
INITIAL_MIN_VOLUME = 10
INITIAL_MIN_MONEYNES = 0.85
INITIAL_MAX_MONEYNES = 1.15

# -----
# 5. Monte Carlo settings
# -----

RUN_MONTE_CARLO = True

# Simulated terminal stock prices per tested contract.
MONTE_CARLO_SIMULATIONS = 10_000

# Whether to display the individual Monte Carlo distributions.
PLOT_MONTE_CARLO = False

# -----
# 6. Scenario-engine settings
# -----

# Number of random stress scenarios. The unchanged BASE state
# is added separately, producing SCENARIO_COUNT + 1 scenarios.
SCENARIO_COUNT = 1_000

# Maximum calendar time allowed to pass in a random scenario.
MAX_DAYS_FORWARD = 30

# Reproduces the same simulations and shocks when inputs match.
RANDOM_SEED = 42

BASE_SCENARIO_ID = "BASE"

# -----
# 7. Synthetic portfolio loss limits
# -----

# Absolute loss levels used by the breach reports.
MAX_PORTFOLIO_LOSS = 100_000.0
MAX_TICKER_LOSS = 25_000.0

```

```

# -----
# 8. Standardised contract analysis
# -----

# Initial labels retained as comparison groups.
CONTRACT_ANALYSIS_ACTIONS = (
    "Buy",
    "Positive Edge",
    "Do Not Buy",
)

# Lower-tail probability used for the fifth percentile and
# expected shortfall. Keep at 0.05 unless the output-column
# names containing "5pct" are also generalised.
CONTRACT_TAIL_PROBABILITY = 0.05

# -----
# 9. Local Greek-attribution subset
# -----

# Maximum absolute shocks used to judge the fixed-Greek
# approximation in a smaller local region.
LOCAL_SPOT_LIMIT = 0.05
LOCAL_VOLATILITY_LIMIT = 0.03
LOCAL_DAYS_LIMIT = 10
LOCAL_RATE_LIMIT = 0.005

# Minimum observations required for a local reliability label.
MINIMUM_LOCAL_SCENARIOS = 20

# -----
# 10. Final liquidity gates
# -----

# Maximum spread relative to the option midpoint.
FINAL_MAX_SPREAD_PCT = 0.15

# Minimum reported daily volume and open interest.
FINAL_MIN_VOLUME = 1
FINAL_MIN_OPEN_INTEREST = 100

# -----
# 11. Final recommendation thresholds
# -----

CANDIDATE_SCORE = 70.0
WATCHLIST_SCORE = 55.0

# Candidate must gain in at least this share of generated
# non-BASE scenarios.
CANDIDATE_PROFIT_FREQUENCY = 0.50

# Candidate and Watchlist rows require positive mean return.
MINIMUM_MEAN_RETURN = 0.0

# Bottom share of liquid contracts considered by the relative

```

```

# lower-tail risk gate.
RISK_TAIL_QUANTILE = 0.10

# A contract in that weak tail is rejected when its gain
# frequency is also below this level.
RISK_PROFIT_FREQUENCY = 0.35

# -----
# 12. Final-score weights
# These must sum to exactly 1.0.
# -----

FINAL_SCORE_WEIGHTS = {
    "Forecast_Edge_Score": 0.25,
    "Mean_Return_Score": 0.20,
    "Median_Return_Score": 0.10,
    "Profit_Frequency_Score": 0.15,
    "Tail_Risk_Score": 0.20,
    "Liquidity_Score": 0.10,
}

# -----
# Construct the run configuration
# -----

CONFIG = ProjectConfig(
    symbols=SYMBOLS,
    target_dte=TARGET_DTE,
    history_period=HISTORY_PERIOD,
    history_interval=HISTORY_INTERVAL,
    risk_free_rate=RISK_FREE_RATE,
    forecast_lookback=FORECAST_LOOKBACK,
    buy_edge=INITIAL_BUY_EDGE,
    parity_tolerance=PARITY_TOLERANCE,
    monte_carlo_simulations=MONTE_CARLO_SIMULATIONS,
    scenario_count=SCENARIO_COUNT,
    max_days_forward=MAX_DAYS_FORWARD,
    random_seed=RANDOM_SEED,
    max_portfolio_loss=MAX_PORTFOLIO_LOSS,
    max_ticker_loss=MAX_TICKER_LOSS,
    run_forecast_benchmarks=RUN_FORECAST_BENCHMARKS,
    run_monte_carlo=RUN_MONTE_CARLO,
    plot_monte_carlo=PLOT_MONTE_CARLO,
    database_path=DATABASE_PATH,
)

VALUATION_DATE = (
    pd.Timestamp.today().normalize()
    if VALUATION_DATE_OVERRIDE is None
    else pd.Timestamp(
        VALUATION_DATE_OVERRIDE
    ).normalize()
)

pd.set_option("display.max_columns", None)
pd.set_option("display.width", 180)

ensure_project_directories()

```

```

configure_logging(
    logging.INFO,
    LOG_DIR / "notebook_run.log",
)

if not np.isclose(
    sum(FINAL_SCORE_WEIGHTS.values()),
    1.0,
):
    raise ValueError(
        "FINAL_SCORE_WEIGHTS must sum to 1.0."
    )

print(CONFIG)
print("Valuation date:", VALUATION_DATE.date())
print(
    "Final-score weight total:",
    sum(FINAL_SCORE_WEIGHTS.values()),
)

```

```

ProjectConfig(symbols=('AAPL', 'MSFT', 'META', 'WMT', 'GOOGL', 'AMZN', 'HC
A'), target_dte=45, history_period='5y', history_interval='1d', risk_free_
rate=0.0375, forecast_lookback=1260, buy_edge=0.05, parity_tolerance=1e-0
6, monte_carlo_simulations=10000, scenario_count=1000, max_days_forward=3
0, random_seed=42, max_portfolio_loss=100000.0, max_ticker_loss=25000.0, r
un_forecast_benchmarks=True, run_monte_carlo=True, plot_monte_carlo=False,
database_path=PosixPath('/Users/sam/Progressive-Projects/Projects/Option P
ricer N Risk Engine/src/data/options_risk_engine.db'))

```

Valuation date: 2026-08-02

Final-score weight total: 1.0000000000000002

2. Data lineage and SQLite snapshot store

A pricing engine should be able to explain which data and assumptions produced a particular result. Each execution is therefore assigned a unique run ID, and the major intermediate and final tables are stored with that identifier.

SQLite is appropriate here because the project data is relational, the individual tables are easily queried by date, ticker or scenario, and the database can be included directly in the project repository. The run-based design also prevents a new execution from silently overwriting an older market snapshot.

```

In [2]: store = SQLiteStore(CONFIG.database_path, SCHEMA_PATH)
store.initialise()
run_id = store.start_run(
    valuation_date=str(VALUATION_DATE.date()),
    notes="Notebook pricing and risk-engine run",
)
print("Run ID:", run_id)
display(store.list_tables())

```

2026-08-02 13:36:35,925 - options_risk_engine.database - INFO - SQLite dat
abase initialised at /Users/sam/Progressive-Projects/Projects/Option Price
r N Risk Engine/src/data/options_risk_engine.db

Run ID: c15636bc18ca4e76bbe3785801a01545

	name
0	equity_prices
1	model_evaluations
2	option_chain_snapshots
3	option_chain_snapshots_full
4	option_pricing_results
5	portfolio_positions
6	portfolio_risk
7	project_runs
8	volatility_features

Database structure

The database was initialised successfully and contains tables for historical prices, engineered features, option-chain snapshots, pricing results, portfolio positions, scenario definitions, scenario P&L and Greek attribution.

This provides a complete chain from the raw market inputs to the final risk outputs. A reviewer can therefore inspect not only the final result but also the exact data used to obtain it.

The very large position-by-scenario tables are still acceptable for this project. For a materially larger universe, a columnar format such as Parquet or a query engine such as DuckDB would be more efficient than placing every high-dimensional result in SQLite.

3. Underlying configuration and forecast horizon

Each `OptionTicker` object stores the market and pricing state associated with one underlying, including its target maturity, interest rate, dividend yield, selected expiry, spot price and volatility inputs.

The object is deliberately used as a state container rather than containing the entire pricing workflow. Separating the stored state from the pricing and validation functions makes the numerical methods easier to test independently.

```
In [3]: from options_risk_engine.domain import OptionTicker

tickers = [
    OptionTicker(
        symbol=symbol,
        target_dte=CONFIG.target_dte,
        risk_free_rate=CONFIG.risk_free_rate,
        forecast_lookback=CONFIG.forecast_lookback,
        buy_edge=CONFIG.buy_edge,
```

```

        parity_tolerance=CONFIG.parity_tolerance,
        valuation_date=VALUATION_DATE,
    )
    for symbol in CONFIG.symbols
]

ticker_setup = pd.DataFrame({
    "Ticker": [ticker.symbol for ticker in tickers],
    "Target DTE": [ticker.target_dte for ticker in tickers],
    "Forecast Horizon": [ticker.forecast_horizon for ticker in tickers],
    "Risk-Free Rate": [ticker.risk_free_rate for ticker in tickers],
})
display(ticker_setup)

```

	Ticker	Target DTE	Forecast Horizon	Risk-Free Rate
0	AAPL	45	32	0.0375
1	MSFT	45	32	0.0375
2	META	45	32	0.0375
3	WMT	45	32	0.0375
4	GOOGL	45	32	0.0375
5	AMZN	45	32	0.0375
6	HCA	45	32	0.0375

Horizon alignment

All seven underlyings use a target maturity of 45 calendar days. At the valuation date this corresponds to a 32-trading-day volatility forecast horizon.

This alignment is important because the volatility supplied to the option pricer should represent approximately the same period as the selected option. A forecast over a materially different horizon would make comparisons between forecast volatility, implied volatility and model price less meaningful.

4. Historical adjusted-close data

Historical adjusted closing prices provide the basis for returns, realised volatility and the forecasting features used later in the project.

Before creating any features, the data is checked for missing observations, date coverage and implausible gaps. This is necessary because rolling features and future targets can otherwise contain unexpected missing values or become misaligned across tickers.

```

In [4]: from options_risk_engine.data.market_data import (
        download_historical_market_data,
        equity_prices_to_long,
        fetch_dividend_yields,
    )

```



```

from options_risk_engine.reporting.plots import plot_price_history

close_data = download_historical_market_data(
    CONFIG.symbols,
    period=CONFIG.history_period,
    interval=CONFIG.history_interval,
)
dividend_yields = fetch_dividend_yields(tickers)

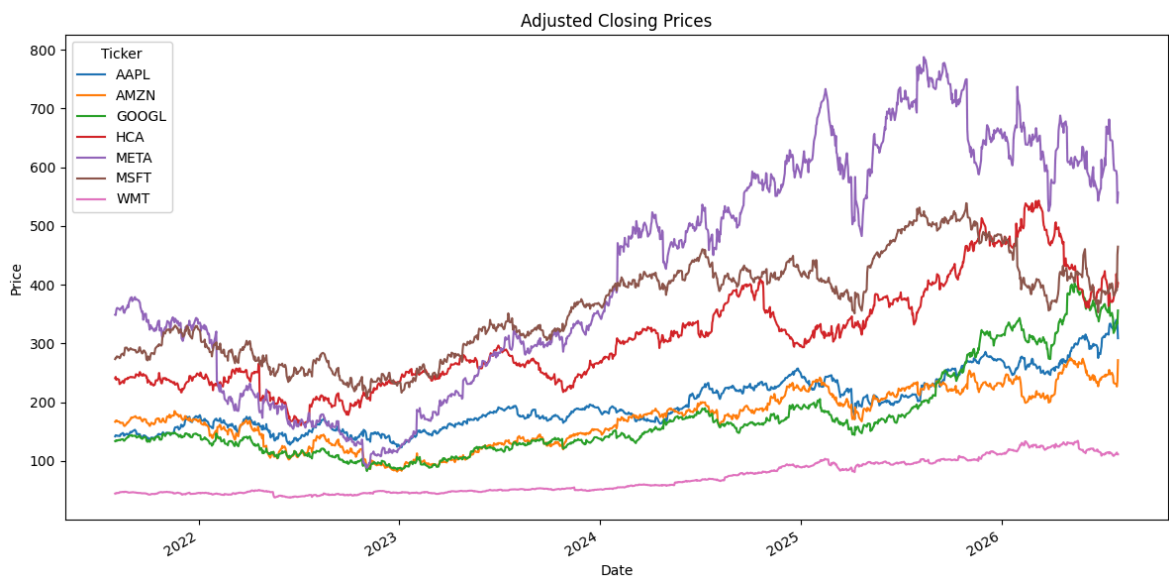
latest_prices = close_data["Close"].tail().round(2)
print("Historical rows:", len(close_data))
print("Missing close observations:", int(close_data.isna().sum().sum()))
display(latest_prices)
plot_price_history(close_data, FIGURE_DIR / "01_price_history.png")
plt.show()

```

/Users/sam/Progressive-Projects/.venv/lib/python3.9/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2.8.3'. See: <http://github.com/urllib3/urllib3/issues/3020>

```
warnings.warn(
2026-08-02 13:36:36,252 - options_risk_engine.data.market_data - INFO - Do
wnloading historical data for 7 symbols
2026-08-02 13:36:37,150 - options_risk_engine.data.market_data - INFO - Hi
storical data downloaded: 1255 rows
Historical rows: 1255
Missing close observations: 0
```

Ticker	AAPL	AMZN	GOOGL	HCA	META	MSFT	WMT
Date							
2026-07-27	336.91	231.39	326.56	389.62	593.87	389.10	111.74
2026-07-28	340.08	230.86	333.71	418.19	593.41	393.35	113.10
2026-07-29	338.19	226.65	336.71	408.18	585.61	390.54	114.22
2026-07-30	333.43	235.50	333.66	395.79	539.03	451.10	111.10
2026-07-31	308.91	271.58	356.13	402.59	556.71	464.72	111.20



Historical-data observations

The download produced 1,255 daily observations for each of the seven underlyings, with no missing adjusted-close values in the aligned dataset. This provides approximately five years of common history.

The price chart shows substantial differences in both price level and long-term path between the companies. Raw prices are therefore not used directly as comparable forecasting features. Percentage returns and annualised volatility measures are created in the next stage.

The final dates also contain several large one-day movements. These recent observations will have a stronger effect on short-window realised volatility than on the longer 252-day measure.

Persisting the historical price snapshot

The wide market-data table is converted into a long relational format with one row per date and ticker. This format is easier to query in SQL and avoids creating a separate database column for every underlying.

```
In [5]: equity_prices_long = equity_prices_to_long(close_data)
store.write_frame(equity_prices_long, "equity_prices", run_id=run_id)
display(equity_prices_long.tail(10))
```

2026-08-02 13:36:38,954 – options_risk_engine.database – INFO – Stored 878 5 rows in equity_prices

	date	ticker	close
8775	2026-07-30	META	539.030029
8776	2026-07-30	MSFT	451.100006
8777	2026-07-30	WMT	111.099998
8778	2026-07-31	AAPL	308.910004
8779	2026-07-31	AMZN	271.579987
8780	2026-07-31	GOOGL	356.130005
8781	2026-07-31	HCA	402.589996
8782	2026-07-31	META	556.710022
8783	2026-07-31	MSFT	464.720001
8784	2026-07-31	WMT	111.199997

5. Return, realised-volatility and forecasting features

The feature-engineering stage converts adjusted prices into daily returns and creates rolling measures intended to describe the recent level, persistence and change in volatility.

The features include realised volatility over short, medium and long windows, lagged returns and volatility-related transformations. The target is future annualised realised volatility over the same trading horizon used for the target option maturity.

Because the target uses future observations, its construction naturally creates missing values at the end of the dataset. These are expected and must not be filled using future information.

```
In [6]: from options_risk_engine.data.market_data import build_volatility_feature
feature_data, returns, feature_columns = build_volatility_feature_dataset(
    close_data=close_data,
    tickers=tickers,
)

feature_summary = pd.DataFrame({
    "Rows": [len(feature_data)],
    "Columns": [feature_data.shape[1]],
    "Features per ticker": [len(feature_columns)],
    "Start": [feature_data.index.min()],
    "End": [feature_data.index.max()],
})
display(feature_summary)
display(feature_data.tail(5))
```

2026-08-02 13:36:38,999 – options_risk_engine.data.market_data – INFO – Feature dataset created: 1255 rows, 21 engineered features per ticker

	Rows	Columns	Features per ticker	Start	End		
0	1255	168	21	2021-08-02	2026-07-31		
Price							
Ticker	AAPL		AMZN	GOOGL	HCA	META	MSFT
Date							
2026-07-27	336.910004	231.389999	326.559998	389.619995	593.869995	389.100006	
2026-07-28	340.079987	230.860001	333.709991	418.190002	593.409973	393.350006	
2026-07-29	338.190002	226.649994	336.709991	408.179993	585.609985	390.540009	
2026-07-30	333.429993	235.500000	333.660004	395.790009	539.030029	451.100006	
2026-07-31	308.910004	271.579987	356.130005	402.589996	556.710022	464.720001	

Feature-dataset observations

The resulting dataset contains 1,255 dates and 21 engineered features for each ticker. The wide matrix contains 168 columns once the price, return, features and target variables are included across all seven underlyings.

The final `Target_RV` observations are missing because the future 32-trading-day window has not yet occurred. This is a consequence of defining a forward-looking target and is not a data-quality error.

The forecasting functions later remove rows without complete features and targets before fitting each model.

Short- and long-horizon realised volatility

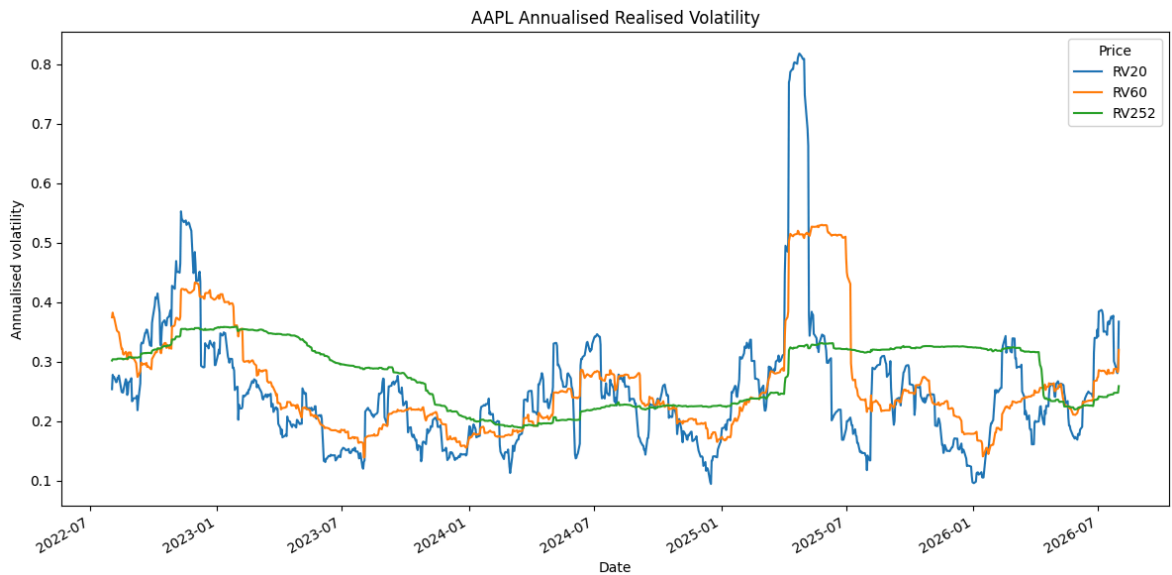
Realised volatility is inspected over 20, 60 and 252 trading days. Comparing multiple windows helps distinguish short-lived market shocks from persistent changes in the underlying volatility regime.

The 20-day series reacts fastest to recent returns, whereas the 252-day series is smoother and represents the longer-run volatility environment.

```
In [7]: from options_risk_engine.reporting.plots import plot_realised_volatility

example_ticker = CONFIG.symbols[0]
latest_volatility = feature_data.xs(example_ticker, axis=1, level="Ticker"
    "RV20", "RV60", "RV252", "Target_RV"
)].tail(10)
display(latest_volatility.round(4))
plot_realised_volatility(
    feature_data, example_ticker, FIGURE_DIR / "02_realised_volatility.png"
)
plt.show()
```

Price	RV20	RV60	RV252	Target_RV
Date				
2026-07-20	0.3759	0.2812	0.2450	NaN
2026-07-21	0.3747	0.2812	0.2450	NaN
2026-07-22	0.3731	0.2807	0.2451	NaN
2026-07-23	0.3774	0.2807	0.2454	NaN
2026-07-24	0.3009	0.2879	0.2477	NaN
2026-07-27	0.2901	0.2881	0.2478	NaN
2026-07-28	0.2839	0.2883	0.2479	NaN
2026-07-29	0.2809	0.2828	0.2480	NaN
2026-07-30	0.2890	0.2833	0.2481	NaN
2026-07-31	0.3674	0.3201	0.2591	NaN



For AAPL, the final 20-day realised volatility is approximately 36.7%, compared with 32.0% over 60 days and 25.9% over 252 days. The ordering indicates that recent price movements were more volatile than the longer-run history.

The graph also shows that short-window volatility can change rapidly and produce temporary spikes, while long-window volatility adjusts slowly. This supports testing models that combine several horizons, such as HAR-RV, while also demonstrating why a single recent volatility number may be unstable.

Persisting the feature matrix

The complete feature matrix is stored so that the exact forecasting inputs can be recovered without repeating the market-data download. MultiIndex column names are flattened before insertion into SQLite.

SQLite is suitable for this project-sized snapshot. Parquet would be a better choice if the feature universe expanded to hundreds of assets or intraday observations.

```
In [8]: feature_snapshot = feature_data.copy()
feature_snapshot.columns = [f"{a}_{b}" for a, b in feature_snapshot.columns]
feature_snapshot = feature_snapshot.reset_index().rename(columns={"Date": "date"})
store.write_frame(feature_snapshot, "volatility_features", run_id=run_id)
print("Stored feature rows:", len(feature_snapshot))
```

```
2026-08-02 13:36:39,511 - options_risk_engine.database - INFO - Stored 1255 rows in volatility_features
Stored feature rows: 1255
```

6. Volatility forecast benchmark

The forecasting experiment asks whether engineered historical features can predict future realised volatility more accurately than simple benchmark forecasts.

Thirty-four configurations are tested, including:

- a training-mean forecast;
- the current 20-day realised-volatility baseline;
- ordinary least squares;
- Ridge and Lasso across several regularisation strengths;
- PCA versions of the linear models;
- Random Forest and Gradient Boosting;
- HAR-RV; and
- GARCH(1,1).

Performance is measured using purged walk-forward validation. For every fold, the model is trained only on observations that occur before the test block, and a purge equal to the forecast horizon is placed between training and test data. This prevents the forward realised-volatility target in the training set from overlapping the test period.

Scaling and PCA are fitted using the training fold only. This avoids allowing information from the future test block to influence feature transformations.

```
In [9]: from options_risk_engine.forecasting.volatility import (
        evaluate_historical_mean_holdout,
        evaluate_symbol_models,
    )

    comparison_tables = {}
    mean_holdout_tables = {}

    if CONFIG.run_forecast_benchmarks:
        for ticker in tickers:
            ticker_df = feature_data.xs(ticker.symbol, axis=1, level="Ticker")
            mean_holdout_tables[ticker.symbol] = evaluate_historical_mean_holdout(
                ticker_df=ticker_df,
                ticker=ticker,
                test_fraction=FORECAST_TEST_FRACTION,
            )
            comparison_tables[ticker.symbol] = evaluate_symbol_models(
                ticker_df=ticker_df,
                feature_cols=feature_columns,
                ticker=ticker,
            )

        model_comparison = pd.concat(
            [table.assign(Ticker=ticker) for ticker, table in comparison_tables.items()],
            ignore_index=True,
        )
        store.write_frame(model_comparison, "model_evaluations", run_id=run_id)
        display(model_comparison.groupby("Ticker").head(5))
    else:
        print("Forecast benchmarks skipped. Set CONFIG.run_forecast_benchmark")
```

2026-08-02 13:36:41,175 - options_risk_engine.forecasting.volatility - INFO - AAPL starting model evaluation
2026-08-02 13:36:41,177 - options_risk_engine.forecasting.volatility - INFO - AAPL has 971 usable rows and 21 features
2026-08-02 13:36:41,178 - options_risk_engine.forecasting.volatility - INFO - Creating model specifications
2026-08-02 13:36:41,178 - options_risk_engine.forecasting.volatility - INFO - Created 34 model configurations
2026-08-02 13:36:41,179 - options_risk_engine.forecasting.volatility - INFO - AAPL walk-forward starts at row 284 with window=30, step=30, purge=32
2026-08-02 13:36:57,517 - options_risk_engine.forecasting.volatility - INFO - AAPL completed 22 walk-forward folds
2026-08-02 13:36:57,518 - options_risk_engine.forecasting.volatility - INFO - AAPL calculating model comparison metrics
2026-08-02 13:36:57,520 - options_risk_engine.forecasting.volatility - INFO - AAPL Train Mean alpha=None RMSE=0.10960 NRMSE=1.04761 R2=-0.09748 IC=-0.48458
2026-08-02 13:36:57,522 - options_risk_engine.forecasting.volatility - INFO - AAPL RV20 Baseline alpha=None RMSE=0.13733 NRMSE=1.31268 R2=-0.72312 IC=0.22482
2026-08-02 13:36:57,524 - options_risk_engine.forecasting.volatility - INFO - AAPL OLS alpha=None RMSE=0.14282 NRMSE=1.36520 R2=-0.86378 IC=-0.02723
2026-08-02 13:36:57,526 - options_risk_engine.forecasting.volatility - INFO - AAPL OLS PCA alpha=None RMSE=0.12644 NRMSE=1.20862 R2=-0.46075 IC=0.00080
2026-08-02 13:36:57,527 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge alpha=0.0001 RMSE=0.14282 NRMSE=1.36519 R2=-0.86376 IC=-0.02723
2026-08-02 13:36:57,529 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge alpha=0.001 RMSE=0.14281 NRMSE=1.36512 R2=-0.86355 IC=-0.02721
2026-08-02 13:36:57,531 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge alpha=0.01 RMSE=0.14273 NRMSE=1.36435 R2=-0.86146 IC=-0.02704
2026-08-02 13:36:57,533 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge alpha=0.1 RMSE=0.14201 NRMSE=1.35747 R2=-0.84273 IC=-0.02493
2026-08-02 13:36:57,534 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge alpha=1 RMSE=0.13805 NRMSE=1.31959 R2=-0.74131 IC=-0.00726
2026-08-02 13:36:57,536 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge alpha=10 RMSE=0.12879 NRMSE=1.23109 R2=-0.51558 IC=0.03284
2026-08-02 13:36:57,537 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge alpha=100 RMSE=0.11727 NRMSE=1.12097 R2=-0.25658 IC=0.08764
2026-08-02 13:36:57,539 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge alpha=1000 RMSE=0.10993 NRMSE=1.05079 R2=-0.10416 IC=0.14536
2026-08-02 13:36:57,541 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge PCA alpha=0.0001 RMSE=0.12644 NRMSE=1.20862 R2=-0.46075 IC=0.00080
2026-08-02 13:36:57,542 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge PCA alpha=0.001 RMSE=0.12644 NRMSE=1.20861 R2=-0.46075 IC=0.00080
2026-08-02 13:36:57,544 - options_risk_engine.forecasting.volatility - INFO - AAPL Ridge PCA alpha=0.01 RMSE=0.12644 NRMSE=1.20860 R2=-0.46071 IC=0.00081
2026-08-02 13:36:57,547 - options_risk_engine.forecasting.volatility - INFO


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0 - AAPL Ridge PCA alpha=0.1 RMSE=0.12643 NRMSE=1.20846 R2=-0.4
6038 IC=0.00091
2026-08-02 13:36:57,581 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Ridge PCA alpha=1 RMSE=0.12628 NRMSE=1.20711 R2=-0.457
11 IC=0.00188
2026-08-02 13:36:57,587 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Ridge PCA alpha=10 RMSE=0.12498 NRMSE=1.19461 R2=-0.42
710 IC=0.01113
2026-08-02 13:36:57,589 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Ridge PCA alpha=100 RMSE=0.11804 NRMSE=1.12834 R2=-0.2
7315 IC=0.07141
2026-08-02 13:36:57,591 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Ridge PCA alpha=1000 RMSE=0.11031 NRMSE=1.05443 R2=-0.
11183 IC=0.14089
2026-08-02 13:36:57,593 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso alpha=0.0001 RMSE=0.14003 NRMSE=1.33849 R2=-
0.79156 IC=-0.01808
2026-08-02 13:36:57,595 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso alpha=0.001 RMSE=0.12662 NRMSE=1.21036 R2=-
0.46498 IC=0.01207
2026-08-02 13:36:57,598 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso alpha=0.01 RMSE=0.11395 NRMSE=1.08925 R2=-0.
18645 IC=0.04100
2026-08-02 13:36:57,601 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso alpha=0.1 RMSE=0.10960 NRMSE=1.04761 R2=-0.0
9748 IC=-0.48458
2026-08-02 13:36:57,603 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso alpha=1 RMSE=0.10960 NRMSE=1.04761 R2=-0.097
48 IC=-0.48458
2026-08-02 13:36:57,605 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso PCA alpha=0.0001 RMSE=0.12637 NRMSE=1.20793 R2=-
0.45910 IC=0.00095
2026-08-02 13:36:57,607 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso PCA alpha=0.001 RMSE=0.12570 NRMSE=1.20156 R2=-
0.44375 IC=0.00222
2026-08-02 13:36:57,609 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso PCA alpha=0.01 RMSE=0.11686 NRMSE=1.11705 R2=-0.
24780 IC=0.06681
2026-08-02 13:36:57,611 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso PCA alpha=0.1 RMSE=0.10562 NRMSE=1.00955 R2=-0.0
1919 IC=0.05694
2026-08-02 13:36:57,613 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Lasso PCA alpha=1 RMSE=0.10960 NRMSE=1.04761 R2=-0.097
48 IC=-0.48458
2026-08-02 13:36:57,614 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Random Forest alpha=None RMSE=0.13732 NRMSE=1.31265 R2=-0.
72304 IC=-0.04671
2026-08-02 13:36:57,616 - options_risk_engine.forecasting.volatility - INF
0 - AAPL Gradient Boosting alpha=None RMSE=0.12833 NRMSE=1.22667 R2=-0.
50473 IC=-0.02468
2026-08-02 13:36:57,618 - options_risk_engine.forecasting.volatility - INF
0 - AAPL HAR-RV alpha=None RMSE=0.12635 NRMSE=1.20779 R2=-0.
45875 IC=-0.01064
2026-08-02 13:36:57,620 - options_risk_engine.forecasting.volatility - INF
0 - AAPL GARCH alpha=None RMSE=0.11599 NRMSE=1.10875 R2=-0.
22932 IC=0.14354
2026-08-02 13:36:57,622 - options_risk_engine.forecasting.volatility - INF
0 - AAPL evaluation complete - best model=Lasso PCA, alpha=0.1, RMSE=0.105
62, NRMSE=1.00955
2026-08-02 13:36:57,625 - options_risk_engine.forecasting.volatility - INF

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0 - MSFT starting model evaluation
2026-08-02 13:36:57,627 - options_risk_engine.forecasting.volatility - INF
0 - MSFT has 971 usable rows and 21 features
2026-08-02 13:36:57,627 - options_risk_engine.forecasting.volatility - INF
0 - Creating model specifications
2026-08-02 13:36:57,628 - options_risk_engine.forecasting.volatility - INF
0 - Created 34 model configurations
2026-08-02 13:36:57,628 - options_risk_engine.forecasting.volatility - INF
0 - MSFT walk-forward starts at row 284 with window=30, step=30, purge=32
2026-08-02 13:37:13,761 - options_risk_engine.forecasting.volatility - INF
0 - MSFT completed 22 walk-forward folds
2026-08-02 13:37:13,762 - options_risk_engine.forecasting.volatility - INF
0 - MSFT calculating model comparison metrics
2026-08-02 13:37:13,764 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Train Mean          alpha=None RMSE=0.08842 NRMSE=1.18225 R2=-0.39772 IC=-0.35506
2026-08-02 13:37:13,766 - options_risk_engine.forecasting.volatility - INF
0 - MSFT RV20 Baseline       alpha=None RMSE=0.10463 NRMSE=1.39906 R2=-0.95737 IC=0.12329
2026-08-02 13:37:13,768 - options_risk_engine.forecasting.volatility - INF
0 - MSFT OLS                 alpha=None RMSE=0.09843 NRMSE=1.31606 R2=-0.73201 IC=-0.01404
2026-08-02 13:37:13,769 - options_risk_engine.forecasting.volatility - INF
0 - MSFT OLS PCA             alpha=None RMSE=0.09562 NRMSE=1.27859 R2=-0.63480 IC=-0.07146
2026-08-02 13:37:13,771 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge               alpha=0.0001 RMSE=0.09843 NRMSE=1.31606 R2=-0.73201 IC=-0.01406
2026-08-02 13:37:13,772 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge               alpha=0.001 RMSE=0.09843 NRMSE=1.31605 R2=-0.73200 IC=-0.01430
2026-08-02 13:37:13,774 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge               alpha=0.01 RMSE=0.09842 NRMSE=1.31601 R2=-0.73189 IC=-0.01655
2026-08-02 13:37:13,776 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge               alpha=0.1 RMSE=0.09836 NRMSE=1.31518 R2=-0.72969 IC=-0.03094
2026-08-02 13:37:13,777 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge               alpha=1 RMSE=0.09780 NRMSE=1.30774 R2=-0.71019 IC=-0.05072
2026-08-02 13:37:13,779 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge               alpha=10 RMSE=0.09637 NRMSE=1.28861 R2=-0.66052 IC=-0.05240
2026-08-02 13:37:13,781 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge               alpha=100 RMSE=0.09140 NRMSE=1.22216 R2=-0.49368 IC=-0.03740
2026-08-02 13:37:13,782 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge               alpha=1000 RMSE=0.08417 NRMSE=1.12539 R2=-0.26651 IC=-0.03361
2026-08-02 13:37:13,784 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge PCA           alpha=0.0001 RMSE=0.09562 NRMSE=1.27859 R2=-0.63480 IC=-0.07145
2026-08-02 13:37:13,786 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge PCA           alpha=0.001 RMSE=0.09562 NRMSE=1.27859 R2=-0.63480 IC=-0.07145
2026-08-02 13:37:13,788 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge PCA           alpha=0.01 RMSE=0.09562 NRMSE=1.27858 R2=-0.63477 IC=-0.07145
2026-08-02 13:37:13,789 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge PCA           alpha=0.1 RMSE=0.09562 NRMSE=1.27849 R2=-0.6
```

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3454 IC=-0.07142
2026-08-02 13:37:13,791 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge PCA alpha=1 RMSE=0.09555 NRMSE=1.27761 R2=-0.632
27 IC=-0.07108
2026-08-02 13:37:13,793 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge PCA alpha=10 RMSE=0.09492 NRMSE=1.26923 R2=-0.61
095 IC=-0.06792
2026-08-02 13:37:13,794 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge PCA alpha=100 RMSE=0.09097 NRMSE=1.21631 R2=-0.4
7941 IC=-0.04782
2026-08-02 13:37:13,796 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Ridge PCA alpha=1000 RMSE=0.08422 NRMSE=1.12610 R2=-0.
26809 IC=-0.03965
2026-08-02 13:37:13,798 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso alpha=0.0001 RMSE=0.09819 NRMSE=1.31288 R2=-
0.72365 IC=-0.03737
2026-08-02 13:37:13,812 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso alpha=0.001 RMSE=0.09536 NRMSE=1.27507 R2=-
0.62582 IC=-0.08360
2026-08-02 13:37:13,828 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso alpha=0.01 RMSE=0.08695 NRMSE=1.16262 R2=-0.
35169 IC=-0.11584
2026-08-02 13:37:13,842 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso alpha=0.1 RMSE=0.08842 NRMSE=1.18225 R2=-0.3
9772 IC=-0.35506
2026-08-02 13:37:13,844 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso alpha=1 RMSE=0.08842 NRMSE=1.18225 R2=-0.397
72 IC=-0.35506
2026-08-02 13:37:13,845 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso PCA alpha=0.0001 RMSE=0.09551 NRMSE=1.27701 R2=-
0.63075 IC=-0.07138
2026-08-02 13:37:13,850 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso PCA alpha=0.001 RMSE=0.09455 NRMSE=1.26416 R2=-
0.59810 IC=-0.07083
2026-08-02 13:37:13,853 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso PCA alpha=0.01 RMSE=0.08802 NRMSE=1.17691 R2=-0.
38512 IC=-0.07801
2026-08-02 13:37:13,856 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso PCA alpha=0.1 RMSE=0.08625 NRMSE=1.15323 R2=-0.3
2994 IC=-0.29032
2026-08-02 13:37:13,858 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Lasso PCA alpha=1 RMSE=0.08842 NRMSE=1.18225 R2=-0.397
72 IC=-0.35506
2026-08-02 13:37:13,859 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Random Forest alpha=None RMSE=0.10502 NRMSE=1.40419 R2=-0.
97174 IC=-0.06917
2026-08-02 13:37:13,861 - options_risk_engine.forecasting.volatility - INF
0 - MSFT Gradient Boosting alpha=None RMSE=0.10118 NRMSE=1.35281 R2=-0.
83009 IC=-0.06625
2026-08-02 13:37:13,863 - options_risk_engine.forecasting.volatility - INF
0 - MSFT HAR-RV alpha=None RMSE=0.09716 NRMSE=1.29911 R2=-0.
68770 IC=-0.13337
2026-08-02 13:37:13,865 - options_risk_engine.forecasting.volatility - INF
0 - MSFT GARCH alpha=None RMSE=0.08325 NRMSE=1.11312 R2=-0.
23903 IC=0.04315
2026-08-02 13:37:13,866 - options_risk_engine.forecasting.volatility - INF
0 - MSFT evaluation complete - best model=GARCH, alpha=nan, RMSE=0.08325,
NRMSE=1.11312
2026-08-02 13:37:13,869 - options_risk_engine.forecasting.volatility - INF
0 - META starting model evaluation

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2026-08-02 13:37:13,871 - options_risk_engine.forecasting.volatility - INF
0 - META has 971 usable rows and 21 features
2026-08-02 13:37:13,871 - options_risk_engine.forecasting.volatility - INF
0 - Creating model specifications
2026-08-02 13:37:13,872 - options_risk_engine.forecasting.volatility - INF
0 - Created 34 model configurations
2026-08-02 13:37:13,873 - options_risk_engine.forecasting.volatility - INF
0 - META walk-forward starts at row 284 with window=30, step=30, purge=32
2026-08-02 13:37:30,441 - options_risk_engine.forecasting.volatility - INF
0 - META completed 22 walk-forward folds
2026-08-02 13:37:30,442 - options_risk_engine.forecasting.volatility - INF
0 - META calculating model comparison metrics
2026-08-02 13:37:30,444 - options_risk_engine.forecasting.volatility - INF
0 - META Train Mean          alpha=None RMSE=0.14507 NRMSE=1.19096 R2=-0.
41839 IC=-0.12845
2026-08-02 13:37:30,446 - options_risk_engine.forecasting.volatility - INF
0 - META RV20 Baseline      alpha=None RMSE=0.20570 NRMSE=1.68873 R2=-1.
85181 IC=-0.17340
2026-08-02 13:37:30,447 - options_risk_engine.forecasting.volatility - INF
0 - META OLS                alpha=None RMSE=0.20237 NRMSE=1.66145 R2=-1.
76042 IC=-0.12234
2026-08-02 13:37:30,449 - options_risk_engine.forecasting.volatility - INF
0 - META OLS PCA           alpha=None RMSE=0.15624 NRMSE=1.28268 R2=-0.
64527 IC=0.02020
2026-08-02 13:37:30,451 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge             alpha=0.0001 RMSE=0.20235 NRMSE=1.66124 R2=-
1.75973 IC=-0.12232
2026-08-02 13:37:30,452 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge             alpha=0.001 RMSE=0.20212 NRMSE=1.65937 R2=-
1.75352 IC=-0.12216
2026-08-02 13:37:30,454 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge             alpha=0.01 RMSE=0.19996 NRMSE=1.64165 R2=-1.
69502 IC=-0.12046
2026-08-02 13:37:30,456 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge             alpha=0.1 RMSE=0.18614 NRMSE=1.52815 R2=-1.3
3524 IC=-0.10331
2026-08-02 13:37:30,457 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge             alpha=1 RMSE=0.16780 NRMSE=1.37760 R2=-0.897
79 IC=-0.03999
2026-08-02 13:37:30,459 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge             alpha=10 RMSE=0.16220 NRMSE=1.33163 R2=-0.77
324 IC=0.00200
2026-08-02 13:37:30,460 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge             alpha=100 RMSE=0.14859 NRMSE=1.21988 R2=-0.4
8810 IC=-0.00626
2026-08-02 13:37:30,462 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge             alpha=1000 RMSE=0.13927 NRMSE=1.14336 R2=-0.
30727 IC=-0.13577
2026-08-02 13:37:30,464 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge PCA        alpha=0.0001 RMSE=0.15624 NRMSE=1.28268 R2=-
0.64527 IC=0.02020
2026-08-02 13:37:30,465 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge PCA        alpha=0.001 RMSE=0.15624 NRMSE=1.28268 R2=-
0.64527 IC=0.02020
2026-08-02 13:37:30,467 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge PCA        alpha=0.01 RMSE=0.15624 NRMSE=1.28267 R2=-0.
64524 IC=0.02020
2026-08-02 13:37:30,469 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge PCA        alpha=0.1 RMSE=0.15622 NRMSE=1.28257 R2=-0.6
4498 IC=0.02017
```

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2026-08-02 13:37:30,470 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge PCA alpha=1 RMSE=0.15610 NRMSE=1.28154 R2=-0.642
35 IC=0.01986
2026-08-02 13:37:30,472 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge PCA alpha=10 RMSE=0.15492 NRMSE=1.27188 R2=-0.61
769 IC=0.01678
2026-08-02 13:37:30,474 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge PCA alpha=100 RMSE=0.14773 NRMSE=1.21281 R2=-0.4
7090 IC=-0.01070
2026-08-02 13:37:30,476 - options_risk_engine.forecasting.volatility - INF
0 - META Ridge PCA alpha=1000 RMSE=0.13981 NRMSE=1.14781 R2=-0.
31746 IC=-0.13662
2026-08-02 13:37:30,477 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso alpha=0.0001 RMSE=0.18685 NRMSE=1.53401 R2=-
1.35318 IC=-0.12708
2026-08-02 13:37:30,479 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso alpha=0.001 RMSE=0.16370 NRMSE=1.34396 R2=-
0.80623 IC=0.00980
2026-08-02 13:37:30,481 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso alpha=0.01 RMSE=0.15531 NRMSE=1.27504 R2=-0.
62573 IC=0.01042
2026-08-02 13:37:30,482 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso alpha=0.1 RMSE=0.14562 NRMSE=1.19552 R2=-0.4
2928 IC=-0.13743
2026-08-02 13:37:30,484 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso alpha=1 RMSE=0.14507 NRMSE=1.19096 R2=-0.418
39 IC=-0.12845
2026-08-02 13:37:30,496 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso PCA alpha=0.0001 RMSE=0.15615 NRMSE=1.28192 R2=-
0.64332 IC=0.01987
2026-08-02 13:37:30,525 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso PCA alpha=0.001 RMSE=0.15532 NRMSE=1.27517 R2=-
0.62606 IC=0.01679
2026-08-02 13:37:30,527 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso PCA alpha=0.01 RMSE=0.14884 NRMSE=1.22192 R2=-0.
49308 IC=-0.02273
2026-08-02 13:37:30,529 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso PCA alpha=0.1 RMSE=0.14774 NRMSE=1.21292 R2=-0.4
7117 IC=-0.26553
2026-08-02 13:37:30,531 - options_risk_engine.forecasting.volatility - INF
0 - META Lasso PCA alpha=1 RMSE=0.14507 NRMSE=1.19096 R2=-0.418
39 IC=-0.12845
2026-08-02 13:37:30,533 - options_risk_engine.forecasting.volatility - INF
0 - META Random Forest alpha=None RMSE=0.16939 NRMSE=1.39067 R2=-0.
93398 IC=-0.10230
2026-08-02 13:37:30,535 - options_risk_engine.forecasting.volatility - INF
0 - META Gradient Boosting alpha=None RMSE=0.15179 NRMSE=1.24617 R2=-0.
55294 IC=0.02360
2026-08-02 13:37:30,537 - options_risk_engine.forecasting.volatility - INF
0 - META HAR-RV alpha=None RMSE=0.16757 NRMSE=1.37573 R2=-0.
89263 IC=0.01569
2026-08-02 13:37:30,541 - options_risk_engine.forecasting.volatility - INF
0 - META GARCH alpha=None RMSE=0.21776 NRMSE=1.78777 R2=-2.
19611 IC=-0.01975
2026-08-02 13:37:30,543 - options_risk_engine.forecasting.volatility - INF
0 - META evaluation complete - best model=Ridge, alpha=1000.0, RMSE=0.1392
7, NRMSE=1.14336
2026-08-02 13:37:30,547 - options_risk_engine.forecasting.volatility - INF
0 - WMT starting model evaluation
2026-08-02 13:37:30,549 - options_risk_engine.forecasting.volatility - INF
```



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0 - WMT has 971 usable rows and 21 features
2026-08-02 13:37:30,549 - options_risk_engine.forecasting.volatility - INF
0 - Creating model specifications
2026-08-02 13:37:30,550 - options_risk_engine.forecasting.volatility - INF
0 - Created 34 model configurations
2026-08-02 13:37:30,550 - options_risk_engine.forecasting.volatility - INF
0 - WMT walk-forward starts at row 284 with window=30, step=30, purge=32
2026-08-02 13:37:46,579 - options_risk_engine.forecasting.volatility - INF
0 - WMT completed 22 walk-forward folds
2026-08-02 13:37:46,579 - options_risk_engine.forecasting.volatility - INF
0 - WMT calculating model comparison metrics
2026-08-02 13:37:46,581 - options_risk_engine.forecasting.volatility - INF
0 - WMT Train Mean          alpha=None RMSE=0.07943 NRMSE=1.12322 R2=-0.2
6161 IC=0.23052
2026-08-02 13:37:46,583 - options_risk_engine.forecasting.volatility - INF
0 - WMT RV20 Baseline       alpha=None RMSE=0.09972 NRMSE=1.41001 R2=-0.9
8812 IC=0.16391
2026-08-02 13:37:46,584 - options_risk_engine.forecasting.volatility - INF
0 - WMT OLS                 alpha=None RMSE=0.10481 NRMSE=1.48210 R2=-1.1
9663 IC=-0.01525
2026-08-02 13:37:46,586 - options_risk_engine.forecasting.volatility - INF
0 - WMT OLS PCA            alpha=None RMSE=0.08305 NRMSE=1.17429 R2=-0.3
7896 IC=0.03276
2026-08-02 13:37:46,588 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge              alpha=0.0001 RMSE=0.10481 NRMSE=1.48200 R2=-
1.19632 IC=-0.01526
2026-08-02 13:37:46,590 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge              alpha=0.001 RMSE=0.10474 NRMSE=1.48109 R2=-1.
19361 IC=-0.01531
2026-08-02 13:37:46,591 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge              alpha=0.01 RMSE=0.10413 NRMSE=1.47246 R2=-1.1
6813 IC=-0.01579
2026-08-02 13:37:46,593 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge              alpha=0.1 RMSE=0.10027 NRMSE=1.41783 R2=-1.01
024 IC=-0.01895
2026-08-02 13:37:46,595 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge              alpha=1 RMSE=0.09354 NRMSE=1.32271 R2=-0.7495
6 IC=-0.01860
2026-08-02 13:37:46,596 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge              alpha=10 RMSE=0.08761 NRMSE=1.23883 R2=-0.534
70 IC=0.02082
2026-08-02 13:37:46,598 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge              alpha=100 RMSE=0.08149 NRMSE=1.15225 R2=-0.32
768 IC=0.07523
2026-08-02 13:37:46,602 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge              alpha=1000 RMSE=0.07914 NRMSE=1.11904 R2=-0.2
5225 IC=0.11964
2026-08-02 13:37:46,627 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge PCA          alpha=0.0001 RMSE=0.08305 NRMSE=1.17429 R2=-
0.37896 IC=0.03276
2026-08-02 13:37:46,641 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge PCA          alpha=0.001 RMSE=0.08305 NRMSE=1.17429 R2=-0.
37896 IC=0.03276
2026-08-02 13:37:46,643 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge PCA          alpha=0.01 RMSE=0.08305 NRMSE=1.17429 R2=-0.3
7895 IC=0.03277
2026-08-02 13:37:46,645 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge PCA          alpha=0.1 RMSE=0.08304 NRMSE=1.17425 R2=-0.37
887 IC=0.03280
2026-08-02 13:37:46,649 - options_risk_engine.forecasting.volatility - INF
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0 - WMT Ridge PCA          alpha=1 RMSE=0.08302 NRMSE=1.17389 R2=-0.3780
1 IC=0.03316
2026-08-02 13:37:46,650 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge PCA          alpha=10 RMSE=0.08278 NRMSE=1.17049 R2=-0.370
06 IC=0.03654
2026-08-02 13:37:46,654 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge PCA          alpha=100 RMSE=0.08141 NRMSE=1.15112 R2=-0.32
509 IC=0.06005
2026-08-02 13:37:46,657 - options_risk_engine.forecasting.volatility - INF
0 - WMT Ridge PCA          alpha=1000 RMSE=0.07945 NRMSE=1.12342 R2=-0.2
6208 IC=0.11297
2026-08-02 13:37:46,658 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso              alpha=0.0001 RMSE=0.09299 NRMSE=1.31489 R2=-
0.72894 IC=-0.00161
2026-08-02 13:37:46,660 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso              alpha=0.001 RMSE=0.08682 NRMSE=1.22763 R2=-0.
50708 IC=0.01922
2026-08-02 13:37:46,662 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso              alpha=0.01 RMSE=0.08319 NRMSE=1.17635 R2=-0.3
8381 IC=-0.06812
2026-08-02 13:37:46,664 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso              alpha=0.1 RMSE=0.07943 NRMSE=1.12322 R2=-0.26
161 IC=0.23052
2026-08-02 13:37:46,665 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso              alpha=1 RMSE=0.07943 NRMSE=1.12322 R2=-0.2616
1 IC=0.23052
2026-08-02 13:37:46,667 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso PCA          alpha=0.0001 RMSE=0.08296 NRMSE=1.17314 R2=-
0.37626 IC=0.03383
2026-08-02 13:37:46,668 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso PCA          alpha=0.001 RMSE=0.08229 NRMSE=1.16364 R2=-0.
35405 IC=0.04331
2026-08-02 13:37:46,670 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso PCA          alpha=0.01 RMSE=0.08131 NRMSE=1.14975 R2=-0.3
2193 IC=0.08995
2026-08-02 13:37:46,672 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso PCA          alpha=0.1 RMSE=0.07997 NRMSE=1.13082 R2=-0.27
875 IC=0.18300
2026-08-02 13:37:46,674 - options_risk_engine.forecasting.volatility - INF
0 - WMT Lasso PCA          alpha=1 RMSE=0.07943 NRMSE=1.12322 R2=-0.2616
1 IC=0.23052
2026-08-02 13:37:46,675 - options_risk_engine.forecasting.volatility - INF
0 - WMT Random Forest      alpha=None RMSE=0.08532 NRMSE=1.20639 R2=-0.4
5539 IC=-0.06956
2026-08-02 13:37:46,677 - options_risk_engine.forecasting.volatility - INF
0 - WMT Gradient Boosting  alpha=None RMSE=0.08816 NRMSE=1.24655 R2=-0.5
5388 IC=-0.07606
2026-08-02 13:37:46,679 - options_risk_engine.forecasting.volatility - INF
0 - WMT HAR-RV             alpha=None RMSE=0.08569 NRMSE=1.21167 R2=-0.4
6814 IC=-0.19011
2026-08-02 13:37:46,680 - options_risk_engine.forecasting.volatility - INF
0 - WMT GARCH              alpha=None RMSE=0.08412 NRMSE=1.18944 R2=-0.4
1476 IC=0.12352
2026-08-02 13:37:46,681 - options_risk_engine.forecasting.volatility - INF
0 - WMT evaluation complete - best model=Ridge, alpha=1000.0, RMSE=0.0791
4, NRMSE=1.11904
2026-08-02 13:37:46,684 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL starting model evaluation
2026-08-02 13:37:46,686 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL has 971 usable rows and 21 features

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2026-08-02 13:37:46,687 - options_risk_engine.forecasting.volatility - INF
0 - Creating model specifications
2026-08-02 13:37:46,687 - options_risk_engine.forecasting.volatility - INF
0 - Created 34 model configurations
2026-08-02 13:37:46,688 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL walk-forward starts at row 284 with window=30, step=30, purge=32
2026-08-02 13:38:03,230 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL completed 22 walk-forward folds
2026-08-02 13:38:03,231 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL calculating model comparison metrics
2026-08-02 13:38:03,233 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Train Mean          alpha=None RMSE=0.07436 NRMSE=1.07006 R2=-
0.14504 IC=-0.20185
2026-08-02 13:38:03,235 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL RV20 Baseline      alpha=None RMSE=0.11222 NRMSE=1.61477 R2=-
1.60747 IC=-0.03534
2026-08-02 13:38:03,236 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL OLS                alpha=None RMSE=0.09815 NRMSE=1.41226 R2=-
0.99448 IC=0.12447
2026-08-02 13:38:03,238 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL OLS PCA           alpha=None RMSE=0.08116 NRMSE=1.16786 R2=-
0.36390 IC=-0.01548
2026-08-02 13:38:03,240 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge             alpha=0.0001 RMSE=0.09813 NRMSE=1.41208 R2=
-0.99396 IC=0.12444
2026-08-02 13:38:03,242 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge             alpha=0.001 RMSE=0.09802 NRMSE=1.41043 R2=-
0.98931 IC=0.12419
2026-08-02 13:38:03,243 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge             alpha=0.01 RMSE=0.09697 NRMSE=1.39539 R2=-
0.94713 IC=0.12174
2026-08-02 13:38:03,245 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge             alpha=0.1 RMSE=0.09173 NRMSE=1.31991 R2=-0.
74216 IC=0.10369
2026-08-02 13:38:03,247 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge             alpha=1 RMSE=0.08597 NRMSE=1.23709 R2=-0.53
038 IC=0.07439
2026-08-02 13:38:03,248 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge             alpha=10 RMSE=0.08158 NRMSE=1.17389 R2=-0.3
7801 IC=0.07840
2026-08-02 13:38:03,250 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge             alpha=100 RMSE=0.07810 NRMSE=1.12383 R2=-0.
26299 IC=0.02447
2026-08-02 13:38:03,252 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge             alpha=1000 RMSE=0.07485 NRMSE=1.07697 R2=-
0.15987 IC=-0.12187
2026-08-02 13:38:03,253 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge PCA         alpha=0.0001 RMSE=0.08116 NRMSE=1.16786 R2=
-0.36390 IC=-0.01548
2026-08-02 13:38:03,255 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge PCA         alpha=0.001 RMSE=0.08116 NRMSE=1.16786 R2=-
0.36390 IC=-0.01548
2026-08-02 13:38:03,257 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge PCA         alpha=0.01 RMSE=0.08116 NRMSE=1.16786 R2=-
0.36389 IC=-0.01548
2026-08-02 13:38:03,258 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge PCA         alpha=0.1 RMSE=0.08116 NRMSE=1.16782 R2=-0.
36381 IC=-0.01552
2026-08-02 13:38:03,260 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge PCA         alpha=1 RMSE=0.08114 NRMSE=1.16750 R2=-0.36
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305 IC=-0.01589
2026-08-02 13:38:03,261 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge PCA alpha=10 RMSE=0.08092 NRMSE=1.16437 R2=-0.3
5575 IC=-0.01951
2026-08-02 13:38:03,263 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge PCA alpha=100 RMSE=0.07930 NRMSE=1.14106 R2=-0.
30202 IC=-0.04677
2026-08-02 13:38:03,265 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Ridge PCA alpha=1000 RMSE=0.07532 NRMSE=1.08377 R2=-
0.17455 IC=-0.14719
2026-08-02 13:38:03,266 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso alpha=0.0001 RMSE=0.09202 NRMSE=1.32408 R2=
-0.75318 IC=0.12169
2026-08-02 13:38:03,268 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso alpha=0.001 RMSE=0.08162 NRMSE=1.17443 R2=-
0.37929 IC=0.07169
2026-08-02 13:38:03,269 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso alpha=0.01 RMSE=0.07513 NRMSE=1.08103 R2=-
0.16864 IC=-0.12171
2026-08-02 13:38:03,271 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso alpha=0.1 RMSE=0.07436 NRMSE=1.07006 R2=-0.
14504 IC=-0.20185
2026-08-02 13:38:03,273 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso alpha=1 RMSE=0.07436 NRMSE=1.07006 R2=-0.14
504 IC=-0.20185
2026-08-02 13:38:03,274 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso PCA alpha=0.0001 RMSE=0.08110 NRMSE=1.16696 R2=
-0.36180 IC=-0.01576
2026-08-02 13:38:03,276 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso PCA alpha=0.001 RMSE=0.08058 NRMSE=1.15949 R2=-
0.34442 IC=-0.01884
2026-08-02 13:38:03,279 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso PCA alpha=0.01 RMSE=0.07663 NRMSE=1.10263 R2=-
0.21579 IC=-0.06975
2026-08-02 13:38:03,297 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso PCA alpha=0.1 RMSE=0.07436 NRMSE=1.07006 R2=-0.
14504 IC=-0.20185
2026-08-02 13:38:03,304 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Lasso PCA alpha=1 RMSE=0.07436 NRMSE=1.07006 R2=-0.14
504 IC=-0.20185
2026-08-02 13:38:03,320 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Random Forest alpha=None RMSE=0.09454 NRMSE=1.36043 R2=-
0.85076 IC=-0.17970
2026-08-02 13:38:03,322 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL Gradient Boosting alpha=None RMSE=0.09136 NRMSE=1.31463 R2=-
0.72826 IC=-0.11392
2026-08-02 13:38:03,324 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL HAR-RV alpha=None RMSE=0.08313 NRMSE=1.19621 R2=-
0.43093 IC=-0.17770
2026-08-02 13:38:03,326 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL GARCH alpha=None RMSE=0.07616 NRMSE=1.09596 R2=-
0.20113 IC=-0.29409
2026-08-02 13:38:03,328 - options_risk_engine.forecasting.volatility - INF
0 - GOOGL evaluation complete - best model=Train Mean, alpha=nan, RMSE=0.0
7436, NRMSE=1.07006
2026-08-02 13:38:03,335 - options_risk_engine.forecasting.volatility - INF
0 - AMZN starting model evaluation
2026-08-02 13:38:03,337 - options_risk_engine.forecasting.volatility - INF
0 - AMZN has 971 usable rows and 21 features
2026-08-02 13:38:03,337 - options_risk_engine.forecasting.volatility - INF

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0 - Creating model specifications
2026-08-02 13:38:03,338 - options_risk_engine.forecasting.volatility - INF
0 - Created 34 model configurations
2026-08-02 13:38:03,338 - options_risk_engine.forecasting.volatility - INF
0 - AMZN walk-forward starts at row 284 with window=30, step=30, purge=32
2026-08-02 13:38:19,554 - options_risk_engine.forecasting.volatility - INF
0 - AMZN completed 22 walk-forward folds
2026-08-02 13:38:19,555 - options_risk_engine.forecasting.volatility - INF
0 - AMZN calculating model comparison metrics
2026-08-02 13:38:19,557 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Train Mean          alpha=None RMSE=0.09926 NRMSE=1.16814 R2=-0.
36456 IC=-0.25176
2026-08-02 13:38:19,559 - options_risk_engine.forecasting.volatility - INF
0 - AMZN RV20 Baseline      alpha=None RMSE=0.13576 NRMSE=1.59759 R2=-1.
55229 IC=-0.07732
2026-08-02 13:38:19,560 - options_risk_engine.forecasting.volatility - INF
0 - AMZN OLS                alpha=None RMSE=0.11565 NRMSE=1.36093 R2=-0.
85212 IC=0.09202
2026-08-02 13:38:19,562 - options_risk_engine.forecasting.volatility - INF
0 - AMZN OLS PCA           alpha=None RMSE=0.09816 NRMSE=1.15513 R2=-0.
33432 IC=0.08226
2026-08-02 13:38:19,564 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge             alpha=0.0001 RMSE=0.11564 NRMSE=1.36089 R2=-
0.85202 IC=0.09200
2026-08-02 13:38:19,566 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge             alpha=0.001 RMSE=0.11562 NRMSE=1.36057 R2=-
0.85115 IC=0.09182
2026-08-02 13:38:19,567 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge             alpha=0.01 RMSE=0.11536 NRMSE=1.35750 R2=-0.
84281 IC=0.09007
2026-08-02 13:38:19,569 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge             alpha=0.1 RMSE=0.11345 NRMSE=1.33510 R2=-0.7
8250 IC=0.07910
2026-08-02 13:38:19,570 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge             alpha=1 RMSE=0.10721 NRMSE=1.26166 R2=-0.591
77 IC=0.06181
2026-08-02 13:38:19,572 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge             alpha=10 RMSE=0.10173 NRMSE=1.19716 R2=-0.43
320 IC=0.05639
2026-08-02 13:38:19,573 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge             alpha=100 RMSE=0.09569 NRMSE=1.12611 R2=-0.2
6812 IC=0.05235
2026-08-02 13:38:19,575 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge             alpha=1000 RMSE=0.09323 NRMSE=1.09712 R2=-0.
20367 IC=-0.10137
2026-08-02 13:38:19,577 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge PCA         alpha=0.0001 RMSE=0.09816 NRMSE=1.15513 R2=-
0.33432 IC=0.08226
2026-08-02 13:38:19,579 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge PCA         alpha=0.001 RMSE=0.09816 NRMSE=1.15513 R2=-
0.33432 IC=0.08226
2026-08-02 13:38:19,580 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge PCA         alpha=0.01 RMSE=0.09816 NRMSE=1.15512 R2=-0.
33431 IC=0.08226
2026-08-02 13:38:19,582 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge PCA         alpha=0.1 RMSE=0.09815 NRMSE=1.15507 R2=-0.3
3418 IC=0.08224
2026-08-02 13:38:19,584 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge PCA         alpha=1 RMSE=0.09811 NRMSE=1.15453 R2=-0.332
94 IC=0.08207
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2026-08-02 13:38:19,585 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge PCA alpha=10 RMSE=0.09768 NRMSE=1.14946 R2=-0.32
125 IC=0.08027
2026-08-02 13:38:19,587 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge PCA alpha=100 RMSE=0.09501 NRMSE=1.11810 R2=-0.2
5014 IC=0.05842
2026-08-02 13:38:19,588 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Ridge PCA alpha=1000 RMSE=0.09321 NRMSE=1.09690 R2=-0.
20319 IC=-0.10037
2026-08-02 13:38:19,590 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso alpha=0.0001 RMSE=0.11287 NRMSE=1.32821 R2=-
0.76413 IC=0.06921
2026-08-02 13:38:19,592 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso alpha=0.001 RMSE=0.10223 NRMSE=1.20299 R2=-
0.44717 IC=0.03921
2026-08-02 13:38:19,594 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso alpha=0.01 RMSE=0.10038 NRMSE=1.18125 R2=-0.
39535 IC=-0.10457
2026-08-02 13:38:19,595 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso alpha=0.1 RMSE=0.09926 NRMSE=1.16814 R2=-0.3
6456 IC=-0.25176
2026-08-02 13:38:19,597 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso alpha=1 RMSE=0.09926 NRMSE=1.16814 R2=-0.364
56 IC=-0.25176
2026-08-02 13:38:19,598 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso PCA alpha=0.0001 RMSE=0.09812 NRMSE=1.15463 R2=-
0.33317 IC=0.08120
2026-08-02 13:38:19,602 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso PCA alpha=0.001 RMSE=0.09774 NRMSE=1.15024 R2=-
0.32306 IC=0.07164
2026-08-02 13:38:19,635 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso PCA alpha=0.01 RMSE=0.09568 NRMSE=1.12597 R2=-0.
26780 IC=-0.02421
2026-08-02 13:38:19,642 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso PCA alpha=0.1 RMSE=0.09897 NRMSE=1.16473 R2=-0.3
5660 IC=-0.26160
2026-08-02 13:38:19,643 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Lasso PCA alpha=1 RMSE=0.09926 NRMSE=1.16814 R2=-0.364
56 IC=-0.25176
2026-08-02 13:38:19,645 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Random Forest alpha=None RMSE=0.13022 NRMSE=1.53237 R2=-1.
34816 IC=-0.21254
2026-08-02 13:38:19,647 - options_risk_engine.forecasting.volatility - INF
0 - AMZN Gradient Boosting alpha=None RMSE=0.13195 NRMSE=1.55276 R2=-1.
41107 IC=-0.20027
2026-08-02 13:38:19,650 - options_risk_engine.forecasting.volatility - INF
0 - AMZN HAR-RV alpha=None RMSE=0.10573 NRMSE=1.24425 R2=-0.
54816 IC=-0.11051
2026-08-02 13:38:19,652 - options_risk_engine.forecasting.volatility - INF
0 - AMZN GARCH alpha=None RMSE=0.10847 NRMSE=1.27649 R2=-0.
62942 IC=-0.22516
2026-08-02 13:38:19,654 - options_risk_engine.forecasting.volatility - INF
0 - AMZN evaluation complete - best model=Ridge PCA, alpha=1000.0, RMSE=0.
09321, NRMSE=1.09690
2026-08-02 13:38:19,659 - options_risk_engine.forecasting.volatility - INF
0 - HCA starting model evaluation
2026-08-02 13:38:19,661 - options_risk_engine.forecasting.volatility - INF
0 - HCA has 971 usable rows and 21 features
2026-08-02 13:38:19,661 - options_risk_engine.forecasting.volatility - INF
0 - Creating model specifications
```

```
2026-08-02 13:38:19,662 - options_risk_engine.forecasting.volatility - INF
0 - Created 34 model configurations
2026-08-02 13:38:19,662 - options_risk_engine.forecasting.volatility - INF
0 - HCA walk-forward starts at row 284 with window=30, step=30, purge=32
2026-08-02 13:38:35,973 - options_risk_engine.forecasting.volatility - INF
0 - HCA completed 22 walk-forward folds
2026-08-02 13:38:35,974 - options_risk_engine.forecasting.volatility - INF
0 - HCA calculating model comparison metrics
2026-08-02 13:38:35,976 - options_risk_engine.forecasting.volatility - INF
0 - HCA Train Mean          alpha=None RMSE=0.06185 NRMSE=1.02731 R2=-0.0
5536 IC=-0.00940
2026-08-02 13:38:35,978 - options_risk_engine.forecasting.volatility - INF
0 - HCA RV20 Baseline      alpha=None RMSE=0.10173 NRMSE=1.68971 R2=-1.8
5512 IC=-0.14081
2026-08-02 13:38:35,980 - options_risk_engine.forecasting.volatility - INF
0 - HCA OLS                alpha=None RMSE=0.21718 NRMSE=3.60717 R2=-12.
01167 IC=-0.12365
2026-08-02 13:38:35,981 - options_risk_engine.forecasting.volatility - INF
0 - HCA OLS PCA           alpha=None RMSE=0.06819 NRMSE=1.13266 R2=-0.2
8292 IC=-0.01170
2026-08-02 13:38:35,983 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge             alpha=0.0001 RMSE=0.21700 NRMSE=3.60414 R2=-1
1.98981 IC=-0.12361
2026-08-02 13:38:35,985 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge             alpha=0.001 RMSE=0.21537 NRMSE=3.57714 R2=-1
1.79594 IC=-0.12328
2026-08-02 13:38:35,987 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge             alpha=0.01 RMSE=0.20064 NRMSE=3.33246 R2=-10.
10531 IC=-0.12001
2026-08-02 13:38:35,988 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge             alpha=0.1 RMSE=0.12889 NRMSE=2.14082 R2=-3.58
311 IC=-0.08870
2026-08-02 13:38:35,990 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge             alpha=1 RMSE=0.07162 NRMSE=1.18952 R2=-0.4149
6 IC=0.04604
2026-08-02 13:38:35,992 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge             alpha=10 RMSE=0.06615 NRMSE=1.09873 R2=-0.207
22 IC=0.05241
2026-08-02 13:38:35,993 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge             alpha=100 RMSE=0.06633 NRMSE=1.10161 R2=-0.21
353 IC=0.00062
2026-08-02 13:38:35,995 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge             alpha=1000 RMSE=0.06539 NRMSE=1.08608 R2=-0.1
7958 IC=-0.05587
2026-08-02 13:38:35,997 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge PCA        alpha=0.0001 RMSE=0.06819 NRMSE=1.13266 R2=-
0.28292 IC=-0.01170
2026-08-02 13:38:35,998 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge PCA        alpha=0.001 RMSE=0.06819 NRMSE=1.13266 R2=-0.
28292 IC=-0.01170
2026-08-02 13:38:36,000 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge PCA        alpha=0.01 RMSE=0.06819 NRMSE=1.13266 R2=-0.2
8291 IC=-0.01170
2026-08-02 13:38:36,001 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge PCA        alpha=0.1 RMSE=0.06819 NRMSE=1.13263 R2=-0.28
284 IC=-0.01170
2026-08-02 13:38:36,003 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge PCA        alpha=1 RMSE=0.06817 NRMSE=1.13233 R2=-0.2821
7 IC=-0.01171
2026-08-02 13:38:36,007 - options_risk_engine.forecasting.volatility - INF
```

```

0 - HCA Ridge PCA          alpha=10 RMSE=0.06801 NRMSE=1.12953 R2=-0.275
84 IC=-0.01180
2026-08-02 13:38:36,030 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge PCA          alpha=100 RMSE=0.06702 NRMSE=1.11310 R2=-0.23
900 IC=-0.01562
2026-08-02 13:38:36,046 - options_risk_engine.forecasting.volatility - INF
0 - HCA Ridge PCA          alpha=1000 RMSE=0.06548 NRMSE=1.08756 R2=-0.1
8278 IC=-0.05877
2026-08-02 13:38:36,048 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso              alpha=0.0001 RMSE=0.15974 NRMSE=2.65317 R2=-
6.03931 IC=-0.10930
2026-08-02 13:38:36,049 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso              alpha=0.001 RMSE=0.06882 NRMSE=1.14311 R2=-0.
30670 IC=-0.03642
2026-08-02 13:38:36,052 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso              alpha=0.01 RMSE=0.06821 NRMSE=1.13295 R2=-0.2
8358 IC=-0.13721
2026-08-02 13:38:36,056 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso              alpha=0.1 RMSE=0.06185 NRMSE=1.02731 R2=-0.05
536 IC=-0.00940
2026-08-02 13:38:36,060 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso              alpha=1 RMSE=0.06185 NRMSE=1.02731 R2=-0.0553
6 IC=-0.00940
2026-08-02 13:38:36,062 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso PCA          alpha=0.0001 RMSE=0.06815 NRMSE=1.13190 R2=-
0.28119 IC=-0.01299
2026-08-02 13:38:36,064 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso PCA          alpha=0.001 RMSE=0.06776 NRMSE=1.12540 R2=-0.
26654 IC=-0.02438
2026-08-02 13:38:36,065 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso PCA          alpha=0.01 RMSE=0.06825 NRMSE=1.13353 R2=-0.2
8489 IC=-0.14369
2026-08-02 13:38:36,067 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso PCA          alpha=0.1 RMSE=0.06269 NRMSE=1.04120 R2=-0.08
409 IC=-0.03019
2026-08-02 13:38:36,068 - options_risk_engine.forecasting.volatility - INF
0 - HCA Lasso PCA          alpha=1 RMSE=0.06185 NRMSE=1.02731 R2=-0.0553
6 IC=-0.00940
2026-08-02 13:38:36,070 - options_risk_engine.forecasting.volatility - INF
0 - HCA Random Forest      alpha=None RMSE=0.08116 NRMSE=1.34807 R2=-0.8
1731 IC=-0.03139
2026-08-02 13:38:36,072 - options_risk_engine.forecasting.volatility - INF
0 - HCA Gradient Boosting  alpha=None RMSE=0.08197 NRMSE=1.36138 R2=-0.8
5336 IC=-0.06110
2026-08-02 13:38:36,074 - options_risk_engine.forecasting.volatility - INF
0 - HCA HAR-RV             alpha=None RMSE=0.06672 NRMSE=1.10811 R2=-0.2
2790 IC=-0.04267
2026-08-02 13:38:36,075 - options_risk_engine.forecasting.volatility - INF
0 - HCA GARCH              alpha=None RMSE=0.06575 NRMSE=1.09207 R2=-0.1
9261 IC=-0.08205
2026-08-02 13:38:36,076 - options_risk_engine.forecasting.volatility - INF
0 - HCA evaluation complete - best model=Train Mean, alpha=nan, RMSE=0.061
85, NRMSE=1.02731
2026-08-02 13:38:36,084 - options_risk_engine.database - INFO - Stored 238
rows in model_evaluations

```

	Model	Alpha	RMSE	MAE	R2	NRMSE	IC	N	1
0	Lasso PCA	0.10	0.105615	0.064111	-0.019185	1.009547	0.056935	660	
1	Train Mean	NaN	0.109597	0.068618	-0.097480	1.047607	-0.484583	660	
2	Lasso PCA	1.00	0.109597	0.068618	-0.097480	1.047607	-0.484583	660	
3	Lasso	1.00	0.109597	0.068618	-0.097480	1.047607	-0.484583	660	
4	Lasso	0.10	0.109597	0.068618	-0.097480	1.047607	-0.484583	660	
34	GARCH	NaN	0.083249	0.063023	-0.239032	1.113118	0.043152	660	
35	Ridge	1000.00	0.084167	0.062990	-0.266511	1.125394	-0.033607	660	
36	Ridge PCA	1000.00	0.084220	0.063059	-0.268093	1.126096	-0.039650	660	
37	Lasso PCA	0.10	0.086249	0.072394	-0.329943	1.153232	-0.290325	660	
38	Lasso	0.01	0.086951	0.064767	-0.351695	1.162624	-0.115842	660	
68	Ridge	1000.00	0.139268	0.114069	-0.307271	1.143360	-0.135766	660	
69	Ridge PCA	1000.00	0.139809	0.114934	-0.317461	1.147807	-0.136623	660	
70	Train Mean	NaN	0.145066	0.120506	-0.418393	1.190963	-0.128445	660	
71	Lasso PCA	1.00	0.145066	0.120506	-0.418393	1.190963	-0.128445	660	
72	Lasso	1.00	0.145066	0.120506	-0.418393	1.190963	-0.128445	660	
102	Ridge	1000.00	0.079138	0.060190	-0.252246	1.119038	0.119644	660	
103	Train Mean	NaN	0.079434	0.056635	-0.261614	1.123216	0.230525	660	
104	Lasso PCA	1.00	0.079434	0.056635	-0.261614	1.123216	0.230525	660	
105	Lasso	1.00	0.079434	0.056635	-0.261614	1.123216	0.230525	660	
106	Lasso	0.10	0.079434	0.056635	-0.261614	1.123216	0.230525	660	
136	Train Mean	NaN	0.074365	0.060426	-0.145037	1.070064	-0.201853	660	G
137	Lasso PCA	1.00	0.074365	0.060426	-0.145037	1.070064	-0.201853	660	G
138	Lasso PCA	0.10	0.074365	0.060426	-0.145037	1.070064	-0.201853	660	G
139	Lasso	1.00	0.074365	0.060426	-0.145037	1.070064	-0.201853	660	G
140	Lasso	0.10	0.074365	0.060426	-0.145037	1.070064	-0.201853	660	G
170	Ridge PCA	1000.00	0.093210	0.067769	-0.203190	1.096900	-0.100369	660	,

	Model	Alpha	RMSE	MAE	R2	NRMSE	IC	N
171	Ridge	1000.00	0.093229	0.067791	-0.203671	1.097119	-0.101366	660
172	Ridge PCA	100.00	0.095012	0.066484	-0.250142	1.118097	0.058425	660
173	Lasso PCA	0.01	0.095680	0.067764	-0.267801	1.125967	-0.024215	660
174	Ridge	100.00	0.095693	0.067231	-0.268123	1.126110	0.052346	660
204	Train Mean	NaN	0.061852	0.049952	-0.055361	1.027307	-0.009398	660
205	Lasso PCA	1.00	0.061852	0.049952	-0.055361	1.027307	-0.009398	660
206	Lasso	1.00	0.061852	0.049952	-0.055361	1.027307	-0.009398	660
207	Lasso	0.10	0.061852	0.049952	-0.055361	1.027307	-0.009398	660
208	Lasso PCA	0.10	0.062688	0.050997	-0.084092	1.041197	-0.030193	660

The model universe includes both simple statistical benchmarks and more flexible machine-learning or volatility-specific approaches. Testing the simpler models is important: a complex model should only be selected if it provides a genuine out-of-sample improvement rather than simply fitting the historical sample more closely.

7. Live option-chain acquisition

For each underlying, the listed expiry closest to the 45-day target is downloaded. Both calls and puts are retained so that the pricing model can be evaluated across strikes and validated using put-call parity.

Option-chain data is a live and imperfect market snapshot. Bid, ask, volume, open interest and last-trade timestamps are therefore retained rather than using only the implied volatility supplied by the data provider.

```
In [10]: from options_risk_engine.pricing.chains import download_option_chains

download_option_chains(tickers)
chain_inventory = pd.DataFrame({
    "Ticker": [ticker.symbol for ticker in tickers],
    "Expiry": [ticker.expiry for ticker in tickers],
    "Calls": [0 if ticker.call_chain is None else len(ticker.call_chain)
    "Puts": [0 if ticker.put_chain is None else len(ticker.put_chain) for
    })
display(chain_inventory)
```



```

2026-08-02 13:38:36,105 - options_risk_engine.pricing.chains - INFO - Down
loading option chains for 7 ticker objects
2026-08-02 13:38:36,106 - options_risk_engine.pricing.chains - INFO - Down
loading option chain for AAPL nearest to 2026-09-16
2026-08-02 13:38:36,362 - options_risk_engine.pricing.chains - INFO - AAPL
chain loaded for 2026-09-18: 93 calls and 74 puts
2026-08-02 13:38:36,362 - options_risk_engine.pricing.chains - INFO - Down
loading option chain for MSFT nearest to 2026-09-16
2026-08-02 13:38:36,606 - options_risk_engine.pricing.chains - INFO - MSFT
chain loaded for 2026-09-18: 90 calls and 82 puts
2026-08-02 13:38:36,607 - options_risk_engine.pricing.chains - INFO - Down
loading option chain for META nearest to 2026-09-16
2026-08-02 13:38:36,821 - options_risk_engine.pricing.chains - INFO - META
chain loaded for 2026-09-18: 190 calls and 146 puts
2026-08-02 13:38:36,822 - options_risk_engine.pricing.chains - INFO - Down
loading option chain for WMT nearest to 2026-09-16
2026-08-02 13:38:37,035 - options_risk_engine.pricing.chains - INFO - WMT
chain loaded for 2026-09-18: 34 calls and 28 puts
2026-08-02 13:38:37,036 - options_risk_engine.pricing.chains - INFO - Down
loading option chain for GOOGL nearest to 2026-09-16
2026-08-02 13:38:37,202 - options_risk_engine.pricing.chains - INFO - GOOG
L chain loaded for 2026-09-18: 134 calls and 98 puts
2026-08-02 13:38:37,203 - options_risk_engine.pricing.chains - INFO - Down
loading option chain for AMZN nearest to 2026-09-16
2026-08-02 13:38:37,364 - options_risk_engine.pricing.chains - INFO - AMZN
chain loaded for 2026-09-18: 58 calls and 52 puts
2026-08-02 13:38:37,365 - options_risk_engine.pricing.chains - INFO - Down
loading option chain for HCA nearest to 2026-09-16
2026-08-02 13:38:37,552 - options_risk_engine.pricing.chains - INFO - HCA
chain loaded for 2026-09-18: 73 calls and 59 puts
2026-08-02 13:38:37,553 - options_risk_engine.pricing.chains - INFO - Opti
on-chain download complete: 7 of 7 tickers loaded

```

	Ticker	Expiry	Calls	Puts
0	AAPL	2026-09-18	93	74
1	MSFT	2026-09-18	90	82
2	META	2026-09-18	190	146
3	WMT	2026-09-18	34	28
4	GOOGL	2026-09-18	134	98
5	AMZN	2026-09-18	58	52
6	HCA	2026-09-18	73	59

All seven option chains were retrieved successfully, and each selected the 18 September 2026 expiry. The number of available contracts varied substantially: META provided the deepest chain, while WMT had the smallest number of strikes.

This difference matters because the quantity and quality of available contracts affect both the breadth of the pricing comparison and the number of positions later entering the synthetic portfolio.

Preserving the original market snapshot

The unmodified option-chain rows are stored before pricing and filtering. This makes it possible to distinguish values downloaded from the market-data source from columns created later by the project.

A snapshot timestamp, ticker, option type and expiry are added to each row so that the source data can be reconstructed independently of the priced table.

```
In [11]: raw_chain_frames = []
for ticker in tickers:
    for option_type, chain in (("Calls", ticker.call_chain), ("Puts", tic
        if chain is None:
            continue
        snapshot = chain.copy().assign(
            ticker=ticker.symbol,
            option_type=option_type,
            expiry=ticker.expiry,
            snapshot_time=pd.Timestamp.utcnow(),
        )
        raw_chain_frames.append(snapshot)

raw_option_chains = pd.concat(raw_chain_frames, ignore_index=True)
store.write_frame(raw_option_chains, "option_chain_snapshots_full", run_i
display(raw_option_chains.head())
```

2026-08-02 13:38:37,592 – options_risk_engine.database – INFO – Stored 121
1 rows in option_chain_snapshots_full

	contractSymbol	lastTradeDate	strike	lastPrice	bid	ask	change
0	AAPL260918C00050000	2026-07-31 13:54:56+00:00	50.0	254.65	257.15	261.00	-27.75
1	AAPL260918C00055000	2026-07-06 15:44:35+00:00	55.0	259.24	252.30	256.10	0.00
2	AAPL260918C00060000	2026-07-01 14:26:11+00:00	60.0	232.92	247.35	251.00	0.00
3	AAPL260918C00065000	2025-08-25 18:54:43+00:00	65.0	164.78	191.15	193.95	0.00
4	AAPL260918C00070000	2026-07-07 13:30:27+00:00	70.0	245.00	237.40	241.15	0.00

8. Option pricing and contract-level validation

Each option is valued using the Black–Scholes model under several volatility assumptions, including downloaded implied volatility, calculated midpoint implied volatility, realised-volatility measures and the selected forecast volatility.

The stage also:

- calculates market midpoint and bid–ask spread;
- checks theoretical option-price bounds;
- solves for bid, midpoint and ask implied volatility;
- calculates analytical Greeks;

- checks put-call parity;
- evaluates quote quality and liquidity; and
- assigns a screening label based on model-implied edge.

The purpose is not merely to generate a theoretical price. It is to establish whether the market data and numerical implementation are sufficiently reliable for the later portfolio risk calculations.

```
In [12]: from options_risk_engine.pricing.chains import (
        filter_highlighted_options,
        price_option_universe,
    )
    from options_risk_engine.reporting.tables import combine_option_tables
    from options_risk_engine.validation import validate_option_universe

    cleaned_options, _, parity_results = price_option_universe(
        tickers=tickers,
        data=feature_data,
    )

    highlighted_options = {
        key: filter_highlighted_options(
            options=frame,
            buy_edge=INITIAL_BUY_EDGE,
            max_spread_pct=INITIAL_MAX_SPREAD_PCT,
            min_ask=INITIAL_MIN_ASK,
            min_open_interest=INITIAL_MIN_OPEN_INTEREST,
            min_volume=INITIAL_MIN_VOLUME,
            min_moneyness=INITIAL_MIN_MONEYNESSESS,
            max_moneyness=INITIAL_MAX_MONEYNESSESS,
        )
        for key, frame in cleaned_options.items()
    }

    large_table = combine_option_tables(
        cleaned_options
    )

    universe_validation = validate_option_universe(
        large_table
    )

    print(universe_validation)
    display(large_table.head(10))
```

```
2026-08-02 13:38:37,610 - options_risk_engine.pricing.chains - INFO - Star
ting option pricing for 7 ticker objects
2026-08-02 13:38:37,610 - options_risk_engine.pricing.chains - INFO - AAPL
beginning option-chain pricing
2026-08-02 13:38:37,612 - options_risk_engine.pricing.chains - INFO - AAPL
volatility inputs - RV20: 0.3674, RV60: 0.3201, RV252: 0.2591, forecast:
0.2623
2026-08-02 13:38:38,281 - options_risk_engine.pricing.chains - INFO - AAPL
implied volatility calculations complete
2026-08-02 13:38:38,286 - options_risk_engine.pricing.chains - INFO - AAPL
put-call parity check: 74 matched strikes, maximum absolute error 0.000000
0000
2026-08-02 13:38:38,316 - options_risk_engine.pricing.chains - INFO - AAPL
pricing complete - 93 total calls, 74 total puts, 0 highlighted calls and
7 highlighted puts
2026-08-02 13:38:38,316 - options_risk_engine.pricing.chains - INFO - MSFT
beginning option-chain pricing
2026-08-02 13:38:38,320 - options_risk_engine.pricing.chains - INFO - MSFT
volatility inputs - RV20: 0.6016, RV60: 0.4622, RV252: 0.3193, forecast:
0.2609
2026-08-02 13:38:39,076 - options_risk_engine.pricing.chains - INFO - MSFT
implied volatility calculations complete
2026-08-02 13:38:39,081 - options_risk_engine.pricing.chains - INFO - MSFT
put-call parity check: 82 matched strikes, maximum absolute error 0.000000
0000
2026-08-02 13:38:39,110 - options_risk_engine.pricing.chains - INFO - MSFT
pricing complete - 90 total calls, 82 total puts, 0 highlighted calls and
6 highlighted puts
2026-08-02 13:38:39,111 - options_risk_engine.pricing.chains - INFO - META
beginning option-chain pricing
2026-08-02 13:38:39,113 - options_risk_engine.pricing.chains - INFO - META
volatility inputs - RV20: 0.5162, RV60: 0.4580, RV252: 0.3978, forecast:
0.4131
2026-08-02 13:38:40,397 - options_risk_engine.pricing.chains - INFO - META
implied volatility calculations complete
2026-08-02 13:38:40,403 - options_risk_engine.pricing.chains - INFO - META
put-call parity check: 130 matched strikes, maximum absolute error 0.000000
00000
2026-08-02 13:38:40,433 - options_risk_engine.pricing.chains - INFO - META
pricing complete - 190 total calls, 146 total puts, 0 highlighted calls an
d 19 highlighted puts
2026-08-02 13:38:40,434 - options_risk_engine.pricing.chains - INFO - WMT
beginning option-chain pricing
2026-08-02 13:38:40,435 - options_risk_engine.pricing.chains - INFO - WMT
volatility inputs - RV20: 0.2209, RV60: 0.2760, RV252: 0.2477, forecast:
0.2065
2026-08-02 13:38:40,672 - options_risk_engine.pricing.chains - INFO - WMT
implied volatility calculations complete
2026-08-02 13:38:40,676 - options_risk_engine.pricing.chains - INFO - WMT
put-call parity check: 28 matched strikes, maximum absolute error 0.000000
0000
2026-08-02 13:38:40,706 - options_risk_engine.pricing.chains - INFO - WMT
pricing complete - 34 total calls, 28 total puts, 0 highlighted calls and
2 highlighted puts
2026-08-02 13:38:40,707 - options_risk_engine.pricing.chains - INFO - GOOG
L beginning option-chain pricing
2026-08-02 13:38:40,708 - options_risk_engine.pricing.chains - INFO - GOOG
L volatility inputs - RV20: 0.4610, RV60: 0.3682, RV252: 0.3212, forecast:
0.3099
2026-08-02 13:38:41,619 - options_risk_engine.pricing.chains - INFO - GOOG
```

```
L implied volatility calculations complete
2026-08-02 13:38:41,624 - options_risk_engine.pricing.chains - INFO - GOOG
L put-call parity check: 96 matched strikes, maximum absolute error 0.0000
000000
2026-08-02 13:38:41,656 - options_risk_engine.pricing.chains - INFO - GOOG
L pricing complete - 134 total calls, 98 total puts, 0 highlighted calls a
nd 12 highlighted puts
2026-08-02 13:38:41,656 - options_risk_engine.pricing.chains - INFO - AMZN
beginning option-chain pricing
2026-08-02 13:38:41,658 - options_risk_engine.pricing.chains - INFO - AMZN
volatility inputs - RV20: 0.6195, RV60: 0.4402, RV252: 0.3520, forecast:
0.3413
2026-08-02 13:38:42,244 - options_risk_engine.pricing.chains - INFO - AMZN
implied volatility calculations complete
2026-08-02 13:38:42,252 - options_risk_engine.pricing.chains - INFO - AMZN
put-call parity check: 52 matched strikes, maximum absolute error 0.000000
0000
2026-08-02 13:38:42,299 - options_risk_engine.pricing.chains - INFO - AMZN
pricing complete - 58 total calls, 52 total puts, 7 highlighted calls and
2 highlighted puts
2026-08-02 13:38:42,300 - options_risk_engine.pricing.chains - INFO - HCA
beginning option-chain pricing
2026-08-02 13:38:42,303 - options_risk_engine.pricing.chains - INFO - HCA
volatility inputs - RV20: 0.5051, RV60: 0.3752, RV252: 0.3018, forecast:
0.2788
2026-08-02 13:38:42,730 - options_risk_engine.pricing.chains - INFO - HCA
implied volatility calculations complete
2026-08-02 13:38:42,735 - options_risk_engine.pricing.chains - INFO - HCA
put-call parity check: 56 matched strikes, maximum absolute error 0.000000
0000
2026-08-02 13:38:42,766 - options_risk_engine.pricing.chains - INFO - HCA
pricing complete - 73 total calls, 59 total puts, 0 highlighted calls and
0 highlighted puts
2026-08-02 13:38:42,767 - options_risk_engine.pricing.chains - INFO - Opti
on pricing finished - 14 full tables, 14 highlighted tables, 7 parity tabl
es and 0 failed tickers
2026-08-02 13:38:42,859 - options_risk_engine.reporting.tables - INFO - Co
mbined 14 option tables into 1211 rows
2026-08-02 13:38:42,860 - options_risk_engine.validation - INFO - Option-u
niverse validation: {'rows': 1211, 'tickers': 7, 'duplicate_contract_symbo
ls': 0, 'invalid_quotes': 417}
{'rows': 1211, 'tickers': 7, 'duplicate_contract_symbols': 0, 'invalid_quo
tes': 417}
```

	contractSymbol	strike	bid	ask	lastPrice	lastTradeDate	implied
0	AAPL260918C00050000	50.0	257.15	261.00	254.65	2026-07-31 13:54:56+00:00	
1	AAPL260918C00055000	55.0	252.30	256.10	259.24	2026-07-06 15:44:35+00:00	
2	AAPL260918C00060000	60.0	247.35	251.00	232.92	2026-07-01 14:26:11+00:00	
3	AAPL260918C00065000	65.0	191.15	193.95	164.78	2025-08-25 18:54:43+00:00	
4	AAPL260918C00070000	70.0	237.40	241.15	245.00	2026-07-07 13:30:27+00:00	
5	AAPL260918C00075000	75.0	222.20	225.90	219.18	2026-05-12 14:03:54+00:00	
6	AAPL260918C00080000	80.0	NaN	NaN	231.73	2026-05-27 18:29:55+00:00	
7	AAPL260918C00085000	85.0	NaN	NaN	192.23	2026-02-11 20:23:39+00:00	
8	AAPL260918C00090000	90.0	217.50	221.05	225.93	2026-07-14 19:18:55+00:00	
9	AAPL260918C00095000	95.0	NaN	NaN	161.70	2026-04-02 14:57:40+00:00	

Option-universe quality

The final universe contains 1,211 unique contracts across all seven tickers, with no duplicated contract symbols. The engine flagged 417 rows as having an invalid quote under its price-bound and bid-ask criteria.

This is approximately one third of the downloaded universe, demonstrating why a live option chain should not be passed directly into a pricing or trading workflow without validation.

Quote quality also differed between calls and puts. Several call chains contained a high proportion of usable quotes, while the HCA and WMT put chains contained relatively few valid observations. These rows remain useful for diagnosing data quality but should not be treated as reliable trade candidates.

```
In [13]: pricing_summary = (
    large_table.groupby(["Ticker", "Option_Type"], observed=True)
    .agg(
        Contracts=("contractSymbol", "size"),
        Valid_Quotes=("Quote Valid", "sum"),
```



```

        Median_Spread=("SpreadPct", "median"),
        Median_IV=("IV_Mid", "median"),
        Median_Forecast_Vol=("ForeV Used", "median"),
        Median_Ask_Edge=("BS_ForeV AskEdge", "median"),
    )
    .reset_index()
)
display(pricing_summary.round(4))

```

	Ticker	Option_Type	Contracts	Valid_Quotes	Median_Spread	Median_IV	Med
0	AAPL	Calls	93	77	0.0530	0.7484	
1	AAPL	Puts	74	30	0.1111	0.3490	
2	AMZN	Calls	58	56	0.0391	0.3577	
3	AMZN	Puts	52	43	0.0723	0.4085	
4	GOOGL	Calls	134	114	0.0408	0.5928	
5	GOOGL	Puts	98	38	0.0662	0.3696	
6	HCA	Calls	73	49	0.0801	0.7478	
7	HCA	Puts	59	12	0.0904	0.1972	
8	META	Calls	190	157	0.1322	0.5966	
9	META	Puts	146	58	0.0535	0.3771	
10	MSFT	Calls	90	86	0.0518	0.7463	
11	MSFT	Puts	82	41	0.0966	0.3712	
12	WMT	Calls	34	25	0.1242	0.9777	
13	WMT	Puts	28	8	0.0890	0.1829	

Put-call parity

Put-call parity provides an internal consistency check for European call and put prices with the same strike and expiry.

The comparison here uses prices generated by the same forecast-volatility model. It therefore tests whether the implemented call and put formulas, discounting and dividend treatment satisfy the required relationship. It is not intended to identify market arbitrage from executable bid and ask quotes.

```

In [14]: parity_summary = pd.DataFrame([
    {
        "Ticker": ticker,
        "Matched Strikes": len(table),
        "Maximum Absolute Error": table["Absolute Parity Error"].max() if
        "Failed": (~table["Within Tolerance"]).sum() if not table.empty else 0
    }
    for ticker, table in parity_results.items()
])
display(parity_summary)

```

	Ticker	Matched Strikes	Maximum Absolute Error	Failed
0	AAPL	74	5.684342e-14	0
1	MSFT	82	1.136868e-13	0
2	META	130	1.421085e-13	0
3	WMT	28	1.421085e-14	0
4	GOOGL	96	7.105427e-14	0
5	AMZN	52	5.684342e-14	0
6	HCA	56	8.526513e-14	0

All matched strikes passed the parity tolerance. Maximum errors were on the order of (10^{-13}) , which is consistent with floating-point rounding rather than an economically meaningful discrepancy.

This provides strong evidence that the call and put pricing formulas use consistent spot, strike, rate, dividend and maturity inputs.

Market price versus forecast-based model price

The scatter plot compares the option-market midpoint with the Black–Scholes value obtained using forecast realised volatility.

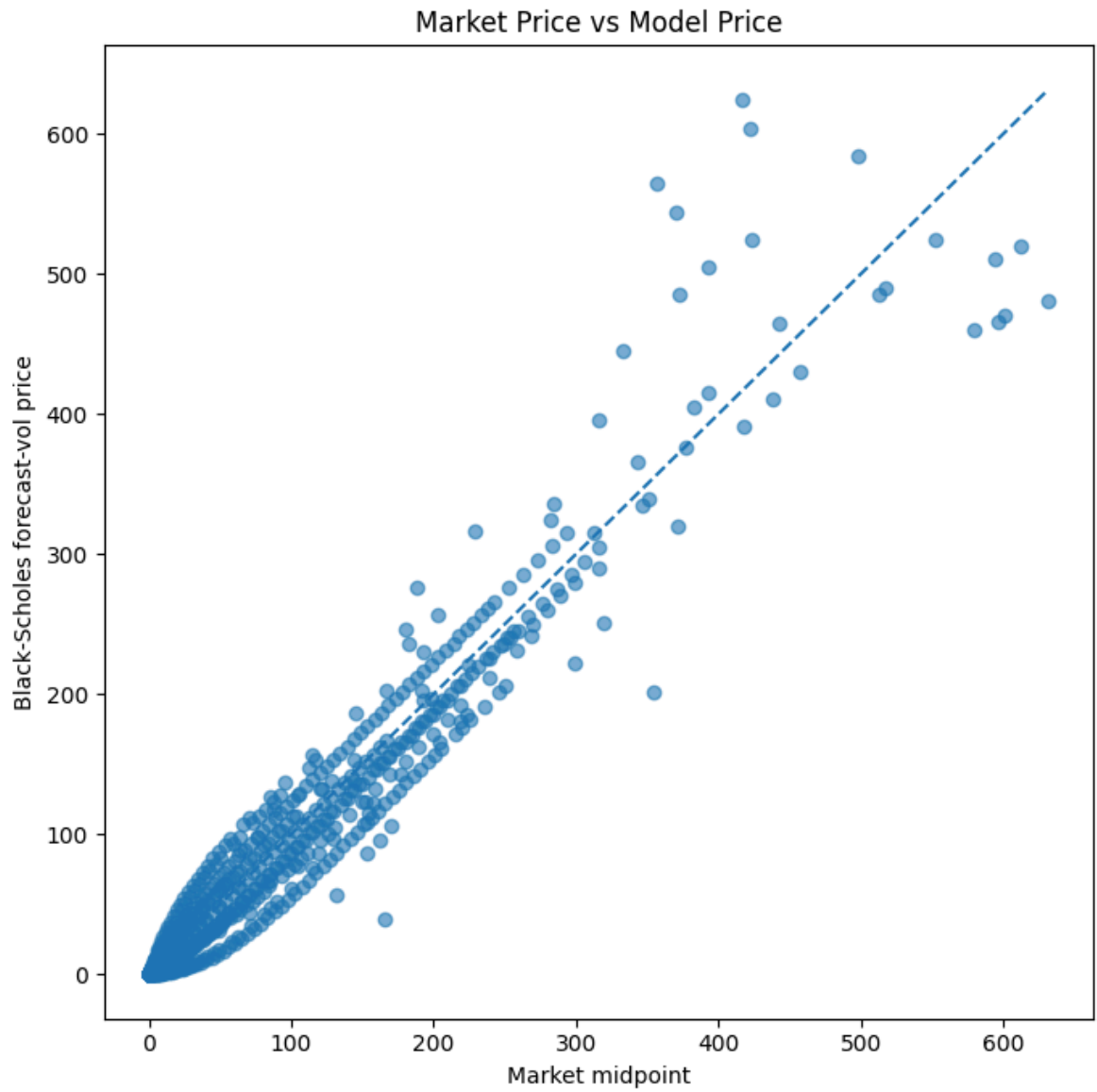
The diagonal line represents exact agreement. Points above the line are valued more highly by the forecast-volatility model than by the market midpoint, while points below the line have a lower model value.

Differences can arise from genuine disagreement about future volatility, but also from the volatility smile, skew, stale quotations, wide spreads and the fact that one forecast-volatility value is currently used across all strikes for an underlying.

```
In [15]: from options_risk_engine.reporting.plots import plot_market_vs_model

plot_market_vs_model(large_table, FIGURE_DIR / "03_market_vs_model.png")
plt.show()

display(large_table[[
    "Ticker", "Option_Type", "contractSymbol", "strike", "MarketMid",
    "BS_ForeV", "BS_ForeV AskEdge", "IV_Mid", "Greek Volatility",
    "Initial Recommended Action",
]].head(20).round(4))
```



	Ticker	Option_Type	contractSymbol	strike	MarketMid	BS_ForeV	BS_F Ask
0	AAPL	Calls	AAPL260918C00050000	50.0	259.075	245.5378	-0.
1	AAPL	Calls	AAPL260918C00055000	55.0	254.200	240.5619	-0.
2	AAPL	Calls	AAPL260918C00060000	60.0	249.175	235.5860	-0
3	AAPL	Calls	AAPL260918C00065000	65.0	192.550	230.6101	0
4	AAPL	Calls	AAPL260918C00070000	70.0	239.275	225.6342	-0.
5	AAPL	Calls	AAPL260918C00075000	75.0	224.050	220.6582	-0.
6	AAPL	Calls	AAPL260918C00080000	80.0	NaN	215.6823	
7	AAPL	Calls	AAPL260918C00085000	85.0	NaN	210.7064	
8	AAPL	Calls	AAPL260918C00090000	90.0	219.275	205.7305	-0.
9	AAPL	Calls	AAPL260918C00095000	95.0	NaN	200.7546	
10	AAPL	Calls	AAPL260918C00100000	100.0	209.350	195.7787	-0
11	AAPL	Calls	AAPL260918C00105000	105.0	204.350	190.8028	-0
12	AAPL	Calls	AAPL260918C00110000	110.0	199.300	185.8268	-0
13	AAPL	Calls	AAPL260918C00115000	115.0	194.325	180.8509	-0
14	AAPL	Calls	AAPL260918C00120000	120.0	189.425	175.8750	-0.
15	AAPL	Calls	AAPL260918C00125000	125.0	184.500	170.8991	-0.
16	AAPL	Calls	AAPL260918C00130000	130.0	179.800	165.9232	-0.
17	AAPL	Calls	AAPL260918C00135000	135.0	174.550	160.9473	-0.
18	AAPL	Calls	AAPL260918C00140000	140.0	169.525	155.9714	-0.
19	AAPL	Calls	AAPL260918C00145000	145.0	164.650	150.9954	-0.

The model and market values have a clear positive relationship, confirming that strike, spot and maturity are being reflected sensibly. However, the dispersion increases for some high-value contracts and several observations sit well away from the equality line.

This is expected because the market does not price all strikes using one constant volatility. The difference between forecast and market price should therefore be interpreted as a screening signal that requires liquidity and quote-quality checks, rather than proof that the market is incorrectly priced.

Analytical Greeks versus finite differences

The analytical Delta, Gamma, Vega, Theta and Rho calculations are checked against numerical finite-difference approximations.

This is an important implementation test because a pricing formula can appear reasonable while one or more derivative formulas contain a sign, scale or unit error. The numerical values are calculated from small changes in the relevant Black–Scholes input and compared using Greek-specific tolerances.

```
In [16]: from options_risk_engine.pricing.greeks import add_numerical_greek_validation

sample_validations = []
for ticker in tickers:
    for option_type, key in (("call", ticker.table_key("Calls")), ("put",
        sample = cleaned_options[key].dropna(subset=["Greek Volatility"])
        if not sample.empty:
            checked = add_numerical_greek_validation(sample, ticker, option_type)
            sample_validations.append(checked.assign(Ticker=ticker.symbol))

greek_validation_sample = pd.concat(sample_validations, ignore_index=True)
validation_columns = [column for column in greek_validation_sample.columns
    display(greek_validation_sample[validation_columns])
```

	contractSymbol	Ticker	Option_Type	Delta Validation Passed	Gamma Validation Passed	Vega Validation Passed
0	AAPL260918C00050000	AAPL	call	True	True	True
1	AAPL260918C00055000	AAPL	call	True	True	True
2	AAPL260918C00060000	AAPL	call	True	True	True
3	AAPL260918C00065000	AAPL	call	True	True	True
4	AAPL260918C00070000	AAPL	call	True	True	True
...	...	...	...	...	...	...
65	HCA260918P00200000	HCA	put	True	True	True
66	HCA260918P00220000	HCA	put	True	True	True
67	HCA260918P00250000	HCA	put	True	True	True
68	HCA260918P00260000	HCA	put	True	True	True
69	HCA260918P00270000	HCA	put	True	True	True

70 rows × 8 columns

The representative sample contains calls and puts from every ticker. The displayed validation flags pass for the tested Greeks across all 70 sample contracts.

This supports the accuracy of both the analytical formulas and their unit conventions, including Vega and Rho per percentage-point change and Theta per calendar day.

Persist the priced option universe

```
In [17]: store.write_frame(large_table, "option_pricing_results", run_id=run_id)
large_table.to_csv(TABLE_DIR / "option_pricing_results.csv", index=False)
print("Stored priced contracts:", len(large_table))
```

2026-08-02 13:38:43,444 – options_risk_engine.database – INFO – Stored 1211 rows in option_pricing_results
Stored priced contracts: 1211

9. Monte Carlo pricing cross-check

Monte Carlo simulation provides a numerical cross-check of the analytical Black–Scholes values. Under the same geometric-Brownian-motion assumptions, the discounted average terminal payoff should converge towards the analytical price as the number of paths increases.

Ten thousand paths are generated for each underlying using the same spot, forecast volatility, rate, dividend yield and maturity as the analytical model. Antithetic random draws are used to reduce simulation noise.

The simulated terminal distribution is also used to estimate contract-level profit probability and expected payoff relative to the option premium.

```
In [18]: from options_risk_engine.pricing.monte_carlo import run_monte_carlo_analy

if CONFIG.run_monte_carlo:
    monte_carlo_results = run_monte_carlo_analysis(
        tickers=tickers,
        cleaned_options=cleaned_options,
        simulations=CONFIG.monte_carlo_simulations,
        plot_results=CONFIG.plot_monte_carlo,
        random_seed=CONFIG.random_seed,
    )
    large_table = combine_option_tables(cleaned_options)
    mc_summary = large_table.groupby(["Ticker", "Option_Type"], observed=
        Median_MC_SE=("MC_SE", "median"),
        Median_MC_minus_BS=("MC - BS ForeV", "median"),
        Median_Probability_Profit=("Probability of Profit", "median"),
        Median_Expected_Return=("Expected Profit Return", "median"),
    ).reset_index()
    display(mc_summary.round(4))
else:
    monte_carlo_results = {}
    print("Monte Carlo skipped")
```



```
2026-08-02 13:38:43,562 - options_risk_engine.pricing.monte_carlo - INFO -
Running 10000 Monte Carlo simulations for AAPL
2026-08-02 13:38:43,563 - options_risk_engine.pricing.monte_carlo - INFO -
AAPL starting Monte Carlo simulation
2026-08-02 13:38:43,564 - options_risk_engine.pricing.monte_carlo - INFO -
AAPL Monte Carlo inputs - S0: 308.91, sigma: 0.2623, T: 0.1288, steps: 34,
simulations: 10000
2026-08-02 13:38:43,584 - options_risk_engine.pricing.monte_carlo - INFO -
AAPL Monte Carlo simulation complete
2026-08-02 13:38:43,640 - options_risk_engine.pricing.monte_carlo - INFO -
AAPL Monte Carlo profit analysis complete
2026-08-02 13:38:43,640 - options_risk_engine.pricing.monte_carlo - INFO -
Running 10000 Monte Carlo simulations for MSFT
2026-08-02 13:38:43,641 - options_risk_engine.pricing.monte_carlo - INFO -
MSFT starting Monte Carlo simulation
2026-08-02 13:38:43,642 - options_risk_engine.pricing.monte_carlo - INFO -
MSFT Monte Carlo inputs - S0: 464.72, sigma: 0.2609, T: 0.1288, steps: 34,
simulations: 10000
2026-08-02 13:38:43,658 - options_risk_engine.pricing.monte_carlo - INFO -
MSFT Monte Carlo simulation complete
2026-08-02 13:38:43,712 - options_risk_engine.pricing.monte_carlo - INFO -
MSFT Monte Carlo profit analysis complete
2026-08-02 13:38:43,713 - options_risk_engine.pricing.monte_carlo - INFO -
Running 10000 Monte Carlo simulations for META
2026-08-02 13:38:43,713 - options_risk_engine.pricing.monte_carlo - INFO -
META starting Monte Carlo simulation
2026-08-02 13:38:43,714 - options_risk_engine.pricing.monte_carlo - INFO -
META Monte Carlo inputs - S0: 556.71, sigma: 0.4131, T: 0.1288, steps: 34,
simulations: 10000
2026-08-02 13:38:43,745 - options_risk_engine.pricing.monte_carlo - INFO -
META Monte Carlo simulation complete
2026-08-02 13:38:43,835 - options_risk_engine.pricing.monte_carlo - INFO -
META Monte Carlo profit analysis complete
2026-08-02 13:38:43,835 - options_risk_engine.pricing.monte_carlo - INFO -
Running 10000 Monte Carlo simulations for WMT
2026-08-02 13:38:43,836 - options_risk_engine.pricing.monte_carlo - INFO -
WMT starting Monte Carlo simulation
2026-08-02 13:38:43,836 - options_risk_engine.pricing.monte_carlo - INFO -
WMT Monte Carlo inputs - S0: 111.20, sigma: 0.2065, T: 0.1288, steps: 34,
simulations: 10000
2026-08-02 13:38:43,849 - options_risk_engine.pricing.monte_carlo - INFO -
WMT Monte Carlo simulation complete
2026-08-02 13:38:43,878 - options_risk_engine.pricing.monte_carlo - INFO -
WMT Monte Carlo profit analysis complete
2026-08-02 13:38:43,879 - options_risk_engine.pricing.monte_carlo - INFO -
Running 10000 Monte Carlo simulations for GOOGL
2026-08-02 13:38:43,880 - options_risk_engine.pricing.monte_carlo - INFO -
GOOGL starting Monte Carlo simulation
2026-08-02 13:38:43,881 - options_risk_engine.pricing.monte_carlo - INFO -
GOOGL Monte Carlo inputs - S0: 356.13, sigma: 0.3099, T: 0.1288, steps: 3
4, simulations: 10000
2026-08-02 13:38:43,899 - options_risk_engine.pricing.monte_carlo - INFO -
GOOGL Monte Carlo simulation complete
2026-08-02 13:38:43,965 - options_risk_engine.pricing.monte_carlo - INFO -
GOOGL Monte Carlo profit analysis complete
2026-08-02 13:38:43,965 - options_risk_engine.pricing.monte_carlo - INFO -
Running 10000 Monte Carlo simulations for AMZN
2026-08-02 13:38:43,966 - options_risk_engine.pricing.monte_carlo - INFO -
AMZN starting Monte Carlo simulation
2026-08-02 13:38:43,967 - options_risk_engine.pricing.monte_carlo - INFO -
```

```

AMZN Monte Carlo inputs - S0: 271.58, sigma: 0.3413, T: 0.1288, steps: 34,
simulations: 10000
2026-08-02 13:38:43,982 - options_risk_engine.pricing.monte_carlo - INFO -
AMZN Monte Carlo simulation complete
2026-08-02 13:38:44,021 - options_risk_engine.pricing.monte_carlo - INFO -
AMZN Monte Carlo profit analysis complete
2026-08-02 13:38:44,022 - options_risk_engine.pricing.monte_carlo - INFO -
Running 10000 Monte Carlo simulations for HCA
2026-08-02 13:38:44,023 - options_risk_engine.pricing.monte_carlo - INFO -
HCA starting Monte Carlo simulation
2026-08-02 13:38:44,023 - options_risk_engine.pricing.monte_carlo - INFO -
HCA Monte Carlo inputs - S0: 402.59, sigma: 0.2788, T: 0.1288, steps: 34,
simulations: 10000
2026-08-02 13:38:44,038 - options_risk_engine.pricing.monte_carlo - INFO -
HCA Monte Carlo simulation complete
2026-08-02 13:38:44,082 - options_risk_engine.pricing.monte_carlo - INFO -
HCA Monte Carlo profit analysis complete
2026-08-02 13:38:44,096 - options_risk_engine.reporting.tables - INFO - Co
mbined 14 option tables into 1211 rows

```

	Ticker	Option_Type	Median_MC_SE	Median_MC_minus_BS	Median_Probability
0	AAPL	Calls	0.2251	0.0306	
1	AAPL	Puts	0.0066	-0.0000	
2	AMZN	Calls	0.2860	0.0477	
3	AMZN	Puts	0.0546	0.0040	
4	GOOGL	Calls	0.2059	0.0499	
5	GOOGL	Puts	0.0074	-0.0000	
6	HCA	Calls	0.0422	0.0371	
7	HCA	Puts	0.3215	-0.0000	
8	META	Calls	0.0401	0.0085	
9	META	Puts	0.5537	-0.0000	
10	MSFT	Calls	0.1957	0.0432	
11	MSFT	Puts	0.2033	-0.0000	
12	WMT	Calls	0.0111	0.0062	
13	WMT	Puts	0.0385	-0.0000	

Monte Carlo observations

Median Monte Carlo prices are very close to the analytical forecast-volatility prices.

The median differences are generally only a few cents, supporting the consistency of the simulation and closed-form implementations.

Monte Carlo standard errors vary across option groups because payoff dispersion depends on strike, volatility and option value. A larger standard error should be interpreted as greater simulation uncertainty rather than a difference in the underlying theoretical model.

Because both methods use the same lognormal assumptions, this comparison is a numerical validation rather than an independent market-pricing model. Likewise, the calculated probability of profit is conditional on the chosen volatility process and premium; it is not a guaranteed real-world probability.

10. Contract screening and model-implied opportunities

The priced universe is grouped by ticker, option type and screening label. Contracts are only labelled as potential buys when the forecast-based value exceeds the ask by the required margin and the quote, spread, volume, open interest and moneyness filters are satisfied.

These labels are intended to demonstrate how the pricing engine can be turned into a systematic research screen. They are not trading recommendations and have not been validated using historical execution or realised trade returns.

```
In [19]: from options_risk_engine.reporting.tables import create_findings_table

findings_table = create_findings_table(large_table)
display(findings_table.round(4))

highlighted = large_table.loc[large_table["Initial Recommended Action"].e
highlight_columns = [
    "Ticker", "Option_Type", "contractSymbol", "strike", "Expiry", "ask",
    "BS_ForeV", "BS_ForeV AskEdge", "SpreadPct", "volume", "openInterest",
    "Probability of Profit", "Expected Profit Return",
]
display(highlighted[highlight_columns].sort_values("BS_ForeV AskEdge", as
```

```
2026-08-02 13:38:44,124 - options_risk_engine.reporting.tables - INFO - Cr
eated findings table with 52 groups
```

	Initial Recommended Action	Option_Table	Option_Count	Median_Lower_Profit	Median_Centr
0	Buy	AAPL Puts	8	-4.3459	
1	Buy	AMZN Calls	14	-2.6478	
2	Buy	AMZN Puts	2	-7.3102	
3	Buy	GOOGL Puts	12	-10.4755	
4	Buy	HCA Puts	8	-5.2754	
5	Buy	META Puts	33	-6.1295	
6	Buy	MSFT Puts	10	-3.5169	
7	Buy	WMT Puts	4	-0.6431	
8	Do Not Buy	AAPL Calls	77	-49.3800	
9	Do Not Buy	AAPL Puts	21	-0.2814	
10	Do Not Buy	AMZN Calls	37	-53.6981	
11	Do Not Buy	AMZN Puts	33	-0.7034	
12	Do Not Buy	GOOGL Calls	114	-29.8186	
13	Do Not Buy	GOOGL Puts	25	-0.6732	
14	Do Not Buy	HCA Calls	49	-32.6573	
15	Do Not Buy	HCA Puts	4	-2.0348	
16	Do Not Buy	META Calls	157	-1.9092	
17	Do Not Buy	META Puts	25	-0.2914	
18	Do Not Buy	MSFT Calls	86	-39.7917	
19	Do Not Buy	MSFT Puts	30	-0.4220	
20	Do Not Buy	WMT Calls	25	-16.7306	
21	Do Not Buy	WMT Puts	4	-0.2110	
22	Invalid Quote	AAPL Calls	1	-7.2753	
23	Invalid Quote	AAPL Puts	21	-38.3495	
24	Invalid Quote	AMZN Calls	1	-53.0673	
25	Invalid Quote	GOOGL Calls	5	-73.6683	
26	Invalid Quote	GOOGL Puts	20	-57.8193	
27	Invalid Quote	HCA Calls	1	-57.3011	
28	Invalid Quote	HCA Puts	28	-30.3905	
29	Invalid Quote	META Calls	7	-244.4181	-
30	Invalid Quote	META Puts	67	-121.1232	
31	Invalid Quote	MSFT Puts	31	-28.3227	
32	Invalid Quote	WMT Puts	11	-3.3961	

	Initial Recommended Action	Option_Table	Option_Count	Median_Lower_Profit	Median_Centr
33	No Data	AAPL Calls	15	-0.4723	
34	No Data	AAPL Puts	23	-0.5125	
35	No Data	AMZN Calls	1	-0.0804	
36	No Data	AMZN Puts	9	-0.3316	
37	No Data	GOOGL Calls	15	-2.1604	
38	No Data	GOOGL Puts	40	-0.1256	
39	No Data	HCA Calls	23	-2.2609	
40	No Data	HCA Puts	19	-3.5169	
41	No Data	META Calls	26	-1.3364	
42	No Data	META Puts	21	-0.2010	
43	No Data	MSFT Calls	4	-31.7204	
44	No Data	MSFT Puts	10	-69.0094	
45	No Data	WMT Calls	9	-0.3517	
46	No Data	WMT Puts	9	-0.2512	
47	Positive Edge	AAPL Puts	1	-1.4269	
48	Positive Edge	AMZN Calls	5	-17.0320	
49	Positive Edge	AMZN Puts	8	-12.7615	
50	Positive Edge	GOOGL Puts	1	-2.8939	
51	Positive Edge	MSFT Puts	1	-1.7484	

	Ticker	Option_Type	contractSymbol	strike	Expiry	ask	BS_ForeV	I
723	WMT	Puts	WMT260918P00100000	100.0	2026-09-18	1.14	3.1195	
722	WMT	Puts	WMT260918P00097500	97.5	2026-09-18	0.77	1.9586	
299	MSFT	Puts	MSFT260918P00425000	425.0	2026-09-18	6.75	17.1002	
297	MSFT	Puts	MSFT260918P00415000	415.0	2026-09-18	4.95	12.1755	
298	MSFT	Puts	MSFT260918P00420000	420.0	2026-09-18	5.90	14.5118	
296	MSFT	Puts	MSFT260918P00410000	410.0	2026-09-18	4.25	10.0917	
721	WMT	Puts	WMT260918P00095000	95.0	2026-09-18	0.51	1.1269	
295	MSFT	Puts	MSFT260918P00405000	405.0	2026-09-18	3.75	8.2568	
1173	HCA	Puts	HCA260918P00370000	370.0	2026-09-18	7.50	16.5061	
1171	HCA	Puts	HCA260918P00360000	360.0	2026-09-18	5.30	11.4797	
294	MSFT	Puts	MSFT260918P00400000	400.0	2026-09-18	3.25	6.6631	
582	META	Puts	META260918P00475000	475.0	2026-09-18	4.70	9.3325	
581	META	Puts	META260918P00470000	470.0	2026-09-18	4.15	8.1817	
584	META	Puts	META260918P00485000	485.0	2026-09-18	6.10	11.9693	
583	META	Puts	META260918P00480000	480.0	2026-09-18	5.40	10.5937	
585	META	Puts	META260918P00490000	490.0	2026-09-18	6.95	13.4630	
580	META	Puts	META260918P00465000	465.0	2026-09-18	3.70	7.1368	
579	META	Puts	META260918P00460000	460.0	2026-09-18	3.25	6.1927	
586	META	Puts	META260918P00495000	495.0	2026-09-18	7.95	15.0779	
578	META	Puts	META260918P00455000	455.0	2026-09-18	2.85	5.3444	

Screening observations

The engine identified 91 contracts in the **Buy** category. The candidates were concentrated heavily in puts, including AAPL, GOOGL, HCA, META, MSFT and WMT puts, with AMZN contributing both calls and puts.

This concentration may indicate that the forecast-volatility model assigns greater value to downside protection than the current asks. However, it may also reflect the inability of a single ticker-level volatility forecast to represent the market's strike-dependent skew.

The output also contains many **Invalid Quote** and **No Data** classifications. This reinforces that a large theoretical edge is not meaningful unless the quote is current, executable and sufficiently liquid.

A historical backtest with realistic bid-ask execution, transaction costs and subsequent realised volatility would be needed before treating the screen as evidence of a profitable strategy.

11. Synthetic portfolio construction

A reproducible synthetic portfolio is constructed to demonstrate the risk engine across a large mixture of calls, puts and ordinary shares.

For every usable option row, side, quantity and entry price are assigned using a fixed random seed. Seven share positions are then added so that the engine can handle linear equity exposure alongside nonlinear options exposure.

This is deliberately an engineering test portfolio. In a production environment, positions would be imported from a portfolio-management system, broker feed or user-supplied file rather than randomly generated.

```
In [20]: from options_risk_engine.risk.portfolio import risk_engine_data_prep, risk_engine_summary

position_options = risk_engine_data_prep(large_table)
risk_free_rates = {ticker.symbol: ticker.risk_free_rate for ticker in tickers}
position_options["Dividend Yield"] = position_options["Ticker"].map(dividend_yield)
position_options["Risk Free Rate"] = position_options["Ticker"].map(risk_free_rates)
portfolio_risk = risk_engine_summary(position_options)

display(position_options.head(15))
display(portfolio_risk.round(4))
```



```
2026-08-02 13:38:44,178 - options_risk_engine.risk.portfolio - INFO - Preparing option universe for the portfolio risk engine: 1211 rows
2026-08-02 13:38:44,182 - options_risk_engine.risk.portfolio - INFO - Risk-engine input cleaned: 959 usable option rows; 252 rows removed
2026-08-02 13:38:44,185 - options_risk_engine.risk.portfolio - INFO - Created reproducible test option positions: 471 long and 488 short
/Users/sam/Progressive-Projects/Projects/Option Pricer N Risk Engine/src/options_risk_engine/risk/portfolio.py:180: FutureWarning: The behavior of DataFrame concatenation with empty or all-NA entries is deprecated. In a future version, this will no longer exclude empty or all-NA columns when determining the result dtypes. To retain the old behavior, exclude the relevant entries before the concat operation.
    position_options = pd.concat(
2026-08-02 13:38:44,197 - options_risk_engine.risk.portfolio - INFO - Added 7 share positions; combined portfolio now contains 966 rows
2026-08-02 13:38:44,200 - options_risk_engine.risk.portfolio - INFO - Portfolio positions prepared: gross market value 17747056.90; current P&L -64853.89
2026-08-02 13:38:44,201 - options_risk_engine.risk.portfolio - INFO - Building current portfolio risk summary from 966 positions
2026-08-02 13:38:44,208 - options_risk_engine.risk.portfolio - INFO - Ticker-level risk calculated for 7 tickers
2026-08-02 13:38:44,209 - options_risk_engine.risk.portfolio - INFO - Portfolio risk summary complete: net value -695290.30; gross value 17747056.90; delta 1592.81
```

	contractSymbol	Ticker	Option_Type	strike	Expiry	Current Stock Price	Time t Expir
0	AAPL260918C00050000	AAPL	Calls	50.0	2026-09-18	308.910004	0.12876
1	AAPL260918C00055000	AAPL	Calls	55.0	2026-09-18	308.910004	0.12876
2	AAPL260918C00060000	AAPL	Calls	60.0	2026-09-18	308.910004	0.12876
3	AAPL260918C00065000	AAPL	Calls	65.0	2026-09-18	308.910004	0.12876
4	AAPL260918C00070000	AAPL	Calls	70.0	2026-09-18	308.910004	0.12876
5	AAPL260918C00075000	AAPL	Calls	75.0	2026-09-18	308.910004	0.12876
6	AAPL260918C00090000	AAPL	Calls	90.0	2026-09-18	308.910004	0.12876
7	AAPL260918C00100000	AAPL	Calls	100.0	2026-09-18	308.910004	0.12876
8	AAPL260918C00105000	AAPL	Calls	105.0	2026-09-18	308.910004	0.12876
9	AAPL260918C00110000	AAPL	Calls	110.0	2026-09-18	308.910004	0.12876
10	AAPL260918C00115000	AAPL	Calls	115.0	2026-09-18	308.910004	0.12876
11	AAPL260918C00120000	AAPL	Calls	120.0	2026-09-18	308.910004	0.12876
12	AAPL260918C00125000	AAPL	Calls	125.0	2026-09-18	308.910004	0.12876
13	AAPL260918C00130000	AAPL	Calls	130.0	2026-09-18	308.910004	0.12876
14	AAPL260918C00135000	AAPL	Calls	135.0	2026-09-18	308.910004	0.12876

	Ticker	Number_of_Positions	Net_Market_Value	Gross_Market_Value	Positio
0	AAPL	125	-142781.9998	2.162154e+06	-143
1	AMZN	100	77914.8005	1.190960e+06	-2000
2	GOOGL	177	-322374.4498	4.135160e+06	-224
3	HCA	81	59862.2499	9.915272e+05	381
4	META	284	-353127.4978	6.609912e+06	-4678
5	MSFT	154	-19684.9000	2.449665e+06	-26
6	WMT	45	4901.4999	2.076795e+05	206
7	PORTFOLIO	966	-695290.2971	1.774706e+07	-6485

Portfolio construction observations

Of the 1,211 priced contracts, 959 option rows contained all fields required by the portfolio engine. A further seven share holdings produced a combined portfolio of 966 positions.

The gross market value is approximately £17.75 million, while the signed net market value is approximately -£695,000. Current synthetic P&L is around -£64,900, and aggregate Delta is approximately +1,592.

The large difference between gross and net value shows substantial offsetting long and short positions. This makes the portfolio useful for testing aggregation and nonlinear risk, but it is much larger and more complex than a typical personal portfolio. Its results should be interpreted as a stress-engine demonstration rather than a proposed allocation.

Current portfolio risk factors

Position-level Greeks are multiplied by direction, quantity and contract multiplier before being aggregated by ticker.

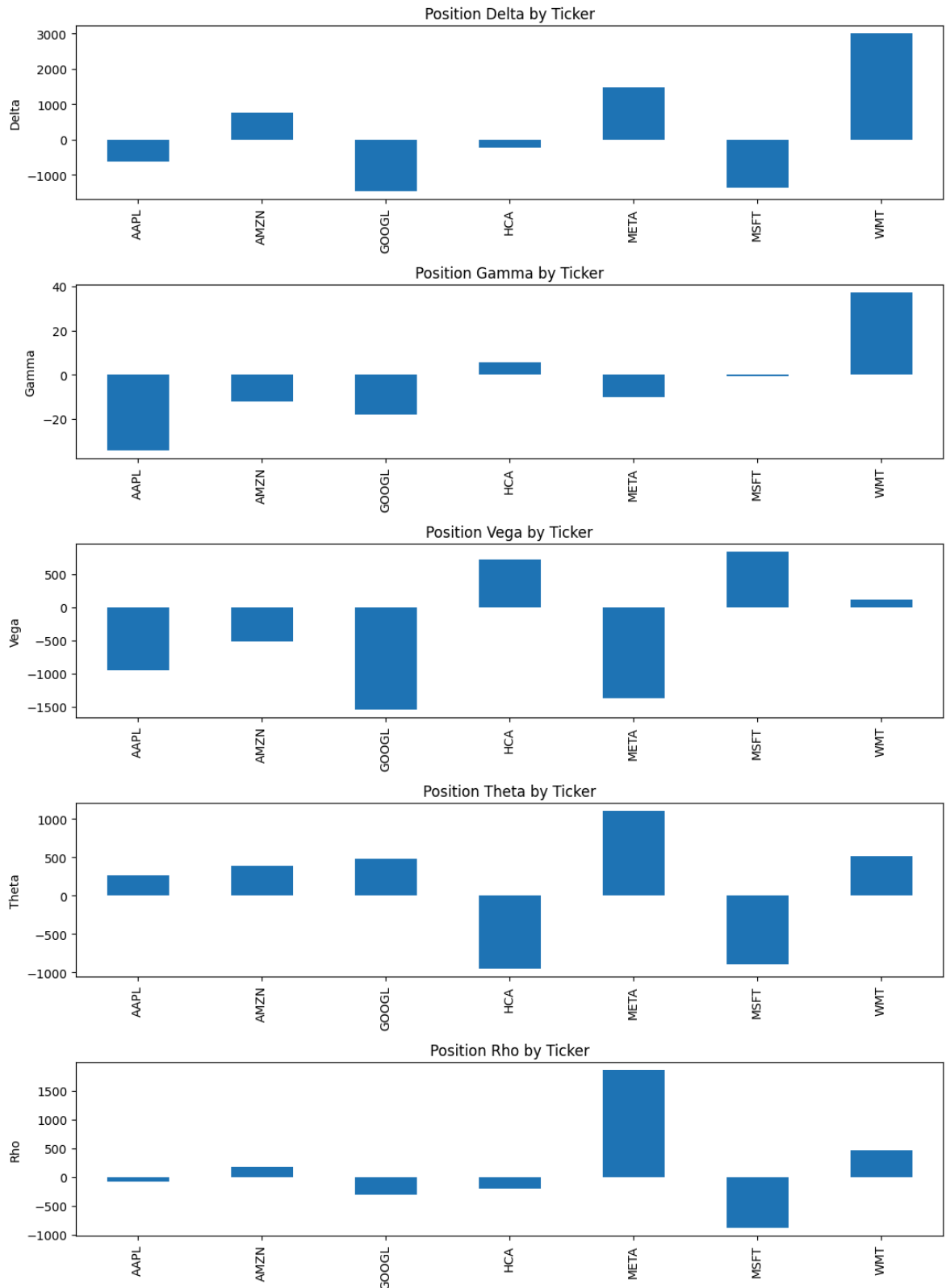
The five charts separate:

- Delta: first-order directional exposure;
- Gamma: change in Delta and convexity;
- Vega: sensitivity to a one-percentage-point volatility change;
- Theta: approximate one-day time decay; and
- Rho: sensitivity to a one-percentage-point rate change.

Because these Greeks have different units and scales, they are displayed in separate panels rather than combined into a single magnitude comparison.

```
In [21]: from options_risk_engine.reporting.plots import plot_portfolio_greeks
```

```
plot_portfolio_greeks(portfolio_risk, FIGURE_DIR / "04_portfolio_greeks.p
plt.show()
```



Greek-exposure observations

The portfolio is directionally net long, but this exposure is created by offsetting ticker positions. WMT and META provide the largest positive Delta contributions, while GOOGL and MSFT provide substantial negative Delta.

The Gamma profile is especially important. Several large ticker books, including AAPL, GOOGL, AMZN and META, have negative Gamma, while WMT provides the largest positive Gamma offset. The portfolio is therefore vulnerable to some large underlying moves even where first-order Delta exposures appear to offset.

Vega is negative for several major books, particularly GOOGL, META and AAPL, but positive for HCA and MSFT. The portfolio can therefore lose from a volatility increase depending on which positions dominate the scenario.

Theta and Rho are also mixed across tickers. These offsets demonstrate why portfolio-level risk cannot be inferred from position count or market value alone. They also foreshadow the importance of Gamma in the later stress-test results.

Persist positions and current risk

```
In [22]: store.write_frame(position_options, "portfolio_positions", run_id=run_id)
store.write_frame(portfolio_risk, "portfolio_risk", run_id=run_id)
```

```
2026-08-02 13:38:45,076 - options_risk_engine.database - INFO - Stored 966
rows in portfolio_positions
2026-08-02 13:38:45,081 - options_risk_engine.database - INFO - Stored 8 r
ows in portfolio_risk
2026-08-02 13:38:45,081 - options_risk_engine.database - INFO - Stored 8 r
ows in portfolio_risk
```

12. Transparent multi-factor stress scenarios

The portfolio is tested under 1,000 reproducible market scenarios plus an unchanged BASE scenario.

Each scenario contains:

- a percentage spot-price shock;
- an absolute volatility shock;
- a number of elapsed calendar days; and
- an absolute interest-rate shock.

The distributions are intentionally broad and transparent. They are designed for stress testing and engine demonstration rather than being calibrated as a formal historical or Monte Carlo Value-at-Risk model.

```
In [23]: from options_risk_engine.risk.scenarios import generate_random_scenarios

scenario_shocks = generate_random_scenarios(
    number_of_scenarios=CONFIG.scenario_count,
    max_days_forward=CONFIG.max_days_forward,
    seed=CONFIG.random_seed,
)
display(scenario_shocks.head(10))
display(scenario_shocks.drop(columns="Scenario ID").describe().T.round(4))
```

2026-08-02 13:38:45,088 – options_risk_engine.risk.scenarios – INFO – Generating 1000 random scenarios with up to 30 days forward; seed=42
 2026-08-02 13:38:45,092 – options_risk_engine.risk.scenarios – INFO – Scenario generation complete: 1001 total rows including BASE

	Spot Shock	Volatility Shock	Days Forward	Rate Shock	Scenario ID
0	0.000000	0.000000	0	0.000000	BASE
1	0.024377	-0.002964	22	-0.000011	R0001
2	-0.083199	-0.036464	1	0.004077	R0002
3	0.060036	-0.020724	11	-0.003509	R0003
4	0.075245	0.031696	13	0.003830	R0004
5	-0.156083	0.000150	29	-0.003810	R0005
6	-0.104174	0.017010	16	0.005453	R0006
7	0.010227	0.033504	16	0.011053	R0007
8	-0.025299	-0.018742	4	-0.000610	R0008
9	-0.001344	0.037812	24	-0.010963	R0009

	count	mean	std	min	25%	50%	75%	max
Spot Shock	1001.0	-0.0023	0.0791	-0.2919	-0.0557	0.0005	0.0472	0.2543
Volatility Shock	1001.0	-0.0041	0.0507	-0.1524	-0.0365	-0.0022	0.0301	0.1457
Days Forward	1001.0	14.7642	9.0786	0.0000	7.0000	15.0000	23.0000	30.0000
Rate Shock	1001.0	0.0002	0.0050	-0.0143	-0.0032	0.0002	0.0037	0.0154

Scenario-distribution observations

The generated spot shocks are centred close to zero with a standard deviation of approximately 7.9%, and range from roughly -29.2% to +25.4%. Volatility shocks range from approximately -15.2 to +14.6 percentage points, while the time horizon varies from zero to 30 calendar days.

The broad ranges ensure that the engine is tested under both moderate and severe movements. The BASE scenario provides a direct accounting check: no change in spot, volatility, time or rates should produce zero P&L.

One limitation is that the same percentage spot and volatility shock is currently applied to every ticker. This creates a common systemic stress but does not represent individual equity correlations or idiosyncratic moves.

13. Full-revaluation scenario engine

For every scenario, each option is fully repriced using its shocked spot, volatility, time to expiry and interest rate. Share positions are repriced directly using the shocked stock price.

Full revaluation is used as the primary scenario P&L because options are nonlinear. A local Greek approximation may be accurate for small movements, but it does not update the Greeks as the option moves through a large shock.

Scenario P&L is measured relative to the model value calculated using the unshocked scenario inputs. This ensures that the BASE scenario has zero P&L even where the current market midpoint differs from the selected model value.

```
In [24]: from options_risk_engine.risk.scenarios import run_scenario_engine

results, scenario_ticker, scenario_portfolio = run_scenario_engine(
    position_options=position_options,
    scenarios=scenario_shocks,
    dividend_yields=dividend_yields,
)

print("Position-scenario rows:", len(results))
display(results.head())
display(scenario_portfolio.head())
```

```
2026-08-02 13:38:45,113 - options_risk_engine.risk.scenarios - INFO - Star
ting scenario engine: 966 positions across 1001 scenarios
2026-08-02 13:38:45,116 - options_risk_engine.risk.scenarios - INFO - Scen
ario positions classified: 565 calls, 394 puts and 7 shares
2026-08-02 13:38:45,122 - options_risk_engine.risk.scenarios - INFO - Base
model prices calculated for 966 positions
2026-08-02 13:38:45,123 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 1/1001: BASE
2026-08-02 13:38:45,487 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 100/1001: R0099
2026-08-02 13:38:45,854 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 200/1001: R0199
2026-08-02 13:38:46,251 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 300/1001: R0299
2026-08-02 13:38:46,669 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 400/1001: R0399
2026-08-02 13:38:47,056 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 500/1001: R0499
2026-08-02 13:38:47,438 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 600/1001: R0599
2026-08-02 13:38:47,809 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 700/1001: R0699
2026-08-02 13:38:48,166 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 800/1001: R0799
2026-08-02 13:38:48,516 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 900/1001: R0899
2026-08-02 13:38:48,923 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 1000/1001: R0999
2026-08-02 13:38:48,928 - options_risk_engine.risk.scenarios - INFO - Reva
luing scenario 1001/1001: R1000
2026-08-02 13:38:49,067 - options_risk_engine.risk.scenarios - INFO - Posi
tion-level scenario results created: 966966 rows
Position-scenario rows: 966966
```


	Scenario ID	Contract Symbol	Ticker	Option Type	Current Market Price	Base Model Price	Scenario Price
0	BASE	AAPL260918C00050000	AAPL	Calls	259.075	259.074991	259.0749
1	BASE	AAPL260918C00055000	AAPL	Calls	254.200	254.200000	254.2000
2	BASE	AAPL260918C00060000	AAPL	Calls	249.175	249.175003	249.1750
3	BASE	AAPL260918C00065000	AAPL	Calls	192.550	230.610072	230.6100
4	BASE	AAPL260918C00070000	AAPL	Calls	239.275	239.275000	239.2750

	Scenario ID	Portfolio_PnL	Portfolio_Value
0	BASE	0.000000	-842272.058183
1	R0001	7831.894656	-834440.163527
2	R0002	-11142.751852	-853414.810035
3	R0003	-10400.133826	-852672.192009
4	R0004	-22980.244444	-865252.302628

Scenario-engine observations

The engine repriced 966 positions across 1,001 scenarios, producing 966,966 position-level rows. The portfolio contained 565 calls, 394 puts and seven share positions.

The BASE portfolio P&L is exactly zero, confirming that the scenario engine uses a consistent model-to-model comparison.

The BASE portfolio value differs from the earlier signed market value because the scenario engine values positions on its model basis. For contracts where the market midpoint cannot be reproduced reliably and a volatility fallback is required, the model price may differ from the current market mark. This difference is intentionally excluded from scenario movement P&L.

The position, ticker and portfolio outputs allow the same scenario to be analysed at several levels without rerunning the pricing engine.

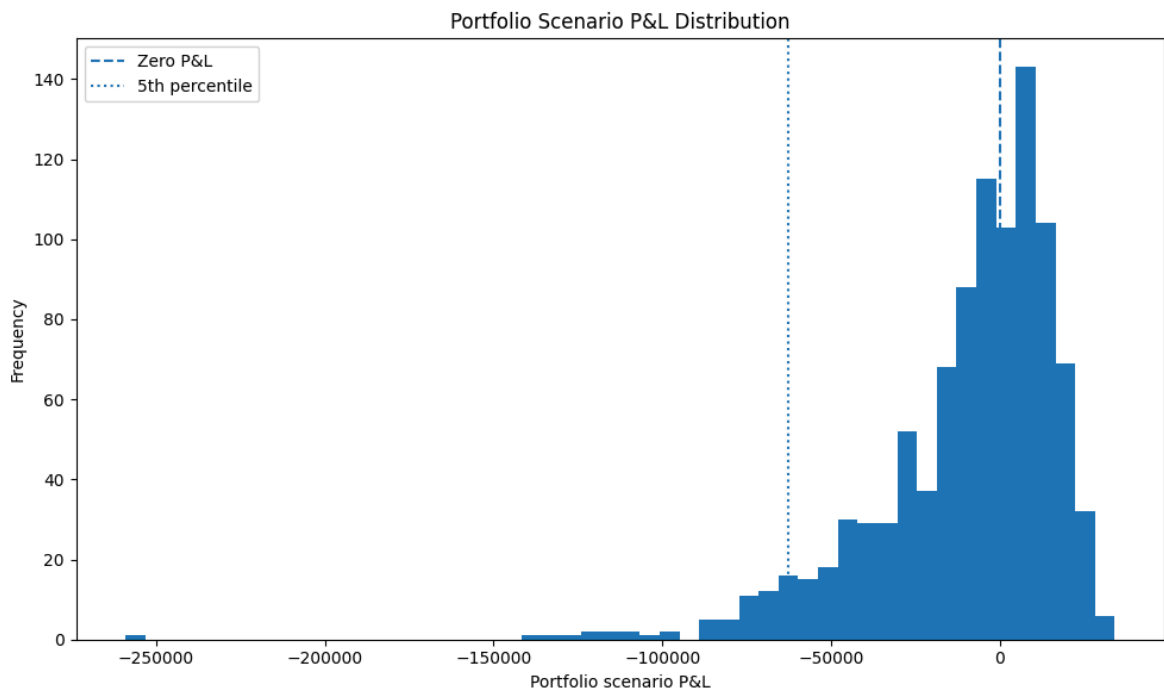
Distribution of portfolio stress P&L

The histogram summarises the non-base portfolio P&L outcomes. The zero line separates profitable and loss-making scenarios, while the fifth-percentile line identifies the lower tail of the assumed stress distribution.

The result should be described as a scenario loss distribution rather than formal VaR because the underlying shocks were selected for transparent stress testing rather than estimated from a calibrated joint market model.

```
In [25]: from options_risk_engine.reporting.plots import plot_scenario_pnl_distribution

plot_scenario_pnl_distribution(
    scenario_portfolio, FIGURE_DIR / "05_scenario_pnl_distribution.png"
)
plt.show()
```



The distribution is visibly left-skewed. Most observations lie relatively close to zero, but a small number of severe scenarios produce losses that are much larger than the maximum gains.

This asymmetry is consistent with a portfolio containing significant negative Gamma and mixed volatility exposure. It indicates that offsetting Delta can reduce ordinary movement risk without eliminating losses from large nonlinear shocks.

14. Tail-loss and risk-limit analysis

The scenario outputs are summarised into central and tail statistics. The analysis also compares losses with predefined portfolio and ticker limits.

The aim is to answer:

- how often the generated scenarios produce a loss;
- how severe the lower tail is;
- which scenarios exceed the chosen loss tolerance; and
- which tickers are responsible for the worst portfolio outcome.

```
In [26]: from options_risk_engine.risk.scenarios import analyse_scenario_results

scenario_analysis = analyse_scenario_results(
    scenarios=scenario_shocks,
    results=results,
    scenario_ticker=scenario_ticker,
```

```

scenario_portfolio=scenario_portfolio,
portfolio_risk=portfolio_risk,
base_scenario_id=BASE_SCENARIO_ID,
max_portfolio_loss=CONFIG.max_portfolio_loss,
max_ticker_loss=CONFIG.max_ticker_loss,
number_to_show=10,
)

display(scenario_analysis["scenario_risk_summary"])
display(scenario_analysis["worst_scenarios"])
display(scenario_analysis["portfolio_breaches"].head(10))

```

```

2026-08-02 13:38:49,680 - options_risk_engine.risk.scenarios - INFO - Analysing scenario outputs: 966966 position rows, 7007 ticker rows and 1001 portfolio rows
2026-08-02 13:38:49,805 - options_risk_engine.risk.scenarios - INFO - Base scenario validation passed: portfolio P&L 0.00000000
2026-08-02 13:38:49,809 - options_risk_engine.risk.scenarios - INFO - All scenario positions were repriced successfully
2026-08-02 13:38:49,813 - options_risk_engine.risk.scenarios - INFO - Scenario distribution analysed: worst P&L -258974.50; 5th percentile -62652.47; probability of loss 56.20%
2026-08-02 13:38:49,866 - options_risk_engine.risk.scenarios - WARNING - 11 portfolio scenarios breached the 100000.00 loss limit
2026-08-02 13:38:49,866 - options_risk_engine.risk.scenarios - WARNING - 1473 ticker-scenario rows breached the 25000.00 loss limit
2026-08-02 13:38:49,867 - options_risk_engine.risk.scenarios - INFO - Scenario analysis complete; worst scenario is R0950

```

	Number of Scenarios	Best PnL	Worst PnL	Mean PnL	Median PnL	Probability of Loss
0	1000	33992.120695	-258974.501469	-10027.264199	-2878.092454	0.56

	Scenario ID	Portfolio_PnL	Portfolio_Value	Spot Shock	Volatility Shock	Days Forward	SI
0	R0950	-258974.501469	-1.101247e+06	-0.291873	0.100048	6	-0.00
1	R0505	-138590.113771	-9.808622e+05	-0.237162	-0.010587	19	0.01
2	R0948	-135009.743342	-9.772818e+05	-0.234448	-0.045455	17	-0.00
3	R0581	-124511.484955	-9.667835e+05	-0.192794	0.059481	2	-0.00
4	R0140	-124106.663085	-9.663787e+05	0.233109	0.000190	0	-0.00
5	R0630	-120299.094089	-9.625712e+05	-0.193022	0.053892	4	-0.00
6	R0830	-113301.103968	-9.555732e+05	-0.213827	-0.011323	16	0.00
7	R0847	-113268.581506	-9.555406e+05	0.254308	0.001635	29	-0.00
8	R0480	-107227.049371	-9.494991e+05	0.207814	0.085293	6	0.01
9	R0249	-106780.068009	-9.490521e+05	0.232405	-0.008578	22	-0.00

	Scenario ID	Portfolio_PnL	Portfolio_Value	Spot Shock	Volatility Shock	Days Forward	SI
0	R0950	-258974.501469	-1.101247e+06	-0.291873	0.100048	6	-0.00
1	R0505	-138590.113771	-9.808622e+05	-0.237162	-0.010587	19	0.01
2	R0948	-135009.743342	-9.772818e+05	-0.234448	-0.045455	17	-0.00
3	R0581	-124511.484955	-9.667835e+05	-0.192794	0.059481	2	-0.00
4	R0140	-124106.663085	-9.663787e+05	0.233109	0.000190	0	-0.00
5	R0630	-120299.094089	-9.625712e+05	-0.193022	0.053892	4	-0.00
6	R0830	-113301.103968	-9.555732e+05	-0.213827	-0.011323	16	0.00
7	R0847	-113268.581506	-9.555406e+05	0.254308	0.001635	29	-0.00
8	R0480	-107227.049371	-9.494991e+05	0.207814	0.085293	6	0.01
9	R0249	-106780.068009	-9.490521e+05	0.232405	-0.008578	22	-0.00

Portfolio stress findings

Of the 1,000 non-base scenarios, 56% produced a portfolio loss. Mean P&L was approximately -£9,959 and median P&L was approximately -£2,826. The fifth percentile was approximately -£62,581.

The best scenario produced a gain of approximately £34,096, whereas the worst scenario produced a loss of approximately £258,560. The much larger downside tail confirms that the portfolio's nonlinear exposure is asymmetric.

Eleven scenarios breached the £100,000 portfolio loss limit. The ticker-level limit was also breached frequently, although several ticker losses can occur within the same portfolio scenario and should not be interpreted as independent events.

The worst scenario, R0950, combined a 29.2% fall in the common spot factor with an increase of approximately 10 volatility points, six elapsed days and a small reduction in rates. This combination exposed the portfolio's negative convexity and volatility sensitivity.

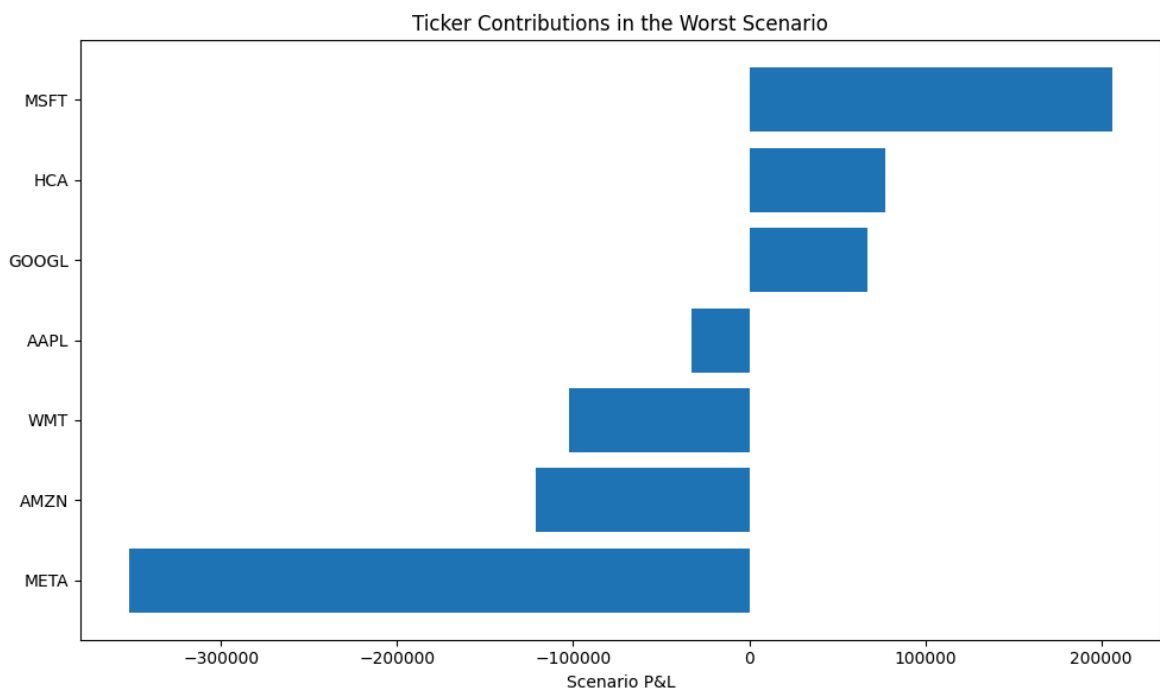
Concentration and offsetting positions

Portfolio P&L can conceal large opposing ticker-level movements. The worst scenario is therefore decomposed by underlying to identify both the principal loss contributors and the positions that reduced the net loss.

```
In [27]: from options_risk_engine.reporting.plots import plot_worst_ticker_contrib
worst_ticker_contributions = scenario_analysis["worst_ticker_contribution"]
display(worst_ticker_contributions)
plot_worst_ticker_contributions(
    worst_ticker_contributions,
```

```
FIGURE_DIR / "06_worst_scenario_tickers.png",
)
plt.show()
```

	Scenario ID	Ticker	Scenario_PnL	Scenario_Value	Spot Shock	Volatility Shock	Days Forward
0	R0950	META	-352145.456273	-962827.059593	-0.291873	0.100048	6
1	R0950	AMZN	-121397.742717	-43316.721405	-0.291873	0.100048	6
2	R0950	WMT	-102611.555044	-124353.601155	-0.291873	0.100048	6
3	R0950	AAPL	-32937.479471	-162148.561715	-0.291873	0.100048	6
4	R0950	GOOGL	66890.896367	-219240.252621	-0.291873	0.100048	6
5	R0950	HCA	77157.596070	143562.282389	-0.291873	0.100048	6
6	R0950	MSFT	206069.239598	267077.354447	-0.291873	0.100048	6



META was the largest loss contributor in the worst scenario, losing approximately £351,975. AMZN and WMT added losses of roughly £121,419 and £102,651 respectively, while AAPL lost approximately £33,002.

These losses were partly offset by gains of around £206,307 from MSFT, £77,257 from HCA and £66,924 from GOOGL.

The result demonstrates both concentration and netting. The final portfolio loss was materially smaller than the sum of the losing ticker books because several positions benefited from the same shock. However, relying on those offsets can be unstable because they depend on the assumed common movement and the specific option exposures held at the valuation date.

15. Greek P&L attribution

Full revaluation measures the scenario result accurately under the pricing model, but it does not by itself explain why the value changed.

The position-level components are aggregated to ticker and portfolio level. The residual is defined as full-revaluation P&L minus the sum of the Greek components.

A small residual indicates that the local Greek approximation represented the scenario well. A large residual can result from large shocks, changing Greeks, higher-order terms, cross-Greek effects or movement near expiry.

```
In [28]: from options_risk_engine.risk.attribution import (
        calculate_greek_profit_loss,
        analyse_greek_attribution,
    )

    # Keep Ticker from the scenario-results table. Selecting Ticker from both
    # tables would create Ticker_x and Ticker_y, while the attribution functi
    # expects a single column named Ticker.
    attribution_position_inputs = position_options[
        [
            "contractSymbol",
            "Current Stock Price",
            "Position Delta",
            "Position Gamma",
            "Position Vega",
            "Position Theta",
            "Position Rho",
        ]
    ].copy()

    expanded_results = results.merge(
        attribution_position_inputs,
        left_on="Contract Symbol",
        right_on="contractSymbol",
        how="left",
        validate="many_to_one",
    )

    if "Ticker" not in expanded_results.columns:
        raise KeyError(
            "Attribution merge did not retain the Ticker column."
        )

    attribution_results, scenario_attribution = (
        calculate_greek_profit_loss(
            expanded_results
        )
    )

    attribution_analysis = analyse_greek_attribution(
        scenario_attribution=scenario_attribution,
        attribution_results=attribution_results,
        scenarios=scenario_shocks,
        base_scenario_id=BASE_SCENARIO_ID,
        number_to_show=10,
    )

    scenario_attribution = (
```

```
    attribution_analysis[
        "scenario_attribution"
    ]
)

display(attribution_results.head())
display(scenario_attribution.head())
```

2026-08-02 13:38:50,312 - options_risk_engine.risk.attribution - INFO - Starting Greek P&L attribution
2026-08-02 13:38:50,768 - options_risk_engine.risk.attribution - INFO - Greek attribution accounting check passed
2026-08-02 13:38:50,769 - options_risk_engine.risk.attribution - INFO - Greek attribution completed for 966966 position-scenario rows and 1001 scenarios
2026-08-02 13:38:50,769 - options_risk_engine.risk.attribution - INFO - Starting Greek attribution analysis
2026-08-02 13:38:50,968 - options_risk_engine.risk.attribution - INFO - Scenario shocks attached to 1001 attribution rows
2026-08-02 13:38:50,970 - options_risk_engine.risk.attribution - INFO - Greek attribution accounting check passed
2026-08-02 13:38:51,117 - options_risk_engine.risk.attribution - INFO - BASE Greek attribution validation passed
2026-08-02 13:38:51,907 - options_risk_engine.risk.attribution - INFO - Greek attribution analysis completed for 1000 scenarios

	Scenario ID	Contract Symbol	Ticker	Option Type	Current Stock Price	Shocked Spot	Scenario P
0	BASE	AAPL260918C00050000	AAPL	Calls	308.910004	308.910004	
1	BASE	AAPL260918C00055000	AAPL	Calls	308.910004	308.910004	-0
2	BASE	AAPL260918C00060000	AAPL	Calls	308.910004	308.910004	-0
3	BASE	AAPL260918C00065000	AAPL	Calls	308.910004	308.910004	
4	BASE	AAPL260918C00070000	AAPL	Calls	308.910004	308.910004	

	Scenario ID	Full_Revaluation_PnL	Delta_PnL	Gamma_PnL	Vega_PnL	T
0	BASE	0.000000	0.000000	0.000000	0.000000	
1	R0001	7831.894656	-1528.348142	-2517.213009	798.271711	1983
2	R0002	-11142.751852	5216.175536	-29321.067422	9820.228508	9
3	R0003	-10400.133826	-3763.985570	-15267.617822	5581.095744	95
4	R0004	-22980.244444	-4717.524657	-23983.012738	-8535.933461	1172

Attribution construction checks

The attribution was calculated for all 966,966 position-scenario rows and aggregated into 1,001 scenario results.

The BASE scenario has zero Delta, Gamma, Vega, Theta, Rho, approximate P&L and residual P&L.

These checks validate the attribution arithmetic before its accuracy is interpreted.

Attribution accuracy summary

```
In [29]: display(attribution_analysis["attribution_summary"])
display(attribution_analysis["small_scenario_summary"])
display(attribution_analysis["worst_attribution_scenarios"])
display(attribution_analysis["base_attribution"])
```

	Number of Scenarios	Mean Absolute Residual	Median Absolute Residual	95th Percentile Absolute Residual	Mean Gross Residual %	Median Gross Residual %	95th Percentile Gross Residual %
0	1000	9529.008343	5550.492032	30341.798895	0.084878	0.074413	0.1816

	Number of Small Scenarios	Mean Absolute Residual	Median Absolute Residual	Mean Gross Residual %	Median Gross Residual %
0	5	221.011172	127.777137	0.01573	0.015333

	Scenario ID	Spot Shock	Volatility Shock	Days Forward	Rate Shock	Full_Revaluation_PnL	Delta_PnL
0	R0316	-0.023726	0.091811	27	-0.004278	8949.144822	1487.4
1	R0259	-0.010706	0.086982	23	-0.002349	5271.964855	671.2
2	R0664	-0.027387	0.089914	27	0.002092	9457.636915	1717.0
3	R0509	-0.044766	0.099876	29	0.011655	9051.191539	2806.0
4	R0199	-0.000410	0.074246	23	-0.001199	5949.518170	25.6
5	R0448	-0.026519	0.075822	22	-0.008835	5922.587352	1662.6
6	R0863	-0.004673	0.070626	24	0.002102	8062.268819	292.9
7	R0240	-0.042355	0.077594	27	0.009206	10263.117523	2655.4
8	R0134	-0.017778	0.069085	16	0.007687	1573.625777	1114.5
9	R0215	-0.026214	-0.132585	20	0.005931	33992.120695	1643.9

	Scenario ID	Full_Revaluation_PnL	Delta_PnL	Gamma_PnL	Vega_PnL	Theta_PnL	Rho_PnL
0	BASE		0.0	0.0	0.0	0.0	0.0

Across all 1,000 non-base scenarios, the mean absolute residual was approximately £9,498 and the median was approximately £5,525. Mean gross residual percentage was approximately 8.5%, with a median of 7.4% and a 95th-percentile value of about 18.1%.

For the small-scenario subset, mean gross residual percentage fell to around 1.6%, and the median absolute residual was only about £130. This is consistent with Greeks being local derivatives that perform best near the original market state.

Only five generated scenarios satisfied the small-shock definition, so this subset is too small for a strong statistical conclusion. It is supporting evidence rather than a definitive estimate of small-move accuracy.

Visual attribution diagnostics

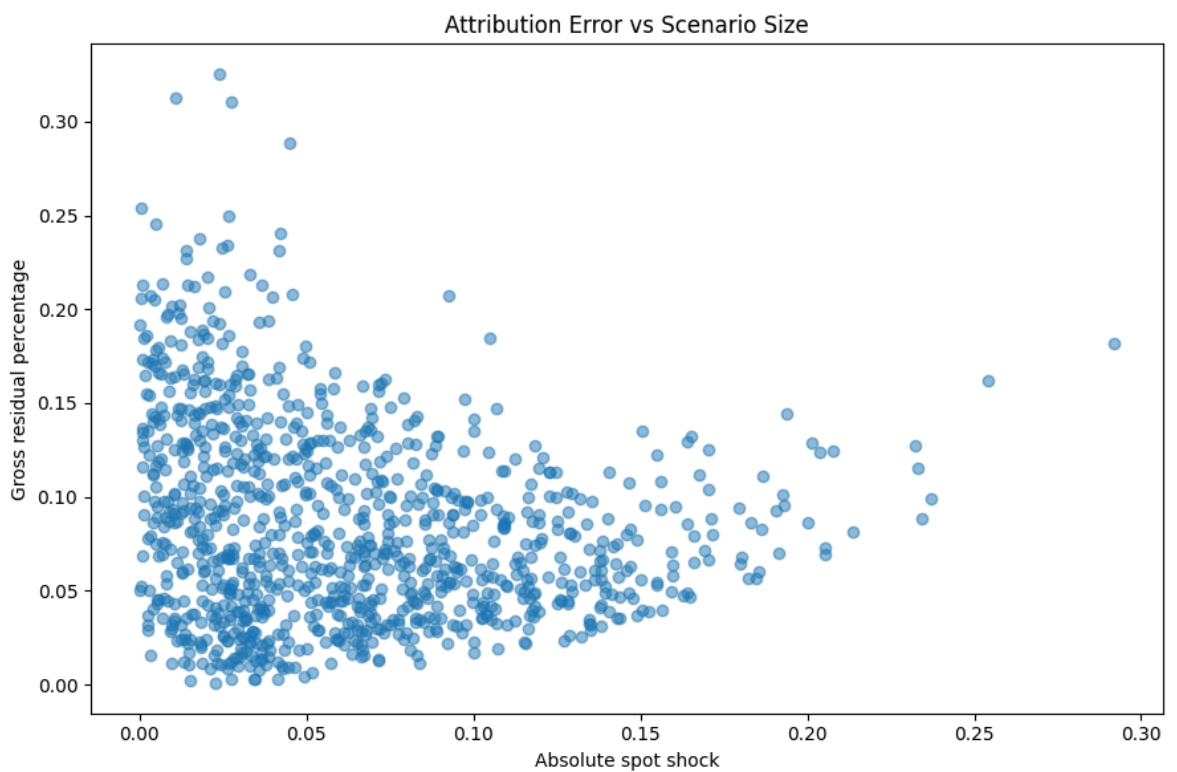
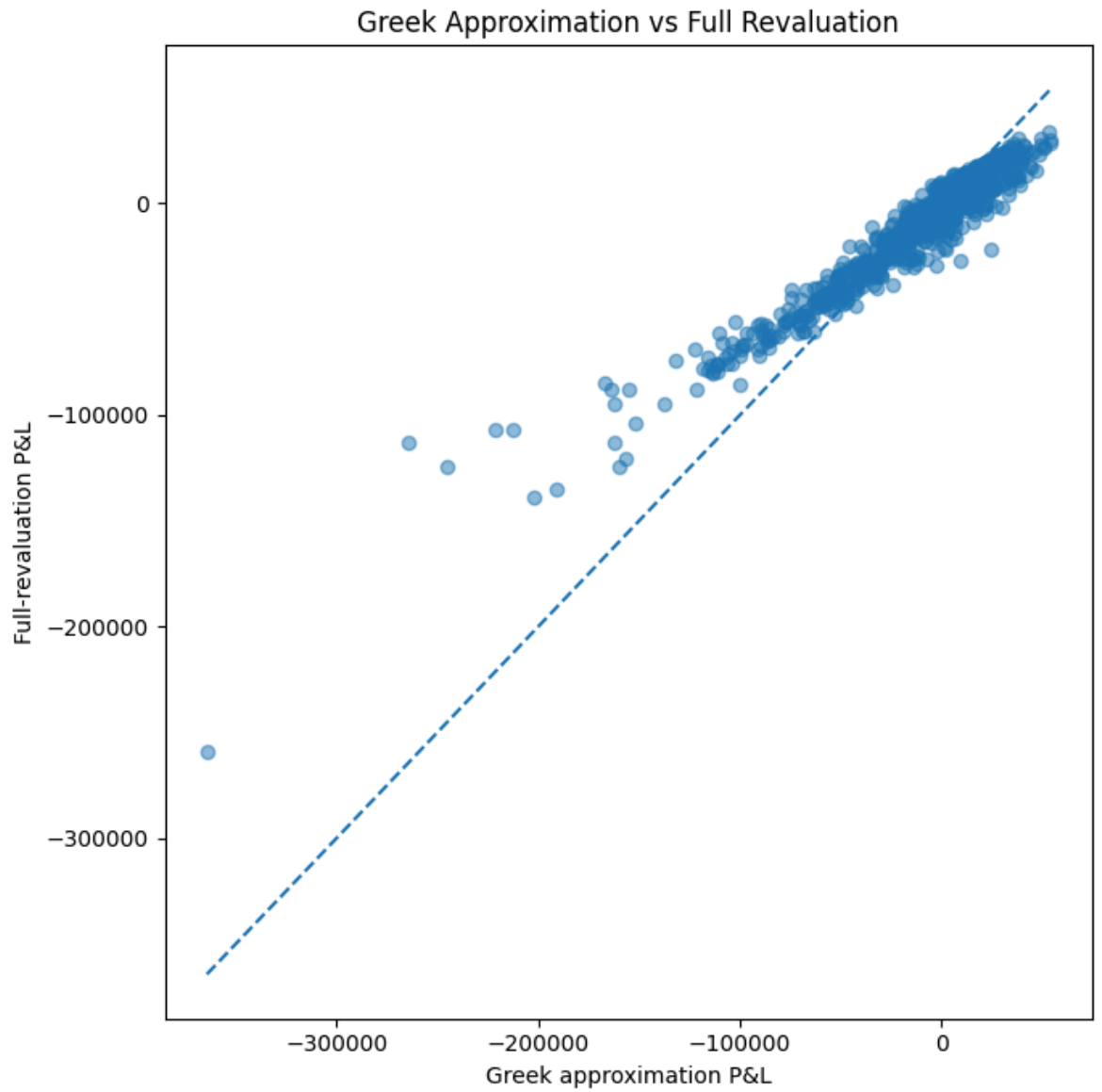
The first graph compares full-revaluation P&L with approximate Greek P&L. A perfect approximation would place every scenario on the diagonal line.

The second graph compares gross residual percentage with absolute spot shock. This helps investigate whether the approximation deteriorates as scenarios become more severe.

```
In [30]: from options_risk_engine.reporting.plots import (
        plot_attribution_components,
        plot_full_vs_approximate_pnl,
        plot_residual_vs_spot_shock,
    )

    plot_full_vs_approximate_pnl(
        scenario_attribution,
        FIGURE_DIR / "07_full_vs_approximate_pnl.png",
    )
    plt.show()

    plot_residual_vs_spot_shock(
        scenario_attribution,
        FIGURE_DIR / "08_residual_vs_spot_shock.png",
    )
    plt.show()
```



Most central scenarios lie reasonably close to the equality line, indicating that the Greeks capture the main direction of ordinary P&L changes. The largest deviations occur in the tails, where the portfolio is moved far from the state at which its Greeks were calculated.

The residual-versus-spot plot does not show a perfectly monotonic relationship. Spot shock is only one part of scenario size; volatility shock, elapsed time, strike location and offsetting position P&L also affect the residual percentage. Small net P&L can additionally make percentage measures appear large.

The appropriate conclusion is therefore that attribution is most reliable for small combined shocks and becomes less dependable under some large nonlinear moves, not that spot movement alone determines approximation error.

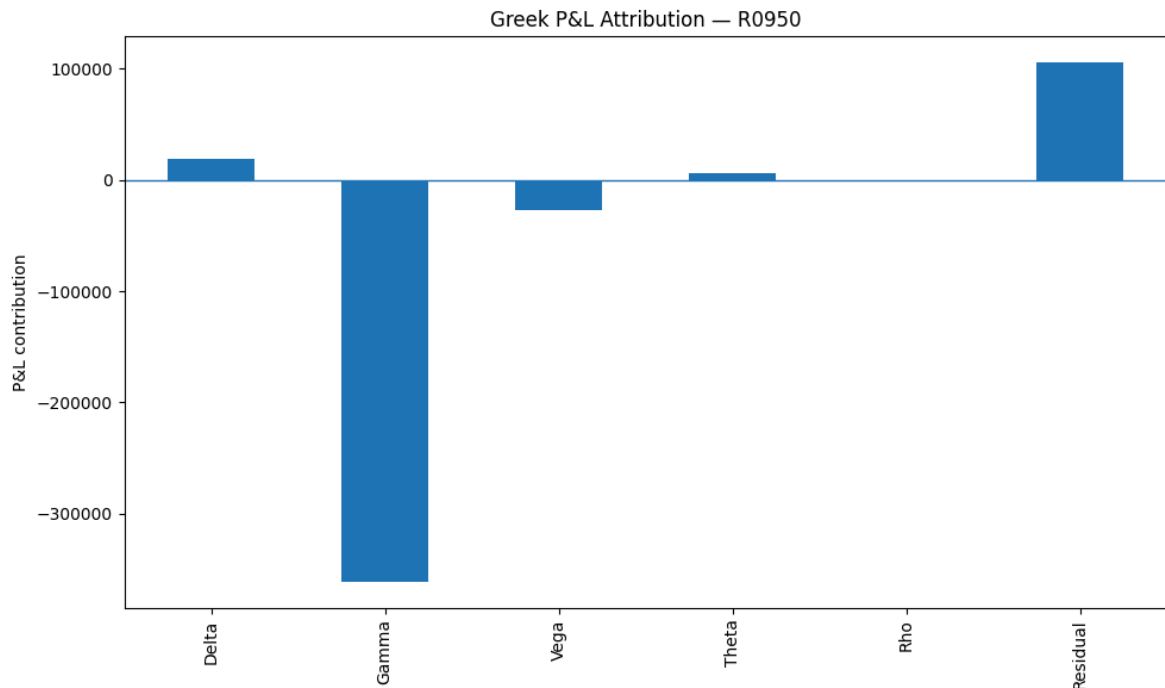
Explaining the worst portfolio loss

The Greek components of R0950 are examined separately to identify the risk factor that dominated the portfolio's largest full-revaluation loss.

```
In [31]: worst_scenario_id = scenario_analysis["worst_scenario_id"]
worst_attribution_row = scenario_attribution.loc[
    scenario_attribution["Scenario ID"].eq(worst_scenario_id)
].iloc[0]

display(worst_attribution_row.to_frame("Value"))
plot_attribution_components(
    worst_attribution_row,
    FIGURE_DIR / "09_worst_scenario_attribution.png",
)
plt.show()
```

	Value
Scenario ID	R0950
Full_Revaluation_PnL	-258974.501469
Delta_PnL	18299.089006
Gamma_PnL	-360856.323175
Vega_PnL	-26944.01053
Theta_PnL	5409.660715
Rho_PnL	-142.594755
Approximate_PnL	-364234.178739
Residual_PnL	105259.67727
Net Residual %	0.406448
Attribution Check	-0.0
Spot Shock	-0.291873
Volatility Shock	0.100048
Days Forward	6
Rate Shock	-0.00142
Gross_Full_PnL	15129834.95547
Gross_Residual	2748137.884965
Gross Residual %	0.181637



Gamma was the dominant risk factor in the worst scenario, contributing an estimated loss of approximately £359,726. Vega added a further loss of around £26,969.

Delta contributed a positive £18,898 and Theta contributed a positive £5,397, partly offsetting the nonlinear loss. Rho was comparatively immaterial.

The total Greek approximation was approximately -£362,542, compared with a full-revaluation loss of approximately -£258,560. The residual was therefore positive £103,982, equivalent to about 40.2% of net full-revaluation P&L and 18.2% on a gross-position basis.

This is economically consistent with the scenario. A 29% spot fall and a 10-point volatility increase are far from the local state at which the Greeks were calculated. Gamma correctly identifies negative convexity as the principal vulnerability, but the fixed-Greek approximation overstates the final loss because the portfolio's Greeks change substantially through the shock.

16. End-to-end risk-engine validation

The final validation checks relationships that span several earlier stages. This is intended to detect integration errors that would not necessarily be found by testing an individual pricing function.

The checks verify that:

- BASE portfolio P&L is zero;
- position P&L aggregates exactly to portfolio P&L;
- full revaluation equals approximation plus residual;
- all scenario prices are present;
- shocked option volatility remains positive;
- shocked option time is nonnegative;
- share prices equal shocked spot values; and
- every position type is recognised.

```
In [32]: from options_risk_engine.validation import validate_risk_engine

validation_checks = validate_risk_engine(
    results=results,
    scenario_portfolio=scenario_portfolio,
    scenario_attribution=scenario_attribution,
)
display(pd.Series(validation_checks, name="Passed").to_frame())
```

```
2026-08-02 13:38:53,150 - options_risk_engine.validation - INFO - All risk
-engine validation checks passed
```

	Passed
base_pnl_zero	True
position_totals_match_portfolio	True
attribution_identity	True
no_missing_scenario_prices	True
option_shocked_volatility_positive	True
option_shocked_time_nonnegative	True
share_prices_match_shocked_spot	True
known_position_types	True

All cross-stage checks passed.

This confirms that the pricing, scenario aggregation, share handling and attribution accounting are internally consistent for this run. It does not prove that the Black-Scholes assumptions or scenario distributions are an accurate representation of future markets, but it provides evidence that the engine implements its stated methodology correctly.

17. Persist final scenario and attribution outputs

The final position-, ticker- and portfolio-level outputs are written to both SQLite and CSV.

Storing multiple aggregation levels allows later analysis to begin with the compact portfolio table while retaining the ability to trace a result back to the individual contracts that produced it.

```
In [33]: outputs_to_store = {
    "scenario_definitions": scenario_shocks,
    "scenario_position_results": results,
    "scenario_ticker_results": scenario_ticker,
    "scenario_portfolio_results": scenario_portfolio,
    "attribution_position_results": attribution_results,
    "scenario_attribution": scenario_attribution,
}
for table_name, frame in outputs_to_store.items():
    store.write_frame(frame, table_name, run_id=run_id)
    frame.to_csv(TABLE_DIR / f"{table_name}.csv", index=False)

print("Stored final tables:")
display(store.list_tables())
```



```
2026-08-02 13:38:53,179 - options_risk_engine.database - INFO - Stored 100
1 rows in scenario_definitions
2026-08-02 13:38:56,881 - options_risk_engine.database - INFO - Stored 966
966 rows in scenario_position_results
2026-08-02 13:39:10,650 - options_risk_engine.database - INFO - Stored 700
7 rows in scenario_ticker_results
2026-08-02 13:39:10,676 - options_risk_engine.database - INFO - Stored 100
1 rows in scenario_portfolio_results
2026-08-02 13:39:15,360 - options_risk_engine.database - INFO - Stored 966
966 rows in attribution_position_results
2026-08-02 13:39:36,608 - options_risk_engine.database - INFO - Stored 100
1 rows in scenario_attribution
Stored final tables:
```

	name
0	attribution_position_results
1	equity_prices
2	model_evaluations
3	option_chain_snapshots
4	option_chain_snapshots_full
5	option_pricing_results
6	portfolio_positions
7	portfolio_risk
8	project_runs
9	scenario_attribution
10	scenario_definitions
11	scenario_portfolio_results
12	scenario_position_results
13	scenario_ticker_results
14	volatility_features

The final run stores 1,001 scenario definitions, 966,966 position-scenario results, 7,007 ticker-scenario results, 1,001 portfolio-scenario results, 966,966 position-attribution rows and 1,001 scenario-attribution rows.

The row counts provide a final completeness check and demonstrate that all positions and scenarios are represented at each required aggregation level.

For a larger production system, the two 966,966-row tables would be natural candidates for partitioned Parquet storage rather than repeated CSV exports.

18. Standardised one-contract scenario analysis

The existing portfolio was designed to test aggregation and portfolio risk. Its directions, quantities and entry prices were assigned using a fixed random seed, so its position-level results cannot fairly compare the quality of individual option recommendations.

This section creates a separate contract-analysis universe in which every eligible option is evaluated on identical terms:

- one long contract;
- quantity one;
- a standard multiplier of 100;
- entry at the current ask; and
- a position scale of +100.

Buy, **Positive Edge** and **Do Not Buy** contracts are retained. Including all three classifications provides comparison groups for testing whether the initial valuation labels correspond to better scenario outcomes.

Three valuation concepts remain separate:

1. **Forecast edge** — forecast-volatility Black–Scholes value minus the ask.
2. **Base mark-to-ask P&L** — market-implied risk-model value minus the ask.
3. **Purchase scenario P&L** — shocked full-revaluation value minus the ask.

This prevents a forecast-based theoretical edge from being presented as an immediately realisable profit.

```
In [34]: from options_risk_engine.recommendations.contracts import (
    attach_standardised_purchase_pnl,
    prepare_standardised_contract_positions,
    validate_standardised_contract_positions,
)

standardised_contracts = (
    prepare_standardised_contract_positions(
        option_universe=large_table,
        risk_free_rates=risk_free_rates,
        dividend_yields=dividend_yields,
        included_actions=(
            CONTRACT_ANALYSIS_ACTIONS
        ),
    ),
)

standardised_position_summary = (
    standardised_contracts
    .groupby(
        [
            "Initial Recommended Action",
            "Option_Type",
        ],
        observed=True,
    )
    .agg(
```

```

        Contracts=(
            "contractSymbol",
            "count",
        ),
        Median_Ask=(
            "ask",
            "median",
        ),
        Median_Forecast_Edge=(
            "BS_ForeV AskEdge",
            "median",
        ),
        Median_Spread=(
            "SpreadPct",
            "median",
        ),
        Median_Volume=(
            "volume",
            "median",
        ),
        Median_Open_Interest=(
            "openInterest",
            "median",
        ),
    )
    .reset_index()
)

print(
    "Standardised contracts:",
    len(standardised_contracts),
)

display(
    standardised_position_summary.round(4)
)

display(
    standardised_contracts[
        [
            "contractSymbol",
            "Ticker",
            "Option_Type",
            "strike",
            "Expiry",
            "Initial Recommended Action",
            "bid",
            "ask",
            "BS_ForeV",
            "BS_ForeV AskEdge",
            "Forecast Edge Per Contract",
            "Greek Volatility",
            "Greek Volatility Source",
            "Premium Paid",
        ]
    ]
    .head(20)
    .round(4)
)

```

2026-08-02 13:39:36,665 – options_risk_engine.recommendations.contracts – INFO – Preparing standardised contract analysis from 1211 option rows

2026-08-02 13:39:36,689 – options_risk_engine.recommendations.contracts – INFO – Prepared 794 standardised long-one-contract positions

Standardised contracts: 794

	Initial Recommended Action	Option_Type	Contracts	Median_Ask	Median_Forecast_Edge	Me
0	Buy	Calls	14	2.635	0.1373	
1	Buy	Puts	77	5.300	0.6359	
2	Do Not Buy	Calls	545	21.350	-0.5284	
3	Do Not Buy	Puts	142	0.420	-0.9567	
4	Positive Edge	Calls	5	16.950	0.0249	
5	Positive Edge	Puts	11	7.350	0.0307	

	contractSymbol	Ticker	Option_Type	strike	Expiry	Recommended Action	Initial
0	AAPL260918C00050000	AAPL	Calls	50.0	2026-09-18	Do Not Buy	257
1	AAPL260918C00055000	AAPL	Calls	55.0	2026-09-18	Do Not Buy	252
2	AAPL260918C00060000	AAPL	Calls	60.0	2026-09-18	Do Not Buy	247
3	AAPL260918C00070000	AAPL	Calls	70.0	2026-09-18	Do Not Buy	237
4	AAPL260918C00075000	AAPL	Calls	75.0	2026-09-18	Do Not Buy	222
5	AAPL260918C00090000	AAPL	Calls	90.0	2026-09-18	Do Not Buy	217
6	AAPL260918C00100000	AAPL	Calls	100.0	2026-09-18	Do Not Buy	207
7	AAPL260918C00105000	AAPL	Calls	105.0	2026-09-18	Do Not Buy	202
8	AAPL260918C00110000	AAPL	Calls	110.0	2026-09-18	Do Not Buy	197
9	AAPL260918C00115000	AAPL	Calls	115.0	2026-09-18	Do Not Buy	192
10	AAPL260918C00120000	AAPL	Calls	120.0	2026-09-18	Do Not Buy	187
11	AAPL260918C00125000	AAPL	Calls	125.0	2026-09-18	Do Not Buy	182
12	AAPL260918C00130000	AAPL	Calls	130.0	2026-09-18	Do Not Buy	178
13	AAPL260918C00135000	AAPL	Calls	135.0	2026-09-18	Do Not Buy	172
14	AAPL260918C00140000	AAPL	Calls	140.0	2026-09-18	Do Not Buy	167
15	AAPL260918C00145000	AAPL	Calls	145.0	2026-09-18	Do Not Buy	162
16	AAPL260918C00150000	AAPL	Calls	150.0	2026-09-18	Do Not Buy	158
17	AAPL260918C00155000	AAPL	Calls	155.0	2026-09-18	Do Not Buy	152
18	AAPL260918C00160000	AAPL	Calls	160.0	2026-09-18	Do Not Buy	148
19	AAPL260918C00165000	AAPL	Calls	165.0	2026-09-18	Do Not Buy	142

Observed standardised contract universe

The standardisation process retained 794 of the 1,211 priced option rows.

The retained universe contains:

- 91 contracts initially classified as **Buy** ;
- 16 contracts classified as **Positive Edge** ; and
- 687 contracts classified as **Do Not Buy** .

The initial **Buy** group is strongly concentrated in puts, with 77 puts and 14 calls. The **Do Not Buy** group is instead dominated by calls, with 545 calls and 142 puts. The forecast-volatility valuation difference was therefore not distributed evenly across option type in this market snapshot.

Every retained row now represents the same hypothetical transaction: one long contract purchased at the current ask with a multiplier of 100. Subsequent differences arise from the contract's strike, option type, volatility, Greeks and quote rather than randomly assigned quantities or directions.

The initial classifications remain valuation labels only. They have not yet been tested against the complete scenario distribution.

```
In [35]: standardised_contract_checks = (  
    validate_standardised_contract_positions(  
        standardised_contracts  
    )  
)  
  
display(  
    pd.Series(  
        standardised_contract_checks,  
        name="Passed",  
    )  
    .to_frame()  
)
```

```
2026-08-02 13:39:36,731 - options_risk_engine.recommendations.contracts -  
INFO - All standardised contract validation checks passed
```

	Passed
contracts_available	True
unique_contract_symbols	True
all_positions_long	True
one_contract_each	True
standard_multiplier	True
entry_equals_ask	True
positive_premium	True
complete_pricing_inputs	True
recognised_option_types	True
forecast_edge_matches_existing_column	True

Standardisation validation

All construction checks passed.

The 794 contracts have unique contract symbols, complete pricing and Greek inputs, positive premiums, recognised option types and identical long-one-contract position mechanics.

The independently calculated forecast-edge return also agrees with the existing **BS_ForeV AskEdge** field. The recommendation analysis therefore uses the same definition of initial model edge as the original pricing stage.

```
In [36]: from options_risk_engine.risk.scenarios import (
        run_scenario_engine,
    )

    (
        raw_contract_scenario_results,
        _,
        _,
    ) = run_scenario_engine(
        position_options=standardised_contracts,
        scenarios=scenario_shocks,
        dividend_yields=dividend_yields,
    )

    print(
        "Contract-scenario rows:",
        len(raw_contract_scenario_results),
    )

    print(
        "Contracts:",
        raw_contract_scenario_results[
            "Contract Symbol"
        ].nunique(),
    )
```



```
print(
    "Scenarios:",
    raw_contract_scenario_results[
        "Scenario ID"
    ].nunique(),
)
```

```
2026-08-02 13:39:36,743 - options_risk_engine.risk.scenarios - INFO - Star
ting scenario engine: 794 positions across 1001 scenarios
2026-08-02 13:39:36,745 - options_risk_engine.risk.scenarios - INFO - Scen
ario positions classified: 564 calls, 230 puts and 0 shares
2026-08-02 13:39:36,752 - options_risk_engine.risk.scenarios - INFO - Base
model prices calculated for 794 positions
2026-08-02 13:39:36,753 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 1/1001: BASE
2026-08-02 13:39:37,121 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 100/1001: R0099
2026-08-02 13:39:37,481 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 200/1001: R0199
2026-08-02 13:39:37,844 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 300/1001: R0299
2026-08-02 13:39:38,204 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 400/1001: R0399
2026-08-02 13:39:38,555 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 500/1001: R0499
2026-08-02 13:39:38,925 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 600/1001: R0599
2026-08-02 13:39:39,292 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 700/1001: R0699
2026-08-02 13:39:39,655 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 800/1001: R0799
2026-08-02 13:39:40,027 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 900/1001: R0899
2026-08-02 13:39:40,386 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 1000/1001: R0999
2026-08-02 13:39:40,391 - options_risk_engine.risk.scenarios - INFO - Reva
luating scenario 1001/1001: R1000
2026-08-02 13:39:40,499 - options_risk_engine.risk.scenarios - INFO - Posi
tion-level scenario results created: 794794 rows
Contract-scenario rows: 794794
Contracts: 794
Scenarios: 1001
```

Full-revaluation coverage

The scenario engine repriced 794 contracts under 1,001 scenarios, producing 794,794 contract-scenario observations.

The scenario set contains the unchanged **BASE** state and 1,000 generated stress states. Every contract was therefore exposed to exactly the same spot, volatility, time and interest-rate assumptions.

The ticker and portfolio aggregates returned by the engine are not used here. These contracts are alternative purchase opportunities rather than positions assumed to be held simultaneously in one portfolio.

```

In [37]: contract_scenario_results = (
    attach_standardised_purchase_pnl(
        scenario_results=(
            raw_contract_scenario_results
        ),
        positions=standardised_contracts,
        base_scenario_id=BASE_SCENARIO_ID,
    )
)

base_contract_results = (
    contract_scenario_results.loc[
        contract_scenario_results[
            "Scenario ID"
        ].eq(BASE_SCENARIO_ID)
    ]
    [
        [
            "Contract Symbol",
            "Ticker",
            "Option Type",
            "Initial Recommended Action",
            "Entry Price",
            "Base Model Price",
            "BS_ForeV",
            "Forecast Edge Per Contract",
            "Base Mark-to-Ask PnL",
            "Scenario Movement PnL",
            "Purchase Scenario PnL",
            "Purchase Scenario Return",
        ]
    ]
    .sort_values(
        "Forecast Edge Per Contract",
        ascending=False,
    )
)

display(
    base_contract_results
    .head(20)
    .round(4)
)

```

2026-08-02 13:39:41,153 - options_risk_engine.recommendations.contracts - INFO - Attached purchase-at-ask P&L to 794794 scenario rows

	Contract Symbol	Ticker	Option Type	Initial Recommended Action	Entry Price	Base Model Price	BS_Fore
633	META260918P00565000	META	Puts	Buy	34.55	33.650	50.859
632	META260918P00560000	META	Puts	Buy	32.10	31.100	47.518
631	META260918P00555000	META	Puts	Buy	29.20	28.550	44.292
630	META260918P00550000	META	Puts	Buy	26.75	26.175	41.178
629	META260918P00545000	META	Puts	Buy	24.10	23.675	38.182
628	META260918P00540000	META	Puts	Buy	21.85	21.525	35.305
627	META260918P00535000	META	Puts	Buy	19.70	19.250	32.557
626	META260918P00530000	META	Puts	Buy	17.70	17.250	29.920
625	META260918P00525000	META	Puts	Buy	16.00	15.525	27.417
624	META260918P00520000	META	Puts	Buy	14.35	14.000	25.040
759	MSFT260918P00425000	MSFT	Puts	Buy	6.75	6.475	17.100
623	META260918P00515000	META	Puts	Buy	12.85	12.475	22.792
622	META260918P00510000	META	Puts	Buy	11.35	11.050	20.672
417	HCA260918P00370000	HCA	Puts	Buy	7.50	6.400	16.500
758	MSFT260918P00420000	MSFT	Puts	Buy	5.90	5.575	14.517
621	META260918P00505000	META	Puts	Buy	10.25	9.875	18.687
620	META260918P00500000	META	Puts	Buy	9.10	8.825	16.816
757	MSFT260918P00415000	MSFT	Puts	Buy	4.95	4.750	12.175
619	META260918P00495000	META	Puts	Buy	7.95	7.650	15.077
618	META260918P00490000	META	Puts	Buy	6.95	6.725	13.463

Interpreting the BASE results

The **BASE** scenario contains no spot, volatility, time or rate shock, so **Scenario Movement PnL** is zero for every contract.

Purchase Scenario PnL can still be negative because the hypothetical buyer pays the ask before the contract is marked using the market-IV risk model. This initial result reflects the ask-to-model difference, including the economic cost of crossing the spread.

Forecast edge remains separate. It compares the forecast-volatility Black–Scholes value with the ask and explains why an initial **Buy** or **Positive Edge** label was generated. It is a theoretical model difference, not an immediately realisable trading profit.

```
In [38]: example_scenario_id = (
          scenario_shocks.loc[
```

```

        ~scenario_shocks[
            "Scenario ID"
        ].eq(BASE_SCENARIO_ID),
        "Scenario ID",
    ]
    .iloc[0]
)

example_contract_scenario = (
    contract_scenario_results.loc[
        contract_scenario_results[
            "Scenario ID"
        ].eq(example_scenario_id)
    ]
    [
        [
            "Scenario ID",
            "Contract Symbol",
            "Ticker",
            "Option Type",
            "Initial Recommended Action",
            "Entry Price",
            "Shocked Spot",
            "Shocked Volatility",
            "Days Forward",
            "Scenario Price",
            "Forecast Edge Per Contract",
            "Base Mark-to-Ask PnL",
            "Scenario Movement PnL",
            "Purchase Scenario PnL",
            "Purchase Scenario Return",
            "Purchase PnL Identity Error",
        ]
    ]
    .sort_values(
        "Purchase Scenario PnL",
        ascending=False,
    )
)

display(
    example_contract_scenario
    .head(10)
    .round(4)
)

display(
    example_contract_scenario
    .tail(10)
    .round(4)
)

print(
    "Maximum absolute purchase-P&L identity error:",
    contract_scenario_results[
        "Purchase PnL Identity Error"
    ]
    .abs()
    .max(),
)

```

	Scenario ID	Contract Symbol	Ticker	Option Type	Initial Recommended Action	Entry Price	Sh
1430	R0001	MSFT260918C00225000	MSFT	Calls	Do Not Buy	200.35	476
1160	R0001	HCA260918C00300000	HCA	Calls	Do Not Buy	85.00	412
1168	R0001	HCA260918C00340000	HCA	Calls	Do Not Buy	51.00	412
798	R0001	AAPL260918C00075000	AAPL	Calls	Do Not Buy	225.90	316
1235	R0001	META260918C00440000	META	Calls	Do Not Buy	121.70	57
919	R0001	AMZN260918C00200000	AMZN	Calls	Do Not Buy	73.50	278
923	R0001	AMZN260918C00220000	AMZN	Calls	Do Not Buy	54.05	278
921	R0001	AMZN260918C00210000	AMZN	Calls	Do Not Buy	64.00	278
922	R0001	AMZN260918C00215000	AMZN	Calls	Do Not Buy	59.10	278
901	R0001	AMZN260918C00105000	AMZN	Calls	Do Not Buy	169.00	278

	Scenario ID	Contract Symbol	Ticker	Option Type	Initial Recommended Action	Entry Price	Sh
999	R0001	AMZN260918P00370000	AMZN	Puts	Do Not Buy	130.25	2
1426	R0001	META260918P00560000	META	Puts	Buy	32.10	4
1427	R0001	META260918P00565000	META	Puts	Buy	34.55	4
1150	R0001	GOOGL260918P00470000	GOOGL	Puts	Do Not Buy	154.25	3
900	R0001	AAPL260918P00420000	AAPL	Puts	Do Not Buy	158.95	3
1151	R0001	GOOGL260918P00490000	GOOGL	Puts	Do Not Buy	178.15	3
1212	R0001	HCA260918P00505000	HCA	Puts	Do Not Buy	146.30	4
1149	R0001	GOOGL260918P00455000	GOOGL	Puts	Do Not Buy	154.65	3
1554	R0001	MSFT260918P00800000	MSFT	Puts	Do Not Buy	378.90	4
1223	R0001	META260918C00330000	META	Calls	Do Not Buy	358.15	4

Maximum absolute purchase-P&L identity error: 1.8189894035458565e-12

Purchase-P&L reconciliation

The maximum purchase-P&L identity error was approximately $(1.82 * 10^{-12})$, which is effectively floating-point zero.

The displayed best and worst rows for the example scenario should not be treated as a final ranking. A contract may perform strongly in that state and poorly throughout the remainder of the distribution.

Section 18 result

Section 18 created a fair contract-comparison experiment using identical purchase mechanics and a common scenario set. No recommendation has been changed yet; the output is the raw evidence needed to test whether the initial valuation labels survive broader risk analysis.

19. Aggregate contract-level scenario statistics

Section 18 produced one observation for every contract under every market scenario. This section converts those observations into one risk summary per contract.

The unchanged `BASE` state is excluded from the outcome distribution because it contains no market shock. Its market-IV model value relative to the purchase ask remains stored separately as the initial mark-to-ask result.

For each contract, the analysis calculates:

- average and median purchase P&L;
- purchase-P&L volatility;
- scenario gain and loss frequencies;
- fifth-percentile P&L;
- expected shortfall across the worst 5% of scenarios;
- best and worst outcomes; and
- the average effect of market shocks independently of the initial mark-to-ask difference.

Returns are measured relative to the premium paid for one contract. The reported frequencies describe the generated stress distribution; they are not calibrated real-world probabilities.

```
In [39]: from options_risk_engine.recommendations.contracts import (
        create_contract_scenario_summary,
        create_initial_action_comparison,
        validate_contract_scenario_summary,
    )

    contract_scenario_summary = (
        create_contract_scenario_summary(
            scenario_results=contract_scenario_results,
            positions=standardised_contracts,
            base_scenario_id=BASE_SCENARIO_ID,
            tail_probability=(
                CONTRACT_TAIL_PROBABILITY
            ),
        )
    )

    print(
        "Contracts summarised:",
```

```

        len(contract_scenario_summary),
    )

    print(
        "Scenarios per contract:",
        contract_scenario_summary[
            "Scenario_Count"
        ].unique(),
    )

    contract_summary_display_columns = [
        "Contract Symbol",
        "Ticker",
        "Option Type",
        "strike",
        "Expiry",
        "Initial Recommended Action",
        "ask",
        "Premium Paid",
        "BS_ForeV AskEdge",
        "Forecast Edge Per Contract",
        "Base_Mark_to_Ask_PnL",
        "Mean_Purchase_PnL",
        "Median_Purchase_PnL",
        "Scenario_Profit_Frequency",
        "Fifth_Percentile_Purchase_PnL",
        "Expected_Shortfall_5pct_PnL",
        "Worst_Purchase_PnL",
        "Best_Purchase_PnL",
    ]

    available_contract_summary_columns = [
        column
        for column in contract_summary_display_columns
        if column in contract_scenario_summary.columns
    ]

    display(
        contract_scenario_summary[
            available_contract_summary_columns
        ]
        .head(10)
        .round(4)
    )

    display(
        contract_scenario_summary[
            available_contract_summary_columns
        ]
        .tail(10)
        .round(4)
    )

```

2026-08-02 13:39:41,292 – options_risk_engine.recommendations.contracts –
 INFO – Aggregating 794794 contract-scenario rows
 2026-08-02 13:39:44,078 – options_risk_engine.recommendations.contracts –
 INFO – Created scenario summaries for 794 contracts
 Contracts summarised: 794
 Scenarios per contract: [1000]

	Contract Symbol	Ticker	Option Type	strike	Expiry	Recommended Action	Initial ask	Pren
0	WMT260918P00092500	WMT	Puts	92.5	2026-09-18	Buy	0.34	
1	WMT260918P00095000	WMT	Puts	95.0	2026-09-18	Buy	0.51	
2	WMT260918P00097500	WMT	Puts	97.5	2026-09-18	Buy	0.77	
3	WMT260918P00100000	WMT	Puts	100.0	2026-09-18	Buy	1.14	
4	MSFT260918P00415000	MSFT	Puts	415.0	2026-09-18	Buy	4.95	4
5	MSFT260918P00410000	MSFT	Puts	410.0	2026-09-18	Buy	4.25	4
6	MSFT260918P00425000	MSFT	Puts	425.0	2026-09-18	Buy	6.75	6
7	AAPL260918P00275000	AAPL	Puts	275.0	2026-09-18	Buy	2.31	2
8	MSFT260918P00420000	MSFT	Puts	420.0	2026-09-18	Buy	5.90	5
9	AAPL260918P00270000	AAPL	Puts	270.0	2026-09-18	Buy	1.85	1

	Contract Symbol	Ticker	Option Type	strike	Expiry	Initial Recommended Action	ask
784	AMZN260918P00260000	AMZN	Puts	260.0	2026-09-18	Positive Edge	7.35
785	AMZN260918P00270000	AMZN	Puts	270.0	2026-09-18	Positive Edge	11.45
786	AMZN260918P00275000	AMZN	Puts	275.0	2026-09-18	Positive Edge	13.95
787	AMZN260918P00280000	AMZN	Puts	280.0	2026-09-18	Positive Edge	16.70
788	AMZN260918P00285000	AMZN	Puts	285.0	2026-09-18	Positive Edge	20.25
789	AMZN260918P00290000	AMZN	Puts	290.0	2026-09-18	Positive Edge	23.80
790	AMZN260918C00270000	AMZN	Calls	270.0	2026-09-18	Positive Edge	14.10
791	AMZN260918C00265000	AMZN	Calls	265.0	2026-09-18	Positive Edge	16.95
792	AMZN260918C00260000	AMZN	Calls	260.0	2026-09-18	Positive Edge	20.00
793	AMZN260918C00255000	AMZN	Calls	255.0	2026-09-18	Positive Edge	23.55

Contract-level outcome distributions

Each row now represents the complete non-BASE scenario distribution for one long contract purchased at the ask.

`Scenario_Profit_Frequency` is the proportion of generated scenarios in which the shocked model value exceeds the purchase price. It is more informative than inspecting one favourable scenario but remains dependent on the shocks selected by the generator.

The downside measures serve different purposes:

- the fifth percentile is the boundary around the worst 5% of outcomes;
- expected shortfall is the average result within that lower tail; and
- worst P&L is the most severe single generated outcome.

Expected shortfall is generally more stable than relying only on one extreme scenario.

```
In [40]: contract_summary_checks = (
    validate_contract_scenario_summary(
        summary=contract_scenario_summary,
        scenario_results=contract_scenario_results,
        base_scenario_id=BASE_SCENARIO_ID,
    )
)
```

```
)

display(
    pd.Series(
        contract_summary_checks,
        name="Passed",
    )
    .to_frame()
)
```

2026-08-02 13:39:44,247 - options_risk_engine.recommendations.contracts - INFO - All contract scenario summary checks passed

Passed	
contracts_available	True
unique_contract_symbols	True
all_scenario_counts_match	True
positive_premiums	True
frequencies_between_zero_and_one	True
frequencies_sum_to_one	True
long_option_loss_bounded_by_premium	True
long_option_return_not_below_minus_one	True
tail_metrics_correctly_ordered	True
mean_return_matches_mean_pnl	True

Contract-summary validation

Every contract was evaluated under the same 1,000 non-BASE scenarios. All frequency identities, premium bounds, return bounds and tail-metric ordering checks passed.

The aggregation therefore preserves the underlying contract-scenario results correctly.

```
In [41]: initial_action_comparison = (
        create_initial_action_comparison(
            contract_scenario_summary
        )
    )

display(
    initial_action_comparison.round(4)
)
```

	Initial Recommended Action	Contracts	Median_Premium_Paid	Median_Forecast_Edge_Return
0	Buy	91	490.0	0.5394
1	Positive Edge	16	1270.0	0.0301
2	Do Not Buy	687	570.0	-0.6349

Observed differences between the initial groups

The initial classification successfully separates contracts by forecast valuation.

The 91 initial **Buy** contracts have a median forecast edge of approximately 53.9% relative to the ask. The 16 **Positive Edge** contracts have a much smaller median edge of approximately 3.0%, while the 687 **Do Not Buy** contracts have a median forecast edge of approximately -63.5%.

Median premium paid also differs between the groups:

- **Buy** : approximately 490 per contract;
- **Positive Edge** : approximately 1,270 per contract; and
- **Do Not Buy** : approximately 570 per contract.

These results establish economically different valuation groups, but they do not yet show that **Buy** contracts dominate on outcome consistency or downside risk.

```
In [42]: highest_mean_return = (
    contract_scenario_summary
    .nlargest(
        15,
        "Mean_Purchase_Return",
    )
)

display(
    highest_mean_return[
        [
            "Contract Symbol",
            "Ticker",
            "Option Type",
            "strike",
            "Initial Recommended Action",
            "Forecast Edge Return",
            "Mean_Purchase_Return",
            "Scenario_Profit_Frequency",
            "Fifth_Percentile_Purchase_Return",
            "Expected_Shortfall_5pct_Return",
            "Worst_Purchase_Return",
        ]
    ]
    .round(4)
)
```

	Contract Symbol	Ticker	Option Type	strike	Initial Recommended Action	Forecast Edge Return	Mean_
0	WMT260918P00092500	WMT	Puts	92.5	Buy	0.7239	
1	WMT260918P00095000	WMT	Puts	95.0	Buy	1.2096	
2	WMT260918P00097500	WMT	Puts	97.5	Buy	1.5436	
3	WMT260918P00100000	WMT	Puts	100.0	Buy	1.7364	
91	WMT260918P00087500	WMT	Puts	87.5	Do Not Buy	-0.2632	
4	MSFT260918P00415000	MSFT	Puts	415.0	Buy	1.4597	
5	MSFT260918P00410000	MSFT	Puts	410.0	Buy	1.3745	
6	MSFT260918P00425000	MSFT	Puts	425.0	Buy	1.5334	
7	AAPL260918P00275000	AAPL	Puts	275.0	Buy	0.3720	
778	AAPL260918P00265000	AAPL	Puts	265.0	Positive Edge	0.0336	
8	MSFT260918P00420000	MSFT	Puts	420.0	Buy	1.4596	
9	AAPL260918P00270000	AAPL	Puts	270.0	Buy	0.1865	
10	MSFT260918P00405000	MSFT	Puts	405.0	Buy	1.2018	
92	AAPL260918P00260000	AAPL	Puts	260.0	Do Not Buy	-0.1637	
11	AAPL260918P00280000	AAPL	Puts	280.0	Buy	0.4767	

Contracts with the highest average scenario returns

The strongest average purchase returns are concentrated in puts.

The leading contracts include WMT puts with strikes from 92.5 to 100, MSFT puts around strikes 405 to 425 and several AAPL puts around strikes 270 to 280. Most were initially classified as **Buy**.

This supports the initial screen when average scenario return is the criterion, but it also reveals concentration in a small number of downside-oriented trades rather than broad support across calls, puts and all seven underlyings.

```
In [43]: highest_profit_frequency = (
    contract_scenario_summary
    .sort_values(
        [
            "Scenario_Profit_Frequency",
            "Mean_Purchase_Return",
        ],
        ascending=[
            False,
            False,
        ],
    )
    .head(15)
)
```

```

display(
  highest_profit_frequency[
    [
      "Contract Symbol",
      "Ticker",
      "Option Type",
      "strike",
      "Initial Recommended Action",
      "Scenario_Profit_Frequency",
      "Mean_Purchase_Return",
      "Median_Purchase_Return",
      "Fifth_Percentile_Purchase_Return",
      "Worst_Purchase_Return",
    ]
  ]
  .round(4)
)

```

	Contract Symbol	Ticker	Option Type	strike	Initial Recommended Action	Scenario_Profit
165	MSFT260918C00225000	MSFT	Calls	225.0	Do Not Buy	
174	HCA260918C00300000	HCA	Calls	300.0	Do Not Buy	
171	HCA260918C00340000	HCA	Calls	340.0	Do Not Buy	
195	AAPL260918C00075000	AAPL	Calls	75.0	Do Not Buy	
269	AMZN260918C00200000	AMZN	Calls	200.0	Do Not Buy	
285	AMZN260918C00220000	AMZN	Calls	220.0	Do Not Buy	
192	AMZN260918P00320000	AMZN	Puts	320.0	Do Not Buy	
336	AMZN260918C00225000	AMZN	Calls	225.0	Do Not Buy	
294	AMZN260918C00230000	AMZN	Calls	230.0	Do Not Buy	
340	AMZN260918C00210000	AMZN	Calls	210.0	Do Not Buy	
188	AMZN260918P00310000	AMZN	Puts	310.0	Do Not Buy	
346	AMZN260918C00215000	AMZN	Calls	215.0	Do Not Buy	
184	AMZN260918P00305000	AMZN	Puts	305.0	Do Not Buy	
311	AMZN260918C00235000	AMZN	Calls	235.0	Do Not Buy	
304	AMZN260918C00240000	AMZN	Calls	240.0	Do Not Buy	

Scenario gain frequency does not reproduce the initial ranking

The contracts with the highest scenario gain frequencies are not the same contracts that have the strongest forecast edges.

The frequency table is led mainly by contracts initially classified as **Do Not Buy**, including several deep-in-the-money calls. Such contracts may retain value in a large

share of generated scenarios while still being unattractive relative to the premium required to purchase them.

Gain frequency therefore cannot be used alone. A contract may gain frequently but offer little model edge, limited upside or an unfavourable purchase price.

```
In [44]: strongest_tail_results = (
    contract_scenario_summary
    .sort_values(
        [
            "Expected_Shortfall_5pct_Return",
            "Mean_Purchase_Return",
        ],
        ascending=[
            False,
            False,
        ],
    )
    .head(15)
)

weakest_tail_results = (
    contract_scenario_summary
    .nsmallest(
        15,
        "Expected_Shortfall_5pct_Return",
    )
)

tail_columns = [
    "Contract Symbol",
    "Ticker",
    "Option Type",
    "strike",
    "Initial Recommended Action",
    "Forecast Edge Return",
    "Mean_Purchase_Return",
    "Scenario_Profit_Frequency",
    "Fifth_Percentile_Purchase_Return",
    "Expected_Shortfall_5pct_Return",
    "Worst_Purchase_Return",
]

display(
    strongest_tail_results[
        tail_columns
    ]
    .round(4)
)

display(
    weakest_tail_results[
        tail_columns
    ]
    .round(4)
)
```

	Contract Symbol	Ticker	Option Type	strike	Recommended Action	Initial Forecast Edge Return	Me
208	GOOGL260918C00050000	GOOGL	Calls	50.0	Do Not Buy	-0.0430	
212	AAPL260918C00050000	AAPL	Calls	50.0	Do Not Buy	-0.0592	
209	GOOGL260918C00060000	GOOGL	Calls	60.0	Do Not Buy	-0.0444	
213	AAPL260918C00055000	AAPL	Calls	55.0	Do Not Buy	-0.0607	
221	META260918C00100000	META	Calls	100.0	Do Not Buy	-0.0671	
215	AAPL260918C00060000	AAPL	Calls	60.0	Do Not Buy	-0.0614	
211	GOOGL260918C00070000	GOOGL	Calls	70.0	Do Not Buy	-0.0460	
268	GOOGL260918C00065000	GOOGL	Calls	65.0	Do Not Buy	-0.0706	
360	GOOGL260918C00040000	GOOGL	Calls	40.0	Do Not Buy	-0.0417	
229	META260918C00120000	META	Calls	120.0	Do Not Buy	-0.0702	
437	META260918C00040000	META	Calls	40.0	Do Not Buy	-0.0593	
195	AAPL260918C00075000	AAPL	Calls	75.0	Do Not Buy	-0.0232	
223	AAPL260918C00070000	AAPL	Calls	70.0	Do Not Buy	-0.0643	
440	META260918C00045000	META	Calls	45.0	Do Not Buy	-0.0599	
276	GOOGL260918C00075000	GOOGL	Calls	75.0	Do Not Buy	-0.0727	

	Contract Symbol	Ticker	Option Type	strike	Recommended Action	Initial Forecast Edge Return	Mean_
1	WMT260918P00095000	WMT	Puts	95.0	Buy	1.2096	
2	WMT260918P00097500	WMT	Puts	97.5	Buy	1.5436	
3	WMT260918P00100000	WMT	Puts	100.0	Buy	1.7364	
6	MSFT260918P00425000	MSFT	Puts	425.0	Buy	1.5334	
8	MSFT260918P00420000	MSFT	Puts	420.0	Buy	1.4596	
23	HCA260918P00370000	HCA	Puts	370.0	Buy	1.2008	
4	MSFT260918P00415000	MSFT	Puts	415.0	Buy	1.4597	
0	WMT260918P00092500	WMT	Puts	92.5	Buy	0.7239	
21	AAPL260918P00305000	AAPL	Puts	305.0	Buy	0.6236	
91	WMT260918P00087500	WMT	Puts	87.5	Do Not Buy	-0.2632	
37	HCA260918P00365000	HCA	Puts	365.0	Buy	0.8233	
5	MSFT260918P00410000	MSFT	Puts	410.0	Buy	1.3745	
20	HCA260918P00360000	HCA	Puts	360.0	Buy	1.1660	
135	WMT260918P00090000	WMT	Puts	90.0	Do Not Buy	-0.3963	
10	MSFT260918P00405000	MSFT	Puts	405.0	Buy	1.2018	

Observed lower-tail trade-off

The strongest lower-tail results are dominated by deep-in-the-money calls initially labelled **Do Not Buy**. Because their premiums contain substantial intrinsic value, their percentage returns are less likely to approach a complete loss under the generated stresses.

The weakest expected-shortfall results are concentrated in puts initially labelled **Buy**, particularly WMT, MSFT and HCA contracts. Several combine large forecast edges and strong mean returns with lower-tail outcomes close to losing the entire premium.

A large theoretical edge and strong upside therefore do not imply controlled downside. The final ranking must balance average return against frequency, expected shortfall and liquidity.

```
In [45]: action_order = [
    "Buy",
    "Positive Edge",
    "Do Not Buy",
]

plt.figure(figsize=(10, 6))

for action in action_order:
    action_rows = (
        contract_scenario_summary.loc[
            contract_scenario_summary[
                "Initial Recommended Action"
            ].eq(action)
        ]
    )

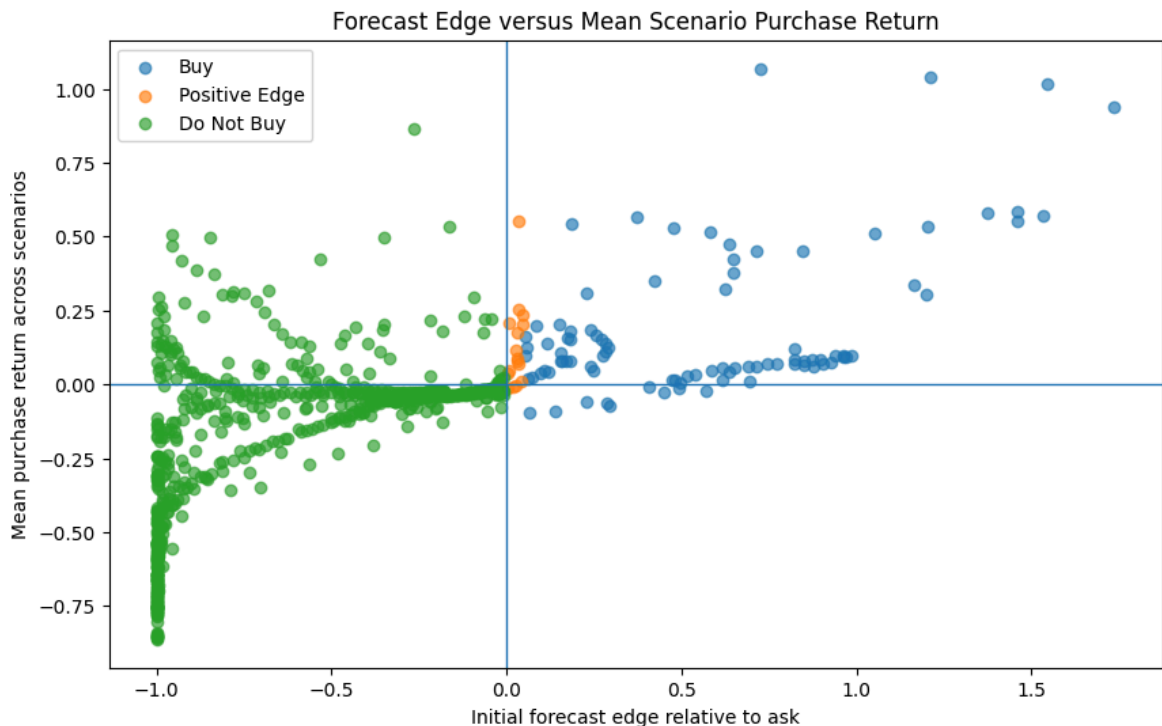
    if action_rows.empty:
        continue

    plt.scatter(
        action_rows[
            "Forecast Edge Return"
        ],
        action_rows[
            "Mean_Purchase_Return"
        ],
        alpha=0.65,
        label=action,
    )

plt.axhline(0, linewidth=1)
plt.axvline(0, linewidth=1)
plt.xlabel(
    "Initial forecast edge relative to ask"
)
plt.ylabel(
    "Mean purchase return across scenarios"
)
```



```
plt.title(
    "Forecast Edge versus Mean Scenario Purchase Return"
)
plt.legend()
plt.show()
```



Forecast edge versus average scenario return

The plot shows a positive but highly dispersed relationship between forecast edge and average scenario purchase return.

Many initial **Buy** contracts appear in the upper-right region. However, several positive-edge contracts still produce weak or negative mean outcomes, while some **Do Not Buy** contracts generate positive average returns.

The forecast signal contains useful cross-sectional information, but strike, option type, purchase premium and exposure to the generated shocks materially change the result after the initial valuation comparison.

```
In [46]: plt.figure(figsize=(10, 6))

for action in action_order:
    action_rows = (
        contract_scenario_summary.loc[
            contract_scenario_summary[
                "Initial Recommended Action"
            ].eq(action)
        ]
    )

    if action_rows.empty:
        continue

    plt.scatter(
```

```

        action_rows[
            "Scenario_Profit_Frequency"
        ],
        action_rows[
            "Expected_Shortfall_5pct_Return"
        ],
        alpha=0.65,
        label=action,
    )

plt.axvline(
    CANDIDATE_PROFIT_FREQUENCY,
    linewidth=1,
)

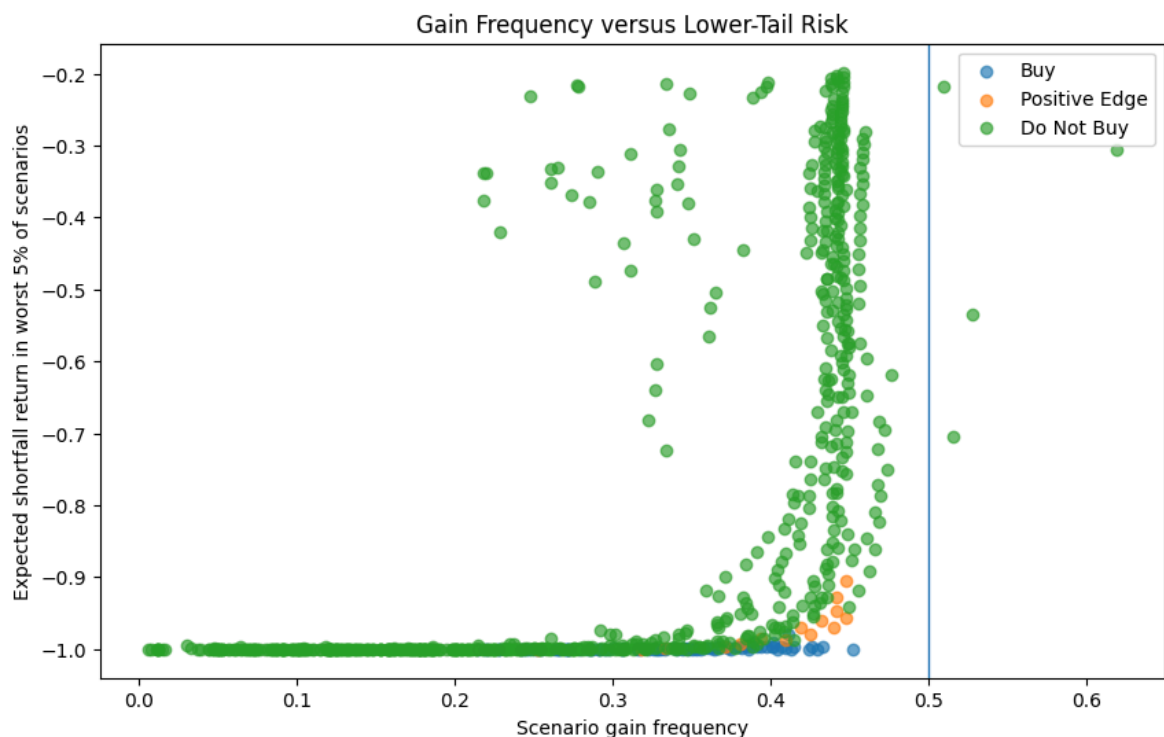
plt.xlabel(
    "Scenario gain frequency"
)

plt.ylabel(
    "Expected shortfall return in worst 5% of scenarios"
)

plt.title(
    "Gain Frequency versus Lower-Tail Risk"
)

plt.legend()
plt.show()

```



Outcome consistency remained the main weakness

The initial **Buy** observations remain below the 50% scenario gain-frequency line later required by the Candidate rule. At the same time, many expected- shortfall

returns are close to -100%, showing that the lower tail can contain near-total premium losses.

This does not invalidate the positive mean returns of several contracts. It shows that those returns arise from asymmetric distributions: large gains in favourable states offset losses in a greater number of generated states.

The final classification must therefore distinguish contracts with interesting upside from contracts with enough consistency to justify the stronger Candidate label.

```
In [47]: store.write_frame(
    contract_scenario_summary,
    "contract_scenario_summary",
    run_id=run_id,
)

store.write_frame(
    initial_action_comparison,
    "initial_action_comparison",
    run_id=run_id,
)

contract_scenario_summary.to_csv(
    TABLE_DIR
    / "contract_scenario_summary.csv",
    index=False,
)

initial_action_comparison.to_csv(
    TABLE_DIR
    / "initial_action_comparison.csv",
    index=False,
)

print(
    "Stored contract summaries:",
    len(contract_scenario_summary),
)

print(
    "Stored recommendation groups:",
    len(initial_action_comparison),
)
```

```
2026-08-02 13:39:44,660 - options_risk_engine.database - INFO - Stored 794
rows in contract_scenario_summary
2026-08-02 13:39:44,665 - options_risk_engine.database - INFO - Stored 3 r
ows in initial_action_comparison
Stored contract summaries: 794
Stored recommendation groups: 3
```

Section 19 result

Section 19 reduced 794,794 scenario rows to one complete outcome profile for each of the 794 contracts.

The initial screen was informative for average return, particularly among selected puts, but did not provide uniform dominance across gain frequency and lower-tail risk.

The main findings are:

- Buy contracts have substantially stronger forecast valuation;
- several Buy puts have the highest average scenario returns;
- some Do Not Buy contracts have high gain frequencies or controlled percentage downside;
- many strong forecast-edge contracts retain near-total premium loss in the lower tail; and
- no initial Buy contract reaches the later 50% Candidate gain-frequency requirement.

The next section explains which Greek exposures generated these patterns.

20. Contract-level Greek attribution and risk explanation

Section 19 measured the outcome distribution of each standardised contract. This section explains which market-risk factors generated those outcomes.

The existing Greek attribution engine decomposes the movement from the base market-IV model value to the shocked scenario value into:

- Delta P&L from changes in the underlying share price;
- Gamma P&L from nonlinear spot exposure;
- Vega P&L from volatility changes;
- Theta P&L from the passage of time; and
- Rho P&L from interest-rate changes.

The initial difference between the ask and the base model value is not included in this decomposition. Any difference between full repricing and the fixed- Greek approximation remains as residual P&L. Large residuals indicate that initial Greeks do not adequately describe the contract under larger shocks.

```
In [48]: from options_risk_engine.recommendations.attribution import (
        create_contract_attribution_results,
        create_contract_attribution_summary,
        create_contract_driver_comparison,
        validate_contract_attribution_summary,
    )

contract_attribution_results = (
    create_contract_attribution_results(
        scenario_results=(
            raw_contract_scenario_results
        ),
        positions=standardised_contracts,
```

```

    )
)

print(
    "Contract-attribution rows:",
    len(contract_attribution_results),
)

print(
    "Contracts:",
    contract_attribution_results[
        "Contract Symbol"
    ].nunique(),
)

print(
    "Scenarios:",
    contract_attribution_results[
        "Scenario ID"
    ].nunique(),
)

```

```

2026-08-02 13:39:44,911 - options_risk_engine.risk.attribution - INFO - Starting Greek P&L attribution
2026-08-02 13:39:45,339 - options_risk_engine.risk.attribution - INFO - Greek attribution accounting check passed
2026-08-02 13:39:45,339 - options_risk_engine.risk.attribution - INFO - Greek attribution completed for 794794 position-scenario rows and 1001 scenarios
2026-08-02 13:39:45,340 - options_risk_engine.recommendations.attribution - INFO - Calculated Greek attribution for 794794 contract-scenario rows
Contract-attribution rows: 794794
Contracts: 794
Scenarios: 1001

```

Scenario PnL is the full-revaluation movement from the base model value.

Approximate PnL is the sum of the five Greek components.

Residual percentage can be unstable when scenario movement is close to zero, so the contract summary also uses gross residual ratios based on absolute P&L.

```

In [49]: contract_attribution_summary = (
    create_contract_attribution_summary(
        attribution_results=(
            contract_attribution_results
        ),
        contract_scenario_summary=(
            contract_scenario_summary
        ),
        base_scenario_id=BASE_SCENARIO_ID,
        local_spot_limit=LOCAL_SPOT_LIMIT,
        local_volatility_limit=(
            LOCAL_VOLATILITY_LIMIT
        ),
        local_days_limit=LOCAL_DAYS_LIMIT,
        local_rate_limit=LOCAL_RATE_LIMIT,
        minimum_local_scenarios=(
            MINIMUM_LOCAL_SCENARIOS
        ),
    )

```

```

    )
)

print(
    "Contracts attributed:",
    len(contract_attribution_summary),
)

attribution_display_columns = [
    "Contract Symbol",
    "Ticker",
    "Option Type",
    "strike",
    "Initial Recommended Action",
    "Forecast Edge Return",
    "Mean_Purchase_Return",
    "Scenario_Profit_Frequency",
    "Expected_Shortfall_5pct_Return",
    "Risk Driver",
    "Dominant Greek",
    "Dominant Driver Share",
    "Mean_Delta_Return",
    "Mean_Gamma_Return",
    "Mean_Vega_Return",
    "Mean_Theta_Return",
    "Mean_Rho_Return",
    "Gross_Residual_Ratio",
    "Local_Scenario_Count",
    "Local_Gross_Residual_Ratio",
    "Attribution Reliability",
]

available_attribution_columns = [
    column
    for column in attribution_display_columns
    if column in contract_attribution_summary.columns
]

display(
    contract_attribution_summary[
        available_attribution_columns
    ]
    .head(10)
    .round(4)
)

display(
    contract_attribution_summary[
        available_attribution_columns
    ]
    .tail(10)
    .round(4)
)

```

2026-08-02 13:39:47,192 - options_risk_engine.recommendations.attribution
 - INFO - Created Greek attribution summaries for 794 contracts
 Contracts attributed: 794

	Contract Symbol	Ticker	Option Type	strike	Initial Recommended Action	Forecast Edge Return	Mean_Pi
0	WMT260918P00092500	WMT	Puts	92.5	Buy	0.7239	
1	WMT260918P00095000	WMT	Puts	95.0	Buy	1.2096	
2	WMT260918P00097500	WMT	Puts	97.5	Buy	1.5436	
3	WMT260918P00100000	WMT	Puts	100.0	Buy	1.7364	
4	MSFT260918P00415000	MSFT	Puts	415.0	Buy	1.4597	
5	MSFT260918P00410000	MSFT	Puts	410.0	Buy	1.3745	
6	MSFT260918P00425000	MSFT	Puts	425.0	Buy	1.5334	
7	AAPL260918P00275000	AAPL	Puts	275.0	Buy	0.3720	
8	MSFT260918P00420000	MSFT	Puts	420.0	Buy	1.4596	
9	AAPL260918P00270000	AAPL	Puts	270.0	Buy	0.1865	

	Contract Symbol	Ticker	Option Type	strike	Initial Recommended Action	Forecast Edge Return	Mean
784	AMZN260918P00260000	AMZN	Puts	260.0	Positive Edge	0.0244	
785	AMZN260918P00270000	AMZN	Puts	270.0	Positive Edge	0.0307	
786	AMZN260918P00275000	AMZN	Puts	275.0	Positive Edge	0.0313	
787	AMZN260918P00280000	AMZN	Puts	280.0	Positive Edge	0.0338	
788	AMZN260918P00285000	AMZN	Puts	285.0	Positive Edge	0.0085	
789	AMZN260918P00290000	AMZN	Puts	290.0	Positive Edge	0.0016	
790	AMZN260918C00270000	AMZN	Calls	270.0	Positive Edge	0.0413	
791	AMZN260918C00265000	AMZN	Calls	265.0	Positive Edge	0.0249	
792	AMZN260918C00260000	AMZN	Calls	260.0	Positive Edge	0.0181	
793	AMZN260918C00255000	AMZN	Calls	255.0	Positive Edge	0.0041	

Interpreting the risk driver

The driver label identifies the largest source of average absolute Greek-attributed movement:

- **Directional** means Delta dominates;
- **Convexity** means Gamma dominates;
- **Volatility** means Vega dominates;
- **Time Decay** means Theta dominates;
- **Interest Rate** means Rho dominates; and
- **Mixed** means no single component clearly dominates.

The label does not determine whether a contract is attractive. It explains what market behaviour the trade depends upon.

```
In [50]: contract_attribution_checks = (
    validate_contract_attribution_summary(
        summary=contract_attribution_summary,
        original_contract_summary=(
            contract_scenario_summary
        ),
    )
)

display(
    pd.Series(
        contract_attribution_checks,
        name="Passed",
    )
    .to_frame()
)
```

2026-08-02 13:39:47,245 - options_risk_engine.recommendations.attribution - INFO - All contract attribution checks passed

Passed	
same_number_of_contracts	True
unique_contract_symbols	True
movement_matches_section_19	True
greek_attribution_identity	True
purchase_pnl_identity	True
driver_shares_sum_to_one	True
driver_shares_valid	True
recognised_risk_drivers	True
recognised_reliability_labels	True

Attribution reconciliation

All contract-attribution checks passed.

For every contract, the initial purchase-price difference, subsequent market movement and Greek explanation remain distinct but reconcile correctly.

The reliability label must still be respected. A low residual under smaller local shocks supports the Greek explanation, while full repricing remains the authoritative result for the larger stresses.

```
In [51]: contract_driver_comparison = (
    create_contract_driver_comparison(
        contract_attribution_summary
    )
)
```



```
display(
    contract_driver_comparison.round(4)
)
```

	Initial Recommended Action	Risk Driver	Contracts	Median_Mean_Purchase_Return	Median_Profit_Frequency
0	Buy	Convexity	3	1.0168	
1	Buy	Directional	36	0.0711	
2	Buy	Mixed	52	0.0979	
3	Do Not Buy	Directional	371	-0.0293	
4	Do Not Buy	Mixed	251	-0.2553	
5	Do Not Buy	Time Decay	65	-0.7343	
6	Positive Edge	Directional	12	0.0575	
7	Positive Edge	Mixed	4	0.2460	

Observed drivers within the initial recommendations

The initial **Buy** group is not dominated by a single volatility label.

Of the 91 initial **Buy** contracts:

- 52 are classified as **Mixed** ;
- 36 are **Directional** ; and
- 3 are **Convexity** .

The **Positive Edge** group contains 12 **Directional** and 4 **Mixed** contracts. The **Do Not Buy** group contains 371 **Directional**, 251 **Mixed** and 65 **Time Decay** contracts.

The original valuation signal is based on forecast volatility, yet the broad scenario outcomes are mainly explained by combinations of Delta, Gamma and Theta or by directional Delta exposure. Vega remains economically relevant, but it is not the largest average absolute component under this run's common shocks.

```
In [52]: initial_buy_attribution = (
    contract_attribution_summary.loc[
        contract_attribution_summary[
            "Initial Recommended Action"
        ].eq("Buy")
    ]
    .sort_values(
        [
            "Mean_Purchase_Return",
            "Scenario_Profit_Frequency",
        ],
        ascending=[
            False,
            False,
        ]
    )
)
```

```
    ],  
  )  
)  
  
display(  
  initial_buy_attribution[  
    [  
      "Contract Symbol",  
      "Ticker",  
      "Option Type",  
      "strike",  
      "Forecast Edge Return",  
      "Mean_Purchase_Return",  
      "Scenario_Profit_Frequency",  
      "Expected_Shortfall_5pct_Return",  
      "Risk Driver",  
      "Dominant Driver Share",  
      "Mean_Delta_Return",  
      "Mean_Gamma_Return",  
      "Mean_Vega_Return",  
      "Mean_Theta_Return",  
      "Gross_Residual_Ratio",  
      "Local_Gross_Residual_Ratio",  
      "Attribution Reliability",  
    ]  
  ]  
  .head(25)  
  .round(4)  
)
```

	Contract Symbol	Ticker	Option Type	strike	Forecast Edge Return	Mean_Purchase_Return
0	WMT260918P00092500	WMT	Puts	92.5	0.7239	1.06
1	WMT260918P00095000	WMT	Puts	95.0	1.2096	1.03
2	WMT260918P00097500	WMT	Puts	97.5	1.5436	1.01
3	WMT260918P00100000	WMT	Puts	100.0	1.7364	0.93
4	MSFT260918P00415000	MSFT	Puts	415.0	1.4597	0.58
5	MSFT260918P00410000	MSFT	Puts	410.0	1.3745	0.58
6	MSFT260918P00425000	MSFT	Puts	425.0	1.5334	0.57
7	AAPL260918P00275000	AAPL	Puts	275.0	0.3720	0.56
8	MSFT260918P00420000	MSFT	Puts	420.0	1.4596	0.55
9	AAPL260918P00270000	AAPL	Puts	270.0	0.1865	0.54
10	MSFT260918P00405000	MSFT	Puts	405.0	1.2018	0.53
11	AAPL260918P00280000	AAPL	Puts	280.0	0.4767	0.52
12	AAPL260918P00285000	AAPL	Puts	285.0	0.5817	0.51
13	MSFT260918P00400000	MSFT	Puts	400.0	1.0502	0.51
14	AAPL260918P00290000	AAPL	Puts	290.0	0.6359	0.47
15	MSFT260918P00395000	MSFT	Puts	395.0	0.8463	0.46
16	MSFT260918P00390000	MSFT	Puts	390.0	0.7146	0.45
17	AAPL260918P00295000	AAPL	Puts	295.0	0.6473	0.42
18	AAPL260918P00300000	AAPL	Puts	300.0	0.6473	0.37
19	MSFT260918P00385000	MSFT	Puts	385.0	0.4205	0.35
20	HCA260918P00360000	HCA	Puts	360.0	1.1660	0.33
21	AAPL260918P00305000	AAPL	Puts	305.0	0.6236	0.32
22	MSFT260918P00380000	MSFT	Puts	380.0	0.2282	0.30
23	HCA260918P00370000	HCA	Puts	370.0	1.2008	0.30
24	GOOGL260918P00320000	GOOGL	Puts	320.0	0.1503	0.20

```
In [53]: driver_plot_data = (
            initial_buy_attribution
            .head(15)
            .set_index(
                "Contract Symbol"
            )
            [
                [
                    "Delta_Driver_Share",
                    "Gamma_Driver_Share",
                    "Vega_Driver_Share",
                    "Theta_Driver_Share",
```

```

        "Rho_Driver_Share",
    ]
]
)

driver_plot_data.plot(
    kind="bar",
    stacked=True,
    figsize=(13, 7),
)

plt.ylabel(
    "Share of average absolute Greek-attributed P&L"
)

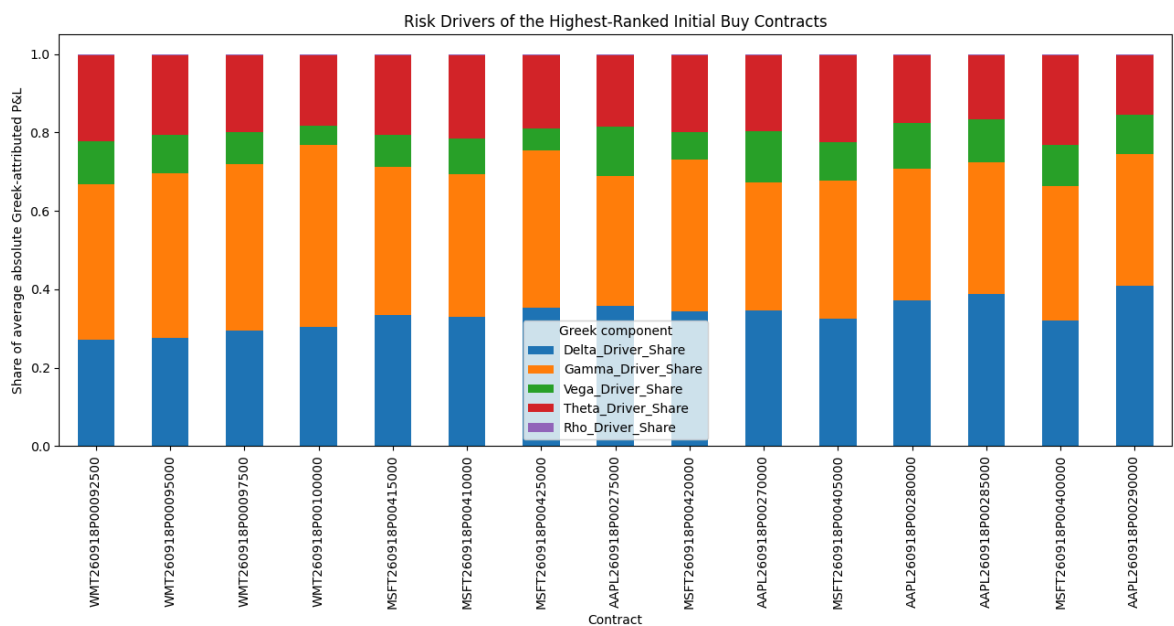
plt.xlabel(
    "Contract"
)

plt.title(
    "Risk Drivers of the Highest-Ranked Initial Buy Contracts"
)

plt.legend(
    title="Greek component"
)

plt.tight_layout()
plt.show()

```



Risk composition of the strongest initial Buy contracts

The highest-ranked initial **Buy** contracts display material exposure to several Greek components rather than one isolated driver.

Delta and Gamma account for the largest portions of most bars, confirming dependence on both the direction and size of the underlying move. Theta also forms

a meaningful part of the risk profile, while Vega contributes a smaller but visible share. Rho is negligible under the tested rate shocks.

A volatility view implemented through an option therefore still creates directional, convexity and timing exposure.

```
In [54]: store.write_frame(
    contract_attribution_summary,
    "contract_attribution_summary",
    run_id=run_id,
)

store.write_frame(
    contract_driver_comparison,
    "contract_driver_comparison",
    run_id=run_id,
)

contract_attribution_summary.to_csv(
    TABLE_DIR
    / "contract_attribution_summary.csv",
    index=False,
)

contract_driver_comparison.to_csv(
    TABLE_DIR
    / "contract_driver_comparison.csv",
    index=False,
)

print(
    "Stored contract attribution summaries:",
    len(contract_attribution_summary),
)

print(
    "Stored driver comparison rows:",
    len(contract_driver_comparison),
)
```

```
2026-08-02 13:39:47,688 - options_risk_engine.database - INFO - Stored 794
rows in contract_attribution_summary
2026-08-02 13:39:47,694 - options_risk_engine.database - INFO - Stored 8 r
ows in contract_driver_comparison
Stored contract attribution summaries: 794
Stored driver comparison rows: 8
```

Section 20 result

Every contract now has both an outcome profile and an explanation of the risks that generated it.

The initial **Buy** contracts were mainly Mixed or Directional rather than purely volatility-driven. Their outcomes depend on the interaction between Delta, Gamma, Vega, Theta and the initial cost of paying the ask.

The three Convexity-driven **Buy** contracts produced the strongest grouped median mean return, while the much larger Mixed and Directional groups produced more moderate outcomes.

Greek attribution provides economic explanation but does not replace full repricing. The final ranking therefore uses the scenario distribution for scoring and the Greek analysis for interpretation.

21. Final risk-aware contract ranking

The initial recommendation classified contracts using forecast-volatility valuation, quote validity and eligibility. Sections 18–20 then evaluated every contract on identical purchase terms, summarised its scenario distribution and identified the risks responsible for its outcomes.

This section combines:

- forecast valuation edge;
- average and median scenario purchase return;
- scenario gain frequency;
- lower-tail risk; and
- liquidity.

Hard rejection rules are applied before selection. Contracts with insufficient market quality become **Liquidity Rejection**. Contracts combining unusually weak lower-tail outcomes with low gain frequency become **Risk Rejection**.

The remaining contracts may become **Candidate**, **Watchlist** or **No Trade**. The score ranks contracts relative to the current universe; it is not a calibrated probability of profit or a substitute for a historical out-of-sample backtest.

```
In [55]: from options_risk_engine.recommendations.ranking import (
        create_final_contract_ranking,
        create_final_recommendation_summary,
        validate_final_contract_ranking,
    )

    final_contract_ranking = (
        create_final_contract_ranking(
            contract_summary=(
                contract_attribution_summary
            ),
            max_spread_pct=(
                FINAL_MAX_SPREAD_PCT
            ),
            min_volume=FINAL_MIN_VOLUME,
            min_open_interest=(
                FINAL_MIN_OPEN_INTEREST
            ),
            candidate_score=CANDIDATE_SCORE,
            watchlist_score=WATCHLIST_SCORE,
```

```
        candidate_profit_frequency=(
            CANDIDATE_PROFIT_FREQUENCY
        ),
        minimum_mean_return=(
            MINIMUM_MEAN_RETURN
        ),
        risk_tail_quantile=(
            RISK_TAIL_QUANTILE
        ),
        risk_profit_frequency=(
            RISK_PROFIT_FREQUENCY
        ),
        score_weights=FINAL_SCORE_WEIGHTS,
    )

)

print(
    "Contracts ranked:",
    len(final_contract_ranking),
)

display(
    final_contract_ranking[
        "Final Recommended Action"
    ]
    .value_counts()
    .rename("Contracts")
    .to_frame()
)
```

2026-08-02 13:39:47,867 - options_risk_engine.recommendations.ranking - INFO - Created final rankings for 794 contracts

Contracts ranked: 794

Contracts	
Final Recommended Action	
Liquidity Rejection	462
No Trade	252
Watchlist	57
Risk Rejection	23

Observed final classifications

The final engine ranked 794 contracts but did not assign the **Candidate** classification to any contract.

The output contains:

- 57 **Watchlist** contracts;
- 252 **No Trade** contracts;
- 23 **Risk Rejection** contracts; and
- 462 **Liquidity Rejection** contracts.

Liquidity is the largest source of exclusion, removing approximately 58.2% of the universe. An attractive theoretical value is therefore not sufficient when spread, volume or open interest does not support a sufficiently tradeable quote.

The absence of Candidates is not a validation failure. It means no initial **Buy** contract simultaneously satisfied the score, liquidity, risk, positive-mean-return and 50% scenario gain-frequency requirements.

```
In [56]: final_ranking_checks = (
    validate_final_contract_ranking(
        ranking=final_contract_ranking,
        max_spread_pct=(
            FINAL_MAX_SPREAD_PCT
        ),
        min_volume=FINAL_MIN_VOLUME,
        min_open_interest=(
            FINAL_MIN_OPEN_INTEREST
        ),
        candidate_score=CANDIDATE_SCORE,
        watchlist_score=WATCHLIST_SCORE,
        candidate_profit_frequency=(
            CANDIDATE_PROFIT_FREQUENCY
        ),
        minimum_mean_return=(
            MINIMUM_MEAN_RETURN
        ),
        risk_tail_quantile=(
            RISK_TAIL_QUANTILE
        ),
        risk_profit_frequency=(
            RISK_PROFIT_FREQUENCY
        ),
        score_weights=FINAL_SCORE_WEIGHTS,
    )
)

display(
    pd.Series(
        final_ranking_checks,
        name="Passed",
    )
    .to_frame()
)
```

2026-08-02 13:39:47,889 - options_risk_engine.recommendations.ranking - INFO - All final ranking checks passed

	Passed
contracts_available	True
unique_contract_symbols	True
scores_between_zero_and_one_hundred	True
weighted_score_matches	True
recognised_final_actions	True
liquidity_gate_matches	True
risk_gate_matches	True
candidate_rules_hold	True
watchlist_rules_hold	True
do_not_buy_not_promoted	True
reasons_complete	True
shortlist_ranks_unique	True

All final-ranking checks passed. The score matches the declared weights, the hard gates are applied consistently and no **Do Not Buy** contract can be promoted to the final shortlist.

```
In [57]: final_recommendation_summary = (
    create_final_recommendation_summary(
        final_contract_ranking
    )
)

display(
    final_recommendation_summary.round(4)
)

initial_to_final_comparison = pd.crosstab(
    final_contract_ranking[
        "Initial Recommended Action"
    ],
    final_contract_ranking[
        "Final Recommended Action"
    ],
    margins=True,
)

display(
    initial_to_final_comparison
)
```

	Final Recommended Action	Contracts	Median_Final_Score	Median_Forecast_Edge	Median_N
0	Watchlist	57	68.3505	0.2903	
1	No Trade	252	59.5572	-0.3495	
2	Risk Rejection	23	59.2412	0.8493	
3	Liquidity Rejection	462	34.2989	-0.9021	

Final Recommended Action	Liquidity Rejection	No Trade	Risk Rejection	Watchlist	All
Initial Recommended Action					
Buy	26	0	17	48	91
Do Not Buy	432	249	6	0	687
Positive Edge	4	3	0	9	16
All	462	252	23	57	794

How the initial labels changed

Of the 91 initial **Buy** contracts:

- 48, approximately 52.7%, became **Watchlist** ;
- 17, approximately 18.7%, became **Risk Rejection** ; and
- 26, approximately 28.6%, became **Liquidity Rejection** .

None became **Candidate** .

Of the 16 **Positive Edge** contracts, 9 reached the Watchlist, 3 became **No Trade** and 4 failed the liquidity gate.

The 687 **Do Not Buy** contracts remained outside the shortlist. The second-stage process therefore did not promote contracts lacking the required initial forecast valuation.

```
In [58]: final_contract_shortlist = (
    final_contract_ranking.loc[
        final_contract_ranking[
            "Final Recommended Action"
        ].isin(
            [
                "Candidate",
                "Watchlist",
            ]
        )
    ].sort_values(
        "Final Shortlist Rank"
    )
```

```
        .copy()
    )

    final_display_columns = [
        "Final Shortlist Rank",
        "Ticker Shortlist Rank",
        "Contract Symbol",
        "Ticker",
        "Option Type",
        "strike",
        "Expiry",
        "Initial Recommended Action",
        "Final Recommended Action",
        "Final Score",
        "ask",
        "Premium Paid",
        "Forecast Edge Return",
        "Mean_Purchase_Return",
        "Median_Purchase_Return",
        "Scenario_Profit_Frequency",
        "Fifth_Percentile_Purchase_Return",
        "Expected_Shortfall_5pct_Return",
        "SpreadPct",
        "volume",
        "openInterest",
        "Risk Driver",
        "Dominant Driver Share",
        "Attribution Reliability",
        "Final Recommendation Reason",
    ]

    available_final_columns = [
        column
        for column in final_display_columns
        if column in final_contract_shortlist.columns
    ]

    display(
        final_contract_shortlist[
            available_final_columns
        ]
        .head(30)
        .round(4)
    )
```

	Final Shortlist Rank	Ticker Shortlist Rank	Contract Symbol	Ticker	Option Type	strike	Expiry	Re
0	1	1	AMZN260918P00280000	AMZN	Puts	280.0	2026-09-18	
1	2	1	GOOGL260918P00350000	GOOGL	Puts	350.0	2026-09-18	
2	3	2	AMZN260918P00265000	AMZN	Puts	265.0	2026-09-18	
3	4	3	AMZN260918P00270000	AMZN	Puts	270.0	2026-09-18	
4	5	4	AMZN260918P00275000	AMZN	Puts	275.0	2026-09-18	
5	6	2	GOOGL260918P00355000	GOOGL	Puts	355.0	2026-09-18	
6	7	5	AMZN260918P00255000	AMZN	Puts	255.0	2026-09-18	
7	8	6	AMZN260918C00270000	AMZN	Calls	270.0	2026-09-18	
8	9	7	AMZN260918P00250000	AMZN	Puts	250.0	2026-09-18	
9	10	3	GOOGL260918P00360000	GOOGL	Puts	360.0	2026-09-18	
10	11	8	AMZN260918C00275000	AMZN	Calls	275.0	2026-09-18	
11	12	4	GOOGL260918P00340000	GOOGL	Puts	340.0	2026-09-18	
12	13	9	AMZN260918P00260000	AMZN	Puts	260.0	2026-09-18	
13	14	1	META260918P00550000	META	Puts	550.0	2026-09-18	
14	15	5	GOOGL260918P00345000	GOOGL	Puts	345.0	2026-09-18	
15	16	2	META260918P00540000	META	Puts	540.0	2026-09-18	

	Final Shortlist Rank	Ticker Shortlist Rank	Contract Symbol	Ticker	Option Type	strike	Expiry	Re
16	17	6	GOOGL260918P00330000	GOOGL	Puts	330.0	2026-09-18	
17	18	10	AMZN260918P00245000	AMZN	Puts	245.0	2026-09-18	
18	19	11	AMZN260918C00280000	AMZN	Calls	280.0	2026-09-18	
19	20	3	META260918P00530000	META	Puts	530.0	2026-09-18	
20	21	4	META260918P00545000	META	Puts	545.0	2026-09-18	
21	22	5	META260918P00535000	META	Puts	535.0	2026-09-18	
22	23	7	GOOGL260918P00335000	GOOGL	Puts	335.0	2026-09-18	
23	24	6	META260918P00520000	META	Puts	520.0	2026-09-18	
24	25	7	META260918P00500000	META	Puts	500.0	2026-09-18	
25	26	8	META260918P00510000	META	Puts	510.0	2026-09-18	
26	27	8	GOOGL260918P00325000	GOOGL	Puts	325.0	2026-09-18	
27	28	12	AMZN260918P00285000	AMZN	Puts	285.0	2026-09-18	
28	29	13	AMZN260918C00285000	AMZN	Calls	285.0	2026-09-18	
29	30	9	GOOGL260918P00320000	GOOGL	Puts	320.0	2026-09-18	

Composition of the final Watchlist

The 57-contract Watchlist is strongly concentrated by ticker and option type:

- 19 META puts;
- 13 GOOGL puts;
- 9 AMZN puts;
- 7 AMZN calls;
- 7 AAPL puts;
- 1 MSFT put; and
- 1 WMT put.

No HCA contract reached the final shortlist.

The highest ranks are led by the AMZN 280 put, GOOGL 350 put and AMZN puts at strikes 265, 270 and 275. The concentration indicates clusters of adjacent strikes with similar exposures rather than 57 independent trading ideas.

Ticker and strike concentration must therefore be considered before treating the Watchlist as a portfolio.

```
In [59]: final_candidates = (
    final_contract_ranking.loc[
        final_contract_ranking[
            "Final Recommended Action"
        ].eq("Candidate")
    ]
    .sort_values(
        "Final Shortlist Rank"
    )
    .copy()
)

print(
    "Final Candidates:",
    len(final_candidates),
)

display(
    final_candidates[
        available_final_columns
    ]
    .head(20)
    .round(4)
)
```

Final Candidates: 0

Final Shortlist Rank	Ticker Shortlist Rank	Contract Symbol	Ticker	Option Type	strike	Expiry	Initial Recommended Action	Recon
----------------------------	-----------------------------	--------------------	--------	----------------	--------	--------	----------------------------------	-------

Why no contract became a Candidate

The empty Candidate table is the correct output under the declared rules.

Several contracts achieved high relative scores, but Candidate status also requires an initial **Buy** label and a scenario gain frequency of at least 50%. The Section 19 results show that the initial **Buy** observations remained below that consistency threshold.

Their positive mean returns are generated by asymmetric upside rather than gains in a majority of generated states. This is sufficient for 48 initial **Buy** contracts to remain on the Watchlist, but not enough for the stronger Candidate classification.

The thresholds should not be relaxed merely to force a positive recommendation. Alternative values belong in a separately labelled sensitivity analysis.

```
In [60]: initial_buys_downgraded = (
    final_contract_ranking.loc[
        final_contract_ranking[
            "Initial Recommended Action"
        ].eq("Buy")
        & ~final_contract_ranking[
            "Final Recommended Action"
        ].eq("Candidate")
    ]
    .sort_values(
        "Final Score",
        ascending=False,
    )
)

display(
    initial_buys_downgraded[
        [
            "Contract Symbol",
            "Ticker",
            "Option Type",
            "strike",
            "Initial Recommended Action",
            "Final Recommended Action",
            "Final Score",
            "Forecast Edge Return",
            "Mean_Purchase_Return",
            "Scenario_Profit_Frequency",
            "Expected_Shortfall_5pct_Return",
            "SpreadPct",
            "Risk Driver",
            "Final Recommendation Reason",
        ]
    ]
    .head(30)
    .round(4)
)
```

	Contract Symbol	Ticker	Option Type	strike	Initial Recommended Action	Final Recommended Action
1	GOOGL260918P00350000	GOOGL	Puts	350.0	Buy	Watchlist
2	AMZN260918P00265000	AMZN	Puts	265.0	Buy	Watchlist
5	GOOGL260918P00355000	GOOGL	Puts	355.0	Buy	Watchlist
6	AMZN260918P00255000	AMZN	Puts	255.0	Buy	Watchlist
9	GOOGL260918P00360000	GOOGL	Puts	360.0	Buy	Watchlist
10	AMZN260918C00275000	AMZN	Calls	275.0	Buy	Watchlist
11	GOOGL260918P00340000	GOOGL	Puts	340.0	Buy	Watchlist
13	META260918P00550000	META	Puts	550.0	Buy	Watchlist
14	GOOGL260918P00345000	GOOGL	Puts	345.0	Buy	Watchlist
15	META260918P00540000	META	Puts	540.0	Buy	Watchlist
16	GOOGL260918P00330000	GOOGL	Puts	330.0	Buy	Watchlist
18	AMZN260918C00280000	AMZN	Calls	280.0	Buy	Watchlist
19	META260918P00530000	META	Puts	530.0	Buy	Watchlist
20	META260918P00545000	META	Puts	545.0	Buy	Watchlist
21	META260918P00535000	META	Puts	535.0	Buy	Watchlist
22	GOOGL260918P00335000	GOOGL	Puts	335.0	Buy	Watchlist

	Contract Symbol	Ticker	Option Type	strike	Initial Recommended Action	Final Recommended Action
23	META260918P00520000	META	Puts	520.0	Buy	Watchlist
24	META260918P00500000	META	Puts	500.0	Buy	Watchlist
25	META260918P00510000	META	Puts	510.0	Buy	Watchlist
26	GOOGL260918P00325000	GOOGL	Puts	325.0	Buy	Watchlist
28	AMZN260918C00285000	AMZN	Calls	285.0	Buy	Watchlist
29	GOOGL260918P00320000	GOOGL	Puts	320.0	Buy	Watchlist
30	GOOGL260918P00365000	GOOGL	Puts	365.0	Buy	Watchlist
31	AMZN260918C00290000	AMZN	Calls	290.0	Buy	Watchlist
32	META260918P00490000	META	Puts	490.0	Buy	Watchlist
33	META260918P00555000	META	Puts	555.0	Buy	Watchlist
35	META260918P00560000	META	Puts	560.0	Buy	Watchlist
36	META260918P00525000	META	Puts	525.0	Buy	Watchlist
37	AMZN260918C00300000	AMZN	Calls	300.0	Buy	Watchlist
38	META260918P00515000	META	Puts	515.0	Buy	Watchlist

```
In [61]: top_score_components = (  
         final_contract_ranking  
         .nlargest(  

```

```

        15,
        "Final Score",
    )
    .set_index(
        "Contract Symbol"
    )
    [
        [
            "Forecast_Edge_Score",
            "Mean_Return_Score",
            "Median_Return_Score",
            "Profit_Frequency_Score",
            "Tail_Risk_Score",
            "Liquidity_Score",
        ]
    ]
)

weighted_score_components = (
    top_score_components
    .mul(
        pd.Series(
            FINAL_SCORE_WEIGHTS
        ),
        axis=1,
    )
)

weighted_score_components.plot(
    kind="bar",
    stacked=True,
    figsize=(14, 7),
)

plt.ylabel(
    "Weighted contribution to final score"
)

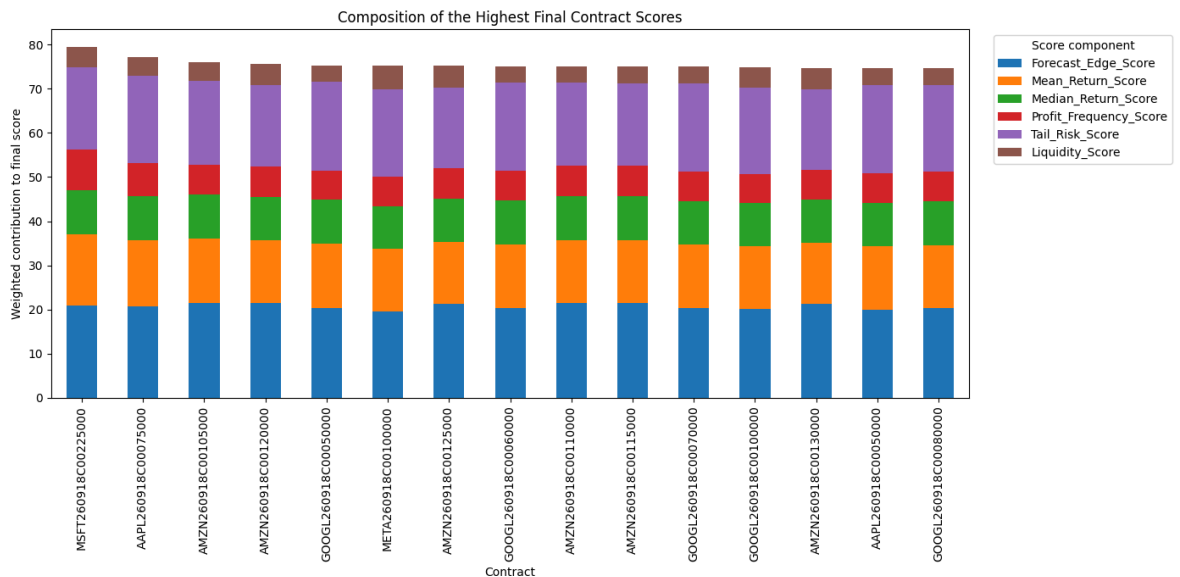
plt.xlabel(
    "Contract"
)

plt.title(
    "Composition of the Highest Final Contract Scores"
)

plt.legend(
    title="Score component",
    bbox_to_anchor=(1.02, 1),
    loc="upper left",
)

plt.tight_layout()
plt.show()

```



Interpreting the highest scores

The highest-scoring contracts receive support from several dimensions rather than forecast edge alone.

A high score does not automatically produce a Candidate. The score measures relative quality across the current universe, whereas Candidate status also applies absolute conditions including positive mean return, sufficient liquidity and at least 50% scenario gain frequency.

The score is therefore a ranking tool; the final classification determines whether the evidence crosses the selected decision threshold.

```
In [62]: store.write_frame(
    final_contract_ranking,
    "final_contract_ranking",
    run_id=run_id,
)

store.write_frame(
    final_recommendation_summary,
    "final_recommendation_summary",
    run_id=run_id,
)

store.write_frame(
    initial_to_final_comparison.reset_index(),
    "initial_to_final_comparison",
    run_id=run_id,
)

final_contract_ranking.to_csv(
    TABLE_DIR
    / "final_contract_ranking.csv",
    index=False,
)

final_recommendation_summary.to_csv(
    TABLE_DIR
```

```

    / "final_recommendation_summary.csv",
    index=False,
)

initial_to_final_comparison.to_csv(
    TABLE_DIR
    / "initial_to_final_comparison.csv",
)

final_contract_shortlist.to_csv(
    TABLE_DIR
    / "final_contract_shortlist.csv",
    index=False,
)

print(
    "Stored final contract rankings:",
    len(final_contract_ranking),
)

print(
    "Stored final shortlist:",
    len(final_contract_shortlist),
)

```

```

2026-08-02 13:39:48,490 - options_risk_engine.database - INFO - Stored 794
rows in final_contract_ranking
2026-08-02 13:39:48,495 - options_risk_engine.database - INFO - Stored 4 r
ows in final_recommendation_summary
2026-08-02 13:39:48,500 - options_risk_engine.database - INFO - Stored 4 r
ows in initial_to_final_comparison
Stored final contract rankings: 794
Stored final shortlist: 57

```

The completed engine applies two distinct recommendation stages.

The first stage evaluates theoretical valuation. A contract receives an initial classification according to whether the forecast-volatility Black-Scholes value exceeds the current ask, after quote-validity and eligibility checks.

The second stage evaluates whether that apparent edge survives purchase at the ask, common scenario repricing, outcome consistency, lower-tail risk, liquidity and the Greek factors responsible for the contract's behaviour.

In this market snapshot, no contract satisfied every Candidate requirement.

The engine produced a 57-contract Watchlist containing 48 initial **Buy** contracts and 9 **Positive Edge** contracts. A further 23 contracts were risk-rejected, 462 were liquidity-rejected and 252 were classified as **No Trade**.

The absence of Candidates is itself a useful decision. It shows that the engine can distinguish an interesting theoretical opportunity from evidence strong enough to support its most demanding classification.

The Watchlist represents contracts requiring further review or monitoring; it should not be interpreted as 57 simultaneous trades.

22. Conclusions, limitations and next steps

Overall conclusion

This project developed an end-to-end options-pricing, portfolio-risk and contract-screening engine.

The completed workflow connects:

- live equity and option-chain data;
- return and realised-volatility feature engineering;
- purged walk-forward volatility forecasting;
- analytical Black–Scholes pricing;
- implied-volatility estimation;
- put-call parity and finite-difference validation;
- Monte Carlo pricing checks;
- synthetic portfolio construction;
- portfolio Greeks;
- full-revaluation stress testing;
- Greek P&L attribution;
- standardised one-contract analysis; and
- final risk-aware contract ranking.

The principal achievement is not the discovery of a universally superior forecasting model or a guaranteed mispriced option. It is the construction of a reproducible and auditable framework that separates forecast evidence, market valuation, executable purchase price, scenario risk and final classification.

Volatility-forecasting findings

The forecasting stage tested 34 configurations using purged walk-forward validation.

The model universe included simple baselines, OLS, Ridge, Lasso, PCA, tree-based methods, HAR-RV and GARCH. Several more complex models achieved small RMSE improvements for individual tickers, but the improvements were not consistent across the seven-underlying universe.

Best-model normalised RMSE generally remained close to or above one, while out-of-sample (R^2) remained negative. The historical mean of the forward realised-volatility target was therefore retained as the final forecast input.

This was a deliberate model-risk decision. Complexity was not selected merely because it fitted the historical sample more closely.

Pricing and numerical validation

The analytical pricing implementation passed the independent numerical checks.

Forecast-volatility call and put values satisfied put-call parity to floating-point precision. Analytical Greeks agreed with finite-difference approximations, and Monte Carlo prices remained close to Black–Scholes values when both methods used the same assumptions.

Comparison with market prices nevertheless showed substantial dispersion, reflecting differences between forecast and implied volatility, variation across strikes, volatility skew and inconsistent quote quality.

Synthetic portfolio-risk findings

The synthetic test portfolio contained 966 option and share positions with gross market value of approximately £17.75 million.

Its aggregate Delta was positive, but the ticker books contained substantial offsetting exposures. Several large books had negative Gamma, meaning first-order Delta understated vulnerability to severe nonlinear movements.

Across 1,000 generated non-BASE stress scenarios:

- approximately 56% produced a portfolio loss;
- fifth-percentile P&L was approximately -£62,581; and
- the worst result was approximately -£258,560.

The worst state combined a common spot fall of approximately 29.2%, an increase of roughly ten volatility percentage points, six elapsed days and a small rate reduction. META was the largest negative contributor, while MSFT provided the largest positive offset.

Greek attribution identified Gamma as the main driver of the worst loss. Fixed initial Greeks explained smaller local shocks more accurately than the largest stresses, confirming that full repricing remains necessary when Greeks change materially.

Standardised contract-analysis findings

The original synthetic positions were unsuitable for comparing individual contracts because they used different directions, quantities and entry prices.

A separate universe was therefore created in which every eligible option was treated as one long contract purchased at the ask with a multiplier of 100.

The standardisation process retained 794 contracts:

- 91 initial Buy ;
- 16 Positive Edge ; and
- 687 Do Not Buy .

Every contract was repriced under the same 1,000 non-BASE scenarios, creating 794,000 comparable stressed outcomes plus 794 BASE observations.

The initial **Buy** group had a median forecast edge of approximately 53.9%, compared with approximately 3.0% for **Positive Edge** and -63.5% for **Do Not Buy**. This confirms that the forecast screen separated contracts by theoretical valuation.

Several WMT, MSFT and AAPL puts generated the largest mean scenario returns, but many high-edge contracts retained expected-shortfall outcomes close to losing the complete premium.

Contracts with the highest gain frequencies or strongest percentage downside were often deep-in-the-money options initially classified as **Do Not Buy**. Valuation, consistency and downside protection therefore describe different properties and cannot be replaced by one metric.

Greek explanation of contract outcomes

Contract-level attribution was completed for all 794,794 contract-scenario rows and all reconciliation checks passed.

Among the 91 initial **Buy** contracts:

- 52 had a Mixed dominant risk classification;
- 36 were Directional; and
- 3 were Convexity-driven.

No initial **Buy** contract was classified as purely Volatility-driven under the dominant-component rule.

Although the initial signal originated from forecast volatility, scenario outcomes were mainly explained by combinations of Delta, Gamma and Theta. Purchasing an option on a volatility view still creates directional, convexity and timing exposure.

Final contract ranking

The final ranking combined forecast edge, average and median scenario return, scenario gain frequency, lower-tail risk and liquidity.

It produced:

- 57 **Watchlist** contracts;
- 252 **No Trade** contracts;
- 23 **Risk Rejection** contracts;
- 462 **Liquidity Rejection** contracts; and
- no final **Candidate** contracts.

Of the 91 initial **Buy** contracts, 48 reached the Watchlist, 17 were rejected for scenario risk and 26 were rejected for liquidity. Nine of the 16 **Positive Edge** contracts also reached the Watchlist.

No contract became a Candidate because none of the initial **Buy** contracts satisfied every absolute rule, particularly the requirement to gain in at least 50% of the generated scenarios.

This is not a failure to select a trade. It demonstrates that the final layer can return no Candidate when the evidence is insufficient.

The Watchlist is concentrated in AMZN, GOOGL and META options, with smaller groups of AAPL contracts and one contract each from MSFT and WMT. It is a ranked research list rather than a diversified portfolio recommendation.

Main limitations

The historical and option-chain inputs represent one live market snapshot. Stored run IDs improve reproducibility, but later executions may receive different prices, strikes, expiries, volumes and open interest.

The volatility forecast uses one value per underlying and cannot reproduce a complete implied-volatility surface, strike skew or term structure.

The generated scenarios are transparent stress distributions rather than calibrated real-world probabilities. All tickers receive the same common percentage spot and volatility shock, which does not represent realistic cross-asset correlations or idiosyncratic company events.

Scenario gain frequency is therefore a frequency within this generated stress set, not a statistically calibrated probability of profit.

The final score is cross-sectional and threshold-dependent. Counts may change when the liquidity gates, scenario distribution, weights or thresholds change. Such changes should be reported as sensitivity tests rather than selected after observing the result.

Black–Scholes also relies on simplified assumptions, including continuous trading, lognormal prices and constant volatility over the pricing interval. The project does not dynamically model early exercise, discrete dividends, market impact, liquidity deterioration or spread widening during stress.

The synthetic portfolio is intentionally large and randomly constructed. Its risk results validate the engine but do not represent a proposed allocation.

Finally, the contract ranking has not yet been tested against subsequently realised option returns.

Next steps

The highest-priority extension is a historical recommendation backtest.

For each historical evaluation date, the engine should:

1. construct the volatility forecast using only information available then;
2. record available option quotes and liquidity;
3. generate initial and final classifications without future information;
4. track selected contracts using executable bid and ask assumptions;
5. compare predicted scenario characteristics with realised option P&L; and
6. test whether Candidates or Watchlist contracts outperform control groups after transaction costs.

Further extensions could include a fitted volatility surface, ticker-specific shocks, empirically estimated correlations, event scenarios, American-option pricing and explicit portfolio construction from the ranked contracts.

Final assessment

The project does not establish that the historical-mean forecast is universally optimal, that every positive model edge represents mispricing or that any current contract should be purchased.

It establishes something more defensible: a validated research and risk framework that can reject weak forecast evidence, distinguish theoretical value from purchase P&L, expose nonlinear option risk and return no final Candidate when the complete evidence does not justify one.

Those distinctions make the project substantially more than a Black–Scholes calculator. It is an auditable options-research and risk-engine workflow with a clear path toward historical validation.